

**TECHNICAL REPORT
MINERAL RESERVES AND RESOURCES,
ANTAMINA DEPOSIT, PERÚ 2010**

**Prepared for Compañía Minera Antamina S.A., owned by:
Teck Resources Limited
BHP Billiton Group
Xstrata PLC
Mitsubishi Cooperation**

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1 Summary

Compañía Minera Antamina S.A. is a Peruvian Sociedad Anonima (a limited liability company) with four shareholders, including BHP Billiton Group with 33.75%, Xstrata PLC with 33.75%, Teck Resources Limited with 22.50% and Mitsubishi Corporation with 10.00%.

The Antamina Deposit is located on the eastern flank of the Western Cordillera in north-central Peru. It is 270 kilometers north of Lima and 130 kilometer from the Pacific coast in an area of rugged peaks, glacial valleys and moderate rainfall. Mining is plans to be conducted between 3683 and 4703 meters above sea level.

The deposit sits at the bottom of a glacial valley surrounded by limestone ridges. It is hosted in a sequence of limestones and sediments, which were strongly deformed by thrusting and folding, then later intruded by intermediate to felsic stocks. The skarn mineralogy consists predominantly of garnet (brown and green) with lesser amounts of diopside, wollastonite, magnetite, chlorite, epidote and calcite zoned around the central intrusion. The general skarn zonation from the intrusive core outward is as follows: pink garnet endoskarn, brown garnet endoskarn, mixed brown and green garnet transitional skarn, mixed brown and green garnet exoskarn, green garnet exoskarn, diopside exoskarn, wollastonite exoskarn, hornfels, marble, limestone. Enriched brecciation zones composed of all skarn types can occur in any lithology type. Before mining, the deposit was partially covered by overburden and lake sediment materials to moderate depths.

Copper occurs mainly as chalcopyrite except for two areas of bornite, representing approximately five percent of the deposit. Zinc generally occurs as sphalerite. Other significant sulphides include molybdenite and pyrite, while trace amounts of silver and bismuth bearing minerals and local areas of galena are found.

Metal zonation is quite distinctive within the deposit. Copper occurs relatively evenly distributed from endoskarn to the limestone contact. Zinc and bismuth tend to occur within 70 meters of the contact of green garnet skarn with limestone/marble/hornfels. Molybdenite is generally located within the intrusive core and the surrounding endoskarn. Silver is present in any of the skarn lithologies. Lead is generally located in green garnet exoskarn, diopside exoskarn and hornfels. However veins and blebs of tennantite and other minerals can be found as rare occurrences in any rock type at Antamina.

Commencing in August 1996, Compañía Minera Antamina S.A. performed an extensive exploration and development program to define the deposit. A resource model was built in 1997 to provide input to the feasibility study. Work performed on the property up to 1999, includes metallurgical testing, check assaying of previous drill hole and underground tunnel pulps, geologic mapping, geotechnical core logging and mapping, 135 kilometers of core drilling, 6 kilometers underground tunnels, and 225 m of drifting for bulk sampling. As well during the development and feasibility work, over 400 tonnes of metallurgical samples, representing all parts of the deposit, were taken and shipped to labs in Canada for various bench scale flotation and milling tests as well as pilot plant tests.

Drilling has continued in the deposit area, with drilling programs conducted in 1999, 2000, 2002, 2003, 2004, 2005 and 2007-2008. This last drilling program was focused on the addition of ore reserves to support the Antamina Expansion Project. As a result a resource model was generated in 2009 and an update of this model was conducted in 2010 which is the base for the reserves and resources report.

To date more than 280,000 samples have been analyzed for some or all of the following: Cu, Zn, Ag, Mo, Bi, As, Pb and Co as well as more than 20 other elements. All analysis was carried out under a strict quality control and assurance program.



Antamina Geology conducted an exercise to update the Resource Model from February through August 2010. The motivation was to enhance the confidence on the estimation and classification of the waste material surrounding the skarn. For the mineralized portion of the deposit this version of the resource model is a replication of the 2009 Resource Model except for the update of the top 5 benches below August 20, 2010 topography that was used in the estimation of the Mid Term Resource Model. The Limestone/marble and Hornfels barren units were divided into 10 rock types in the updated geological model. On the other hand, the 2009 Resource Model is an incremental update to the 2008 Resource Model and included over 150,000 m of drilling completed from September 2007 through December 2008. All estimates in this report are referred to the 2010 Resource Model although the estimation methodology is referred to the exercise for the construction of the 2009 Resource Model.

Additional mine engineering work has established a pit design and a life-of-mine plan based on the Antamina 2010 Resource Model and updated mine parameters. The mineral reserve within the current life-of-mine plan (based on the 2010 reserve pit design) is shown in Table 1.1, the mineral resources is shown in Table 1.2; and are reported as defined in CIM Standards on Mineral Resources and Reserves: Definitions and Guidelines.

It should be noted that zinc is not recovered from copper-only ore and molybdenum is usually not recovered from copper-zinc ore.

Table 1.1: Statement of Mineral Reserve Estimate as of January 1st 2011.

Classification	Ore (Mtonnes)	Cu (%)	Zn (%)	Ag (g/t)	Mo (%)
Proven Copper Ores	100	1.07	0.16	8.3	0.034
Proven Copper Zinc Ores	47	0.84	1.83	15.0	0.007
Probable Copper Ores	492	0.95	0.14	8.9	0.027
Probable Copper Zinc Ores	183	0.83	2.00	14.5	0.006
Total Proven & Probable Reserves	822	0.93	0.66	10.4	0.022

Table 1.2: Statement of Mineral Resource Estimate (Incremental)* as of January 1st 2011.

Classification	Ore (Mtonnes)	Cu (%)	Zn (%)	Ag (g/t)	Mo (%)
Measured Copper Ores	35	0.46	0.12	4.6	0.037
Measured Copper Zinc Ores	15	0.46	0.85	9.1	0.024
Indicated Copper Ores	276	0.93	0.12	9.2	0.023
Indicated Copper Zinc Ores	78	0.87	1.90	13.6	0.006
Meas. + Indicated Resources Total	404	0.86	0.49	9.6	0.021
Inferred Copper Ores	531	0.79	0.11	9.2	0.018
Inferred Copper Zinc Ores	177	0.54	1.33	9.7	0.003
Inferred Resources Total	708	0.73	0.42	9.3	0.014

* None of the Material identified as Mineral Reserve is included in the quantities stated as Mineral Resource.

The resource blocks are further identified into the following ore types: Cu low Bi (type 1), Cu high Bi (type 2), Cu-Zn low Bi (type 3), Cu-Zn high Bi (type 4), Cu bornite (type 5) and bornite with Cu-Zn (type 6). In addition other ore subtypes, such as Cu very high Bi (type 2A) and Cu-Zn very high Bi (type 4A) are identified. The identification of ore types allows mine planners to design ore campaigns for the concentrator that will ensure the best possible concentrate quality. The resource blocks are identified as Measured, Indicated or Inferred per the **CIM Standards of Mineral Resources and Reserves: Definitions and Guidelines**.

The property is being mined by open pit mining methods as a conventional shovel and truck operation for over ten years. Pre-production stripping started in late 1998 and average daily production is currently approximately 400,000 tonnes per day. A new expansion project to upgrade the concentrator production level to 130,000 tpd is under construction and it is scheduled to be fully operative by the end of 2011.



2 Introduction

The technical report is prepared for Compañía Minera Antamina S.A.; the purpose of this report is to update the 2008 technical report based in more recent information including the approved expansion project.

This report is written in support of the formal declaration of Mineral Reserves and Mineral Resources as of January 1st 2011 as required under National Instrument 43-101 of the Canadian Securities Administrators. The format and content of the report conforms to Form 43-101F1.

Primary reference documents used in this report include, but are not limited to: *Antamina Project Feasibility Study*, March 1998; *Antamina Project Updated Feasibility Study*, January 1999; a series of memoranda prepared by Mineral Resources Development Inc. (MRDI) during the estimation of the August 2000 Resource Model; a series of memoranda prepared by AMEC during the estimation of the 2003 Interim Resource Model; the 2005 Resource Model, the 2006 Resource Model, the 2008 Resource Model and the 2009 Resource Model reports prepared by AMEC, the 2010 Resource Model update prepared by Antamina geology team, the Expansion Program Feasibility Report 2009 and a number of Compañía Minera Antamina S.A. internal memoranda.

In the capacity of Development Geology Superintendent, Jhon Espinoza M. is the Mineral Resource Qualified Person. He has been employed on the Antamina site since November 1999. He has intimate knowledge of Antamina deposit, field geology and has participated in the resource model estimation since 2000. He has conducted personal inspection during the construction of the 2010 Resource Model and he provided guidance in the definition of the geologic and statistical framework for the grade estimation of the 2010 and 2009 Resource Models.

In the capacity of Mine Engineering Superintendent, Luis Lozada is the Mineral Reserve Qualified Person. He has been employed on the Antamina site since January 2000. He has intimate knowledge of the Antamina deposit and its mining and ore processing considerations. He has personally inspected the work done for reporting the reserves, including the reported ore tonnage, the life-of-mine plan, reconciliation report and has prepared the mineral reserves and mineral resource report.

3 Reliance on Other Experts

Historical information has been taken from the Antamina Feasibility Study, March 1998 and the Antamina Project Updated Feasibility Study, January 1999.

Underground tunnel assay data was verified against historical underground maps as part of the 2003 Interim Resource Model study. This data had been previously audited and used in the Feasibility Resource Model.

The authors have relied on the Compañía Minera Antamina S.A. Legal Department to prepare sections: 4.1 Area of Property (Surface Rights), 4.3 Mining Concessions, 4.4 Nature of Title, 4.7 Royalties, 4.9 Permits Required for Operation, and Appendix B – Main Permits Required for Operations.

The authors have relied on information provided by the Compañía Minera Antamina S.A. Concentrator department to prepare sections: 16 Mineral Processing and Metallurgical Testing, 23.2.2 Processing, and 23.2.3 Metallurgy.

The authors have relied on information provided by the Compañía Minera Antamina S.A. Environmental Department to prepare Sections: 4.8 Environmental Liabilities, and 23.6 Environmental Considerations.

The authors have relied on information provided by the Compañía Minera Antamina S.A. Finance Department in preparing Sections: 23.7 Taxes, and 23.9 Economic Analysis.



Technical and economic parameters used for reserve definition were developed based on the Antamina Price and Cost Protocols. The protocols are explained in Section 23.

Therefore the authors assume this information is correct and no effort has been made to verify information beyond their personal knowledge of its content. The authors have no reason to believe that any such information is incorrect.

4 Property Description and Location

The following sections describe the location, nature of the Antamina claims, and permits.

4.1 Area of Property (Surface Rights)

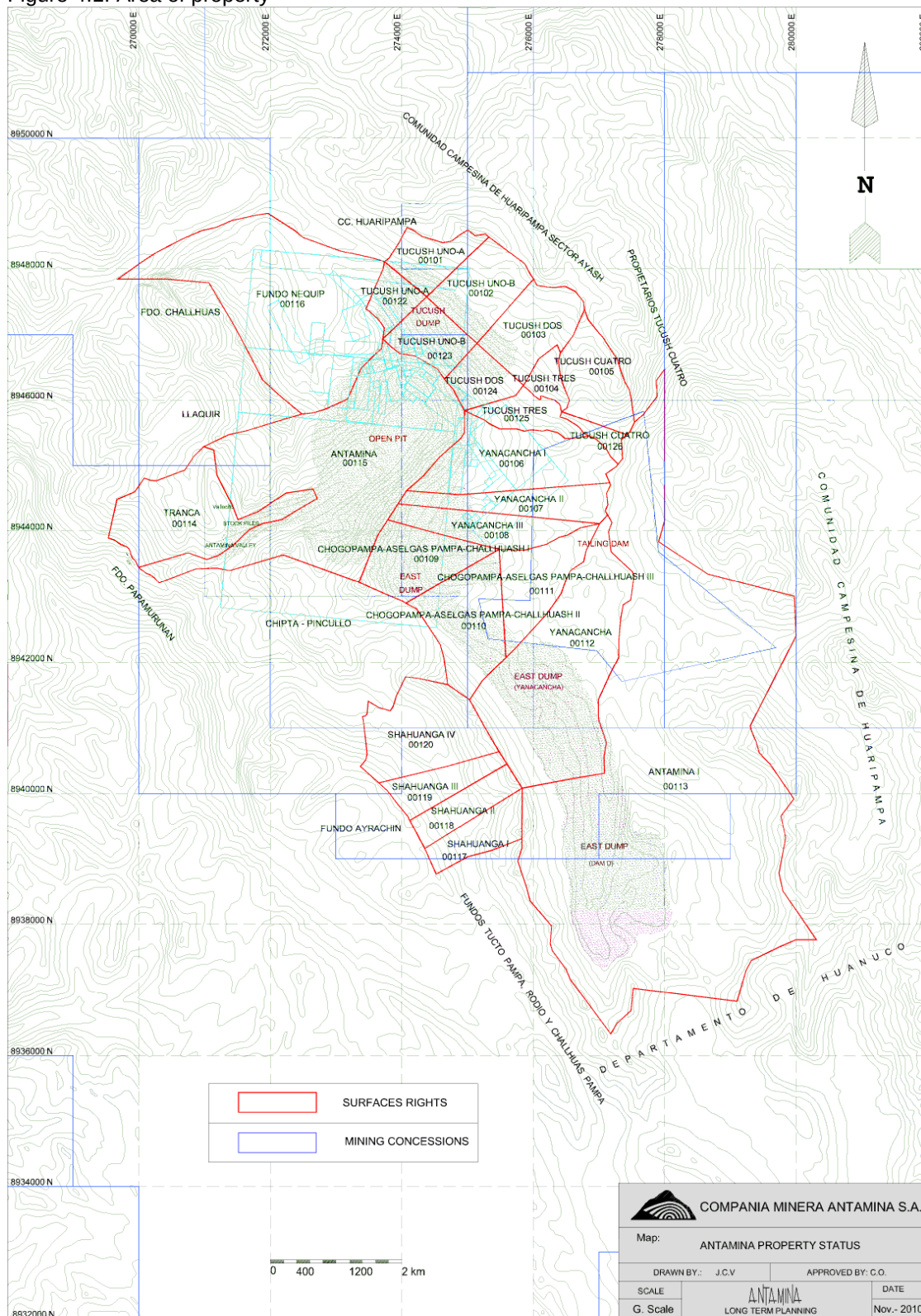
The surface rights to the mine site are listed in Table 4.1 and shown in Figure 4.1.

Table 4.1: Surface Rights

Surface Rights	Area (ha)
Antamina	779
Tranca	260
Yanacancha (Huaripampa)	2,202
Fundo Yanacancha	519
Yanacancha I	234
Yanacancha II	117
Yanacancha III	117
Chogopampa Aselgaspampa Challhuash I	142
Chogopampa Aselgaspampa Challhuash II	189
Chogopampa Aselgaspampa Challhuash III	142
Shahuanga I	83
Shaguanga II	81
Shaguanga III	81
Shaguanga IV	217
Tucush Uno-A (UC 00101)	80
Tucush Uno-A (UC 00122)	52
Tucush Uno-B (UC 00123)	86
Tucush Uno-B (UC 00102)	134
Tucush 2 (UC 00103)	143
Tucush 2 (UC 00124)	63
Tucush 3 (UC 00104)	42
Tucush 3 (UC 00125)	61
Tucush 4 (UC 00126)	22
Tucush 4 (UC 00105)	131
Nequip	611



Figure 4.1: Area of property



To the knowledge of the authors, these properties encompass all the rights necessary above and immediately adjacent to the deposit that are required to effect exploitation of the deposit for which Reserves and Resources are being stated in this report. As well, Antamina has acquired additional surface rights, rights-of-way and easements for establishment of the infrastructure elements necessary for the functional exploitation of the deposit including the pipeline, power supply line, access road and port.

4.2 Property Location

The Antamina Deposit is located in the Central Andes of northern Perú at approximately 9°32'17"S latitude and 77°03'51"W longitude and 4,300 meters altitude. It is 270 kilometers north of Lima and 103 kilometers east of the Pacific Ocean and lies on the eastern side of the Western Cordillera in the upper part of the Rio Marañón basin, a tributary of the Amazon (Figure 4.2). All of the mining concessions are located in San Marcos District, Province of Huari, Ancash Department.

Figure 4.2: Location of Antamina

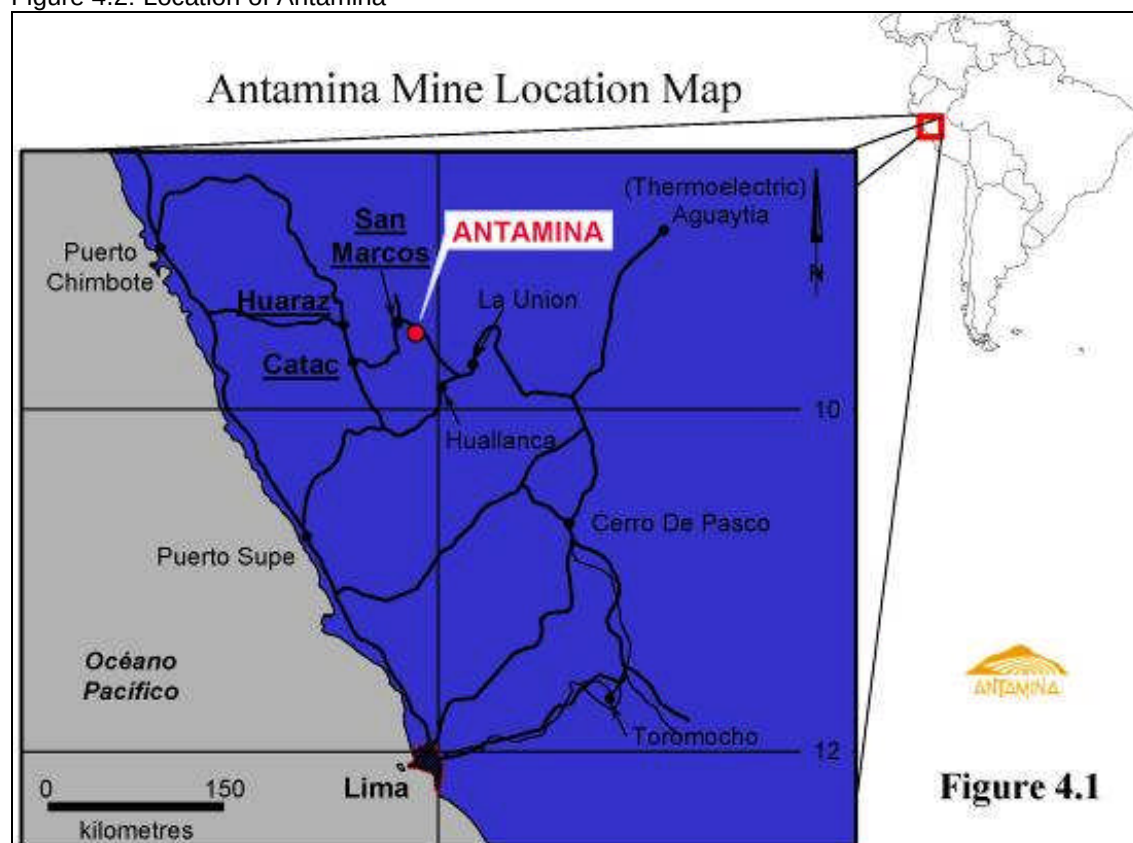


Figure 4.1



4.3 Mining Concessions

A list of the Mining Concessions is shown in Table 4.2.

Table 4.2: List of the Mining Concessions

List of Mining Concessions of the Administrative Economic Unit: Antamina

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	NUEVA ANTAMINA PRIMERA	09003246X01	990	San Marcos	Huari	Ancash
2	ANTAMINA NORTE	09000613Y01	330	San Marcos	Huari	Ancash
3	ANTAMINA NORTE PRIMERA	09000614Y01	495	San Marcos	Huari	Ancash
4	ANTAMINA 30	09001865X01	4	San Marcos	Huari	Ancash
5	ANTAMINA 33	09001868X01	2	San Marcos	Huari	Ancash
6	ANTAMINA 34	09001869X01	1	San Marcos	Huari	Ancash
7	ANTAMINA 37	09001872X01	1	San Marcos	Huari	Ancash
8	ANTAMINA 38	09000520Y01	2	San Marcos	Huari	Ancash
9	ANTAMINA 39	09001874X01	64	San Marcos	Huari	Ancash
10	ANTAMINA 40	09001875X01	77	San Marcos	Huari	Ancash
11	ANTAMINA 41	09001876X01	12	San Marcos	Huari	Ancash
12	ANTAMINA 42	09001877X01	1	San Marcos	Huari	Ancash
13	ANTAMINA 43	09001878X01	2	San Marcos	Huari	Ancash
14	ANTAMINA 44	09001879X01	1	San Marcos	Huari	Ancash
15	ANTAMINA 45	09000506Y01	1	San Marcos	Huari	Ancash
16	ANTAMINA 47	09001882X01	1	San Marcos	Huari	Ancash
17	ANTAMINA CUARENTIOCHO	09001883X01	3	San Marcos	Huari	Ancash
18	ANTAMINA 49	09001884X01	4	San Marcos	Huari	Ancash
19	ANTAMINA 50	09001885X01	3	San Marcos	Huari	Ancash
20	ANTAMINA 51	09001886X01	1	San Marcos	Huari	Ancash
21	ANTAMINA 52	09001887X01	1	San Marcos	Huari	Ancash
22	ANTAMINA N° 53	09001888X01	1	San Marcos	Huari	Ancash
23	ANTAMINA 54	09001889X01	3	San Marcos	Huari	Ancash
24	ANTAMINA 55	09001950X01	1	San Marcos	Huari	Ancash
25	ANTAMINA 69	09001904X01	2	San Marcos	Huari	Ancash
26	ANTAMINA SETENTIUNO	09001906X01	6	San Marcos	Huari	Ancash
27	ANTAMINA 72	09001907X01	2	San Marcos	Huari	Ancash
28	ANTAMINA No 73	09001908X01	5	San Marcos	Huari	Ancash
29	ANTAMINA 83	09003256X01	2	San Marcos	Huari	Ancash
30	CASUALIDAD	09000910X01	10	San Marcos	Huari	Ancash
31	CASUALIDAD II	09000990X01	18	San Marcos	Huari	Ancash
32	EL ESTANQUE-10	09000561Y01	4	San Marcos	Huari	Ancash
33	FORTUNA	09000402Y02	4	San Marcos	Huari	Ancash
34	FORTUNA No. 2	09000928X01	24	San Marcos	Huari	Ancash
35	FORTUNA No. 3	09000080X01	10	San Marcos	Huari	Ancash
36	FORTUNA No. 4	09000974X02	24	San Marcos	Huari	Ancash
37	FORTUNA 5	09001402X01	10	San Marcos	Huari	Ancash
38	FORTUNA 7	09001403X01	20	San Marcos	Huari	Ancash
39	FORTUNA No. 8	09000424Y01	2	San Marcos	Huari	Ancash
40	JULIA ELOISA	09000900X01	8	San Marcos	Huari	Ancash
41	LA RECOMENDADA-12	09003787X01	15	San Marcos	Huari	Ancash
42	LA TRIUNFADORA 117	09000548Y01	1	San Marcos	Huari	Ancash
43	RECOMPENSA	09000431Y01	6	San Marcos	Huari	Ancash
44	RECOMPENSA No. 2	09001061X01	2	San Marcos	Huari	Ancash
45	CHAPI	09000429Y01	6	San Marcos	Huari	Ancash
46	MARGARITA	09000451Y01	2	San Marcos	Huari	Ancash
47	MERCEDES	09000462Y01	10	San Marcos	Huari	Ancash
48	PODEROSA	09000430Y01	8	San Marcos	Huari	Ancash
49	PODEROSA DOS	09000474Y01	2	San Marcos	Huari	Ancash
50	PODEROSA TRES	09000475Y01	4	San Marcos	Huari	Ancash
51	SAN FRANCISCO	09000407Y01	4	San Marcos	Huari	Ancash
52	VIVEZA	09001070X01	2	San Marcos	Huari	Ancash
53	EL TESORO	09000418Y01	10	San Marcos	Huari	Ancash
54	FLOR ANDINA	09000494Y01	6	San Marcos	Huari	Ancash
55	PROLONGACION	09000473Y01	6	San Marcos	Huari	Ancash
56	SARINA	09001173X01	5	San Marcos	Huari	Ancash
57	ÑAHUINPUQUIO	09000962X01	6	San Marcos	Huari	Ancash
58	EL RECUERDO	09000476Y01	12	San Marcos	Huari	Ancash
59	ANTAMINA-ESTE	10909495	300	San Marcos	Huari	Ancash
60	CONTONGA No. 1-A	0900597AY01	88	San Marcos	Huari	Ancash
61	ANTA 96 1	10236496	200	San Marcos	Huari	Ancash
62	ANTA 96 2	10240996	100	San Marcos	Huari	Ancash
63	ANTA 96 3	10241096	100	San Marcos	Huari	Ancash
64	ANTA 97 1	10092597	100	San Marcos	Huari	Ancash
65	RECUAY 11	10486495	1,000	San Marcos	Huari	Ancash
66	RECUAY 12	10486295	1,000	San Marcos	Huari	Ancash
67	RECUAY 13	10396695	1,000	San Marcos	Huari	Ancash
68	RECUAY 20	10489895	1,000	San Marcos	Huari	Ancash
69	RECUAY 21	10128695	600	San Marcos	Huari	Ancash



List of Mining Concessions of the Administrative Economic Unit: Antamina 1

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ROSITA DE ORO	09000405Y01	14	San Marcos	Huari	Ancash
2	RECUAY 22	10523295	1,000	San Marcos	Huari	Ancash
3	RECUAY 23		1,000	San Marcos	Huari	Ancash
4	RECUAY 24	10523695	1,000	San Marcos	Huari	Ancash
5	RECUAY 29	10487195	1,000	San Marcos	Huari	Ancash
6	RECUAY 30-2000	10158600	200	San Marcos	Huari	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 2 - Huarmey

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	HUARMEY 1	10355395	1,000	Huarmey	Huarmey	Ancash
2	HUARMEY 2	10355195	500	Huarmey	Huarmey	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 3

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 30	10528406	1,000	San Marcos	Huari	Ancash
2	ANTA 31	10528506	1,000	San Marcos	Huari	Ancash
3	ANTA 32	10528606	1,000	San Marcos	Huari	Ancash
4	ANTA 33	10528706	1,000	San Marcos	Huari	Ancash
5	ANTA 35	10528806	900	San Marcos	Huari	Ancash
6	ANTA 34	10528906	1,000	San Marcos	Huari	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 4

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 26	10527906	1,000	Llata	Humalies	Huanuco
2	ANTA 27	10528106	1,000	Llata	Humalies	Huanuco
3	ANTA 28	10528206	1,000	Llata	Humalies	Huanuco
4	ANTA 29	10528306	1,000	Llata	Humalies	Huanuco

List of Mining Concessions of the Administrative Economic Unit: Antamina 5

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 24	10527806	1,000	San Marcos	Huari	Ancash
2	ANTA 25	10528006	1,000	San Marcos	Huari	Ancash
3	ANTA 36	10529006	1,000	San Marcos	Huari	Ancash
4	ANTA 37	10529106	500	San Marcos	Huari	Ancash
5	ANTA 38	10529306	600	San Marcos	Huari	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 6

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 21	10527406	1,000	San Marcos	Huari	Ancash
2	ANTA 20	10527506	1,000	San Marcos	Huari	Ancash
3	ANTA 22	10527606	1,000	San Marcos	Huari	Ancash
4	ANTA 23	10527706	1,000	San Marcos	Huari	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 7

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ROBYN 1	10133398	800	San Marcos	Huari	Ancash
2	ROBYN 2	10133498	1,000	San Marcos	Huari	Ancash
3	ROBYN 3	10133598	1,000	San Marcos	Huari	Ancash
4	REBECA	09000452Y01	8	San Marcos	Huari	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 8

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 42	10529606	800	San Pedro de Chana/Miraflores/Puños	HUARI / SANTA / HUAMALIES	ANCASH / HUANOUCO
2	ANTA 43	10529706	800	San Pedro de Chana/Miraflores/Puños	HUARI / SANTA / HUAMALIES	ANCASH / HUANOUCO
3	ANTA 44	10529806	900	San Pedro de Chana/Miraflores/Puños	HUARI / SANTA / HUAMALIES	ANCASH / HUANOUCO

List of Mining Concessions of the Administrative Economic Unit: Antamina 9

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 16	10527106	1,000	Huachis / Ponto / San Pedro de Chana	Huari	Ancash
2	ANTA DIECIOCHO 2007	10527206	1,000	Huachis / Ponto / San Pedro de Chana	Huari	Ancash
3	ANTA 19	10527306	1,000	Huachis / Ponto / San Pedro de Chana	Huari	Ancash
4	ANTA 39	10529206	900	Huachis / Ponto / San Pedro de Chana	Huari	Ancash
5	ANTA 41	10529406	800	Huachis / Ponto / San Pedro de Chana	Huari	Ancash
6	ANTA 40	10529506	800	Huachis / Ponto / San Pedro de Chana	Huari	Ancash



List of Mining Concessions of the Administrative Economic Unit: Antamina 10

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 12	10526506	1,000	Huachis / Ponto/Rahuapampa / San Pedro de Chana	Huari	Ancash
2	ANTA 13	10526606	1,000	Huachis / Ponto/Rahuapampa / San Pedro de Chana	Huari	Ancash
3	ANTA 14	10526806	1,000	Huachis / Ponto/Rahuapampa / San Pedro de Chana	Huari	Ancash
4	ANTA 15	10526906	1,000	Huachis / Ponto/Rahuapampa / San Pedro de Chana	Huari	Ancash
5	ANTA DIECISIETE 2006	10527006	1,000	Huachis / Ponto/Rahuapampa / San Pedro de Chana	Huari	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 11

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 3	10525706	900	San Marcos	Huari	Ancash
2	ANTA 4	10525806	700	San Marcos	Huari	Ancash
3	ANTA 5	10525906	1,000	San Marcos	Huari	Ancash
4	ANTA NUEVE 2006	10526306	1,000	San Marcos	Huari	Ancash
5	ANTA DIEZ 2006	10526406	1,000	San Marcos	Huari	Ancash
6	ANTA 11	10526706	1,000	San Marcos	Huari	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 12

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA UNO 2007	10525506	700	Cajay / Huari / Huachis / Masin / Rahuapampa	Huari	Ancash
2	ANTA 2	10525606	1,000	Cajay / Huari / Huachis / Masin / Rahuapampa	Huari	Ancash
3	ANTA OCHO 2007	10526006	800	Cajay / Huari / Huachis / Masin / Rahuapampa	Huari	Ancash
4	ANTA SEIS 2007	10526106	1,000	Cajay / Huari / Huachis / Masin / Rahuapampa	Huari	Ancash
5	ANTA SIETE 2007	10526206	900	Cajay / Huari / Huachis / Masin / Rahuapampa	Huari	Ancash
6	ANTA 45	10529906	700	Cajay / Huari / Huachis / Masin / Rahuapampa	Huari	Ancash

List of Mining Concessions of the Administrative Economic Unit: Antamina 13 - Huarmey

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	HUARMY 4	10056600	400	Huarmey	Huarmey	Ancash
2	HUARMY 5	10056700	500	Huarmey	Huarmey	Ancash
3	HUARMY 6	10098000	1,000	Huarmey	Huarmey	Ancash
4	HUARMY 3-2000	10190200	100	Huarmey	Huarmey	Ancash
5	VULCANO	10208798	400	Huarmey	Huarmey	Ancash

List of Mining Concessions without an Administrative Economic Unit

No.	Name	Code	Area (ha)	Location		
				District	Province	Region
1	ANTA 46	01-02080-09	1,000	San Marcos	Huari	Ancash
2	ANTA 47	01-02174-09	700	San Marcos	Huari	Ancash

To the knowledge of the authors, the mining concessions listed above are in good standing and constitute all of the mineral rights that are required to permit exploitation of the deposit for which Reserves and Resources are being stated in this report.

4.4 Nature of Title

Annually, the titleholder of mining concessions is obligated to pay an amount of money for validity fees ("derechos de vigencia"), depending on the nature of concessions held.

By means of Legislative Decree No. 913 and after several changes, the cost of validity fees for mining concessions has been established at US \$3.00 per hectare (petitioned or granted). Notwithstanding these changes are not applicable to concessions which are part of UEA Antamina and UEA Antamina 1 and two concessions in Huarmey which still pay US \$2.00 per hectare due to the fact that Compañía Minera Antamina S.A. entered into a stability agreement with the Peruvian State that has stabilized for fifteen years the tax regime in place in the year 1998.

Also Compañía Minera Antamina S.A. pays validity fees for its Beneficiation Concession (Huincush) and it's Transportation Concession (Pipeline). These payments are calculated based on the approved production capacity (104,000 TM) in the first case and the pipeline length (302.0 kilometer) in the latter.

Notwithstanding, since 2010 Compañía Minera Antamina S.A. has been paying validity fees for the Beneficiation Concession (Huincush) based on the production capacity (130,000 TM), which will be reached at the end of 2011.



The validity fees payment must be made by the last working day of June each year and the payment must be recorded before the Public Registry of Mining during July.

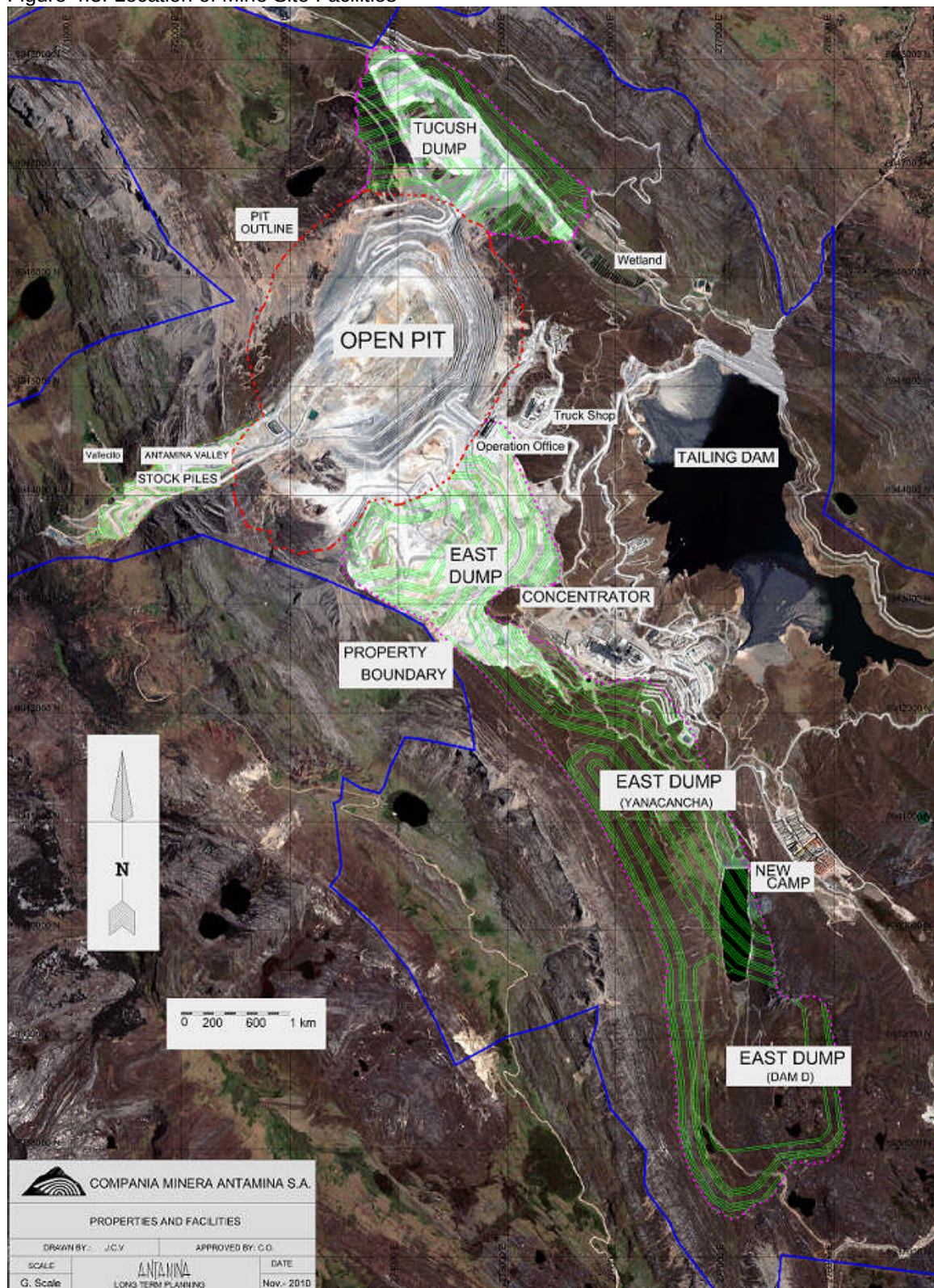
Surface rights purchased by Compañía Minera Antamina S.A. have been duly registered in the Public Registry (Registro de Propiedad Inmueble de la Oficina Registral – Región Chavin).

4.5 Location of Property Boundaries

The property boundaries were located based on extensive surface mapping and geophysical survey, identifying the extent of the mineralized areas and zones with potential for skarn type mineralization. Condemnation drilling was also conducted to define areas for facilities. Figure 4.3 displays the area of operations, including, the mine, tailings storage facility, waste dumps, concentrator and the camp relative to the surface property holdings.

[Due to document format reasons, please find the graphic of this section in the next page]

Figure 4.3: Location of Mine Site Facilities





4.6 Location of Known Mineralized Zones

Figure 4.3 displays the area of operations, including, the mine, tailings storage facility, waste dumps, concentrator and the camp relative to the surface property holdings.

4.7 Royalties

Since the year 2004 and by means of Law 28258 and complementary regulations, the Peruvian State has imposed mining royalties on mining companies. These royalties are defined as a consideration for the exploitation of the mineral resource, and are calculated based upon the value of the concentrate (based on the international market prices). This value is based on various percentages of the concentrate value depending on the total value of the concentrate. The maximum level is 3% for more than US \$120 Million of value of concentrates. Compañía Minera Antamina S.A. is not paying royalties since it has a stability agreement signed with the Peruvian government in which it has been granted tax and administrative stability for fifteen years beginning January 2001. The tax and administrative regime that is stabilized is the one registered September 16, 1998 (the date on which the feasibility study of the project was approved). The stability agreement thus ends in the year 2016, and royalties are considered applicable in the financial analysis done for the period 2016-2020. In addition, there is a "voluntary contribution" amounting to 3.75 percent of after tax profits to go into a development fund for use in the impacted areas of Ancash province.

Of the Antamina's mining property, the concessions Recuay 11, Recuay 12, Recuay 13, Recuay 20, Recuay 21, Recuay 22, Recuay 23, Recuay 24 and Recuay 29, which have not been exploited and the need to is yet unknown, are subject to a 2.5% contractual royalties in favor of Rio Tinto Mining & Exploration Limited. These do not affect the rest of the concessions of the deposit, which are free of any contractual royalties or back-in rights.

4.8 Environmental Liabilities

In the long term, mine closure liabilities will include the following:

- A tailings impoundment containing approximately 1.1 billion tonnes of flotation tailings in storage behind a rock fill dam.
- Waste rock stockpiles containing a total of approximately 3.4 billion tonnes of rock excavated from the open pit, which will be recontoured and revegetated to reclaim the land for alternative use.
- Surface infrastructure, including the concentrator, camp, warehouse, port and other facilities, that will need to be removed, sold or turned over to Peruvian authorities for alternative use.
- A possible need to treat water draining from the Ayash valley in order to reduce trace metals and sediments to acceptable levels for discharge to receiving streams.
- A possible need to treat water in the Antamina Valley prior to discharge.

Compañía Minera Antamina S.A. will address these liabilities to the extent possible during the operation of the mine by implementing a progressive reclamation program. However, these liabilities will not be eliminated until following mine closure.

There is a significant remaining environmental liability for the project associated with Laguna Canrash. Revegetation efforts have largely controlled the erosion of the banks and resulted in much improved water quality. Stabilization work along this section of road continues. However, while stabilization work is ongoing, Compañía Minera Antamina S.A. will continue to operate two



flocculation systems at Laguna Canrash, to minimize sediment inputs to the lake and to the stream draining it.

A closure plan complying with Peruvian law is currently on file with the Ministry of Energy and Mines. This plan considers temporary, progressive, final, and post-closure activities for all facilities; and more specifically, physical, biological, socio-economic, and cultural characteristics of closure. Included are the details for facility dismantling, demolition, post-closure stability, and long term maintenance (water management/treatment systems, socio-economic support, land use/reclamation, schedules and financial provisions). Engineering and designs are required to be provided to a feasibility level and must be updated every five years or within 12 months of any EIA or modification to EIA. A conceptual closure plan is provided in the EIA (Environmental Impact Assessment) that was approved in April 2008, and a final closure plan was completed and submitted to the government in March 6, 2009 and subsequently approved in November 18, 2009. This filing will supplant the current closure plan that had been submitted to the Ministry of Energy and Mines.

4.9 Permits Required for Operation

By the end of April, 2001, Antamina had identified a total of 320 permits required for the construction and operation of the project. Due to changing legal requirements and additional identified permit requirements, Antamina has obtained more than 400 permits to date, see Appendix B with the most important current permits and licenses.

5 Accessibility, Climate, Local Resources, Infrastructure & Physiography

The topography in the area is characterized by steep, sharp limestone ridges and peaks, generally at 4,500 to 4,800 m altitude, but up to a maximum of 5,073 meters. There are short glacial valleys with lakes and deep, steep-sided river canyons and valleys. The dominant ridge and valley trend is northwest, reflecting the regional structural and tectonic fabric with shorter structurally controlled northeast trending valleys such as Antamina. The vegetation is poorly developed and predominantly consists of a highland grass called "ichu" mainly used for pasture and in the construction of shelter.

Access to the mine site is by an all-weather chip sealed road. The mine road connects at the Peruvian National Highway 14 at Conococha Lake. Highway 14 connects to the Pan American highway with the city of Huaraz via Peruvian National Highway 3N. The closest town to the mine site is San Marcos, 38 kilometers by dirt road. Huaraz is the closest city to the mine site, 200 kilometers by paved road or 156 kilometers by partial dirt road.

The Antamina site ambient air temperatures range from an hourly maximum of 15.3°C to an hourly minimum of minus 0.1°C and the rainfall averages 1870 mm per year. These conditions are appropriate to conduct mining operation through the year. Occasional interruptions in the mining activities may be due to strong lightning storms.

Based on the current Life of Mine Plan Compañía Minera Antamina S.A. surface rights are sufficient for mining, tailings disposal, waste disposal, processing and required infrastructure. These facilities are shown in relation to the property boundary in Figure 4.3.

6 History¹

6.1 Early History

The Antamina valley has seen limited mineral production by indigenous peoples for centuries. The first recorded owner and operator at Antamina was Leopold Pflucker in 1850. He built a small copper and lead smelter at Juproc using coal from nearby outcrops. The Italian naturalist Antonio

¹ Bechtel International Inc., *Antamina Project Feasibility Study*, March 1998, vol. 2, pp. 15-16.



Raymondi visited the area in November 1860 and found the smelter to be producing lead ingots of 35 kg containing 20 to 25 ounces of silver.

In 1903 Vicente Lezameta mined at Antamina and produced copper matte at a grade of 32%. Mining was stopped and then resumed in 1912 to 1914 with an unsuccessful attempt to leach copper.

With the start of the World War I in 1914, there was a search for new copper deposits and several geologists visited Antamina, including E. Diez Canseco, D. J. McLaughlin, J. L. Gilden, and A. H. Means.

In 1925 A. H. Means visited Antamina for Northern Perú Copper and recommended a diamond drill program. Eight holes (totaling 780 m) were drilled looking for a porphyry copper deposit and Northern Perú Copper dropped the property after failing to obtain favorable results.

6.2 Cerro de Pasco 1952 – 1971

The Cerro de Pasco Corporation was the first company to carry out exploratory work of any magnitude. Its work was confined to the steep slopes on the East side of the deposit where the topography allowed easy underground access by means of adits, at several levels.

Some 32 diamond drill holes totaling 3,200 meters, were completed, 18 from surface and 14 from underground. In addition, Cerro drifted and crosscut 4,300 meters within the eastern zone and drove raises totaling 220 meters in the heart of the zone. The objective was to prove up a high grade copper deposit and to this end; Cerro defined over one million tonnes averaging better than 3.0% copper and a lower grade reserve of 10 million tonnes.

On October 30, 1970, all of the mining assets owned by Cerro were transferred to the Government of Perú.

6.3 Minero Perú and Geomin 1971 –1981

Following expropriation, 2,200 hectares of mining rights were passed to Minero Perú, the mining administration agency of the Government of Perú, which in 1974 formed the Empresa Minera Especial (EME) in partnership with the Government of Romania mining agency called Geomin.

EME carried out a careful and methodical program of work on the property culminating in a full feasibility study. The caliber of the work done is high and although much of it required updating, the resulting database provided a firm base to build on.

EME completed a series of full feasibility studies of Antamina based on the proven and probable reserves determined from the drilling and underground sampling. The studies included full engineering appraisals of all aspects, including open pit design, mine equipment selection, concentrator design, all surface facilities, local social impact, geotechnical studies, marketing and economic analysis, etc. Bench and pilot plant metallurgical work was done in the period 1975 to 1978 in Romania.

Several studies were completed at different mining rates. The basic mining plan involved an initial open pit producing 10,000 tonnes per day of ore for seven years then 20,000 tonnes per day for 13 years. EME update the initial study in 1978, 1979 and 1982. Lower rates of production were addressed from 2,500 to 5,000 tonnes per day, with the objective of limiting the capital investment.



6.4 1981 - Present Day

Due to its failure to finance the project, EME was disbanded in the 1981-82 period. In the ensuing years, Minero Perú continued its studies to the extent that there were over 100 reports on the project.

In 1992, Minero Perú used the above studies as a basis for an attempt to market Antamina and produced an Investment Compendium that was not widely circulated, and the sales effort failed. Then as socio-economic conditions improved under President Fujimori, the Antamina property was transferred to Centromin and became part of its sale package in 1993.

In 1995 and 1996 Rio Algom Limited and Inmet Mining Corporation, both of Canada, conducted extensive reviews of the project culminating in the formation of a partnership to bid on Antamina and the subsequent successful bid in early 1996. Shortly afterward Rio Algom and Inmet formed Compañía Minera Antamina S.A. as a 50:50 owned company.

In 1998 Inmet sold its interest in Compañía Minera Antamina S.A. to two other Canadian companies and Compañía Minera Antamina S.A. was restructured under an ownership of 37.5% Rio Algom, 37.5% Noranda Inc., and 25% Teck Corporation. In 1999, the ownership was further modified as each of the three partners sold 10% of their interest to Mitsubishi Corporation, resulting in the ownership of 33.75% Rio Algom, 33.75% Noranda, 22.50% Teck, and 10% Mitsubishi. In 2000, Billiton Plc of Great Britain bought 100% of Rio Algom Limited thereby effectively becoming one of the partners. In 2001 BHP Limited merged with Billiton PLC forming BHP Billiton Group. Teck Corporation and Cominco Limited merged in 2001 forming Teck Cominco Limited (now Teck Resources Limited). In 2005 Noranda Inc. amalgamated with Falconbridge Limited with the resulting company called Falconbridge Limited. In November 2006 Xstrata acquired Falconbridge Limited became one of the owners. Currently ownership of Antamina is BHP Billiton Group (33.75%), Xstrata PLC (33.75%), Teck Resources Limited (22.50%), and Mitsubishi Corporation (10%).

7 Geological Setting

7.1 Regional Geology

The Antamina and Usu Pallares deposits are located within the sedimentary Mesozoic belt of the Central Andes. The formations and stratigraphic sequence for the region is shown in Figure 7.1.

Figure 7.1: Stratigraphy of Antamina District

Period	Epoch	Age	Age Ma	Group	Formation	Thick m	Lithology
Cretaceous	Late	Santonian	87.5 - 84	Goyllarisquisga	Celendin	115 - 225	Marls, nodular limestone, thin bedded, fossiliferous.
		Coniacian	88.5 - 87.5		Jumasha	200 - 800	Gray limestone, massive, thick bedded 1 - 2 m, micritic, poorly fossiliferous, dolomitic in parts.
		Turonian	91 - 88.5		Pariatambo	100 - 200	Dark gray marls and limestone, bituminous, thin bedded. Dark gray chert nodules abundant at top.
		Cenomanian	97.5 - 91		Chulec	100 - 200	Thin bedded shelly limestone, marl, yellow-cream color.
	Early	Albian	113 - 97.5		Pariahuanca	54 - 210	Massive shelly limestone 1 - 2 m beds, calcareous sandstone at base. Yellowish color.
		Aptian	119 - 113		Farrat	20	White sandstone.
		Barremian	124 - 119		Carhuaz	1300 - 1500	Shale, silty shale, thin qz-st beds, brown-purplish. Disconformable. Gray shales and coal at base.
		Hauterivian	131 - 124		Santa	150 - 341	Blue-gray shelly limestone, 0.1 - 1 m beds, chert nodules, parts dolomitic.
		Valanginian	138 - 131		Chimu	600 - 685	Quartz-sandstone white. Dark shale and coal at base.
		Berriasian	144 - 138		Oyon	100	Shale, siltstone, sandstone, coal, important. Thrust plane.
		Tithonian	152 - 144		Chicama	> 1500	Gray slates, quartzites, sandstones.
Jurassic	Late						

Bernaides, V.E., 1956. The Cretaceous System in Northern Peru. Bull. Am. Mus. Nat. Hist., 106, 353-494.
Wilson, J., 1963. Cretaceous Stratigraphy of Central Andes of Peru. Bulletin AAPG, 47, 1-34.
Cobbing, J. et al., 1996. Geología de los Cuadrángulos de Huaraz, Recuay, La Unión, Chiquian y Yanahuasca. INGEMMET Boletín No 76.
Decade of North American Geology, 1989. Geologic Time Scale, Geol. Soc. Of America.

Stratigraphy of the Antamina District

Figure 7.1



7.1.1 Plate Tectonic Setting

The Andes mountain belt is situated on the Pacific margin of South America. The oceanic Nazca Plate is being subducted to the east under the continental South American Plate. Plate collision and subduction zone melting have generated the folding and faulting, mountain uplift, volcanism, plutonism and the majority of the mineral deposits of the Andes. Northern Perú, where Antamina is located, is a seismically active zone but does not have any active volcanoes.

7.1.2 Geomorphology

The Andes of Northern Perú is comprised of, from west to east, the Coastal Zone (desert), the Western Cordillera (Cordillera Occidental) and the Eastern Cordillera (Cordillera Oriental). Antamina lies in the eastern part of the Western Cordillera.

The Western Cordillera has two main mountain chains. The Cordillera Negra in the west is separated by the valley of the Santa River (Huaraz valley) from the Cordillera Blanca in the east. Antamina is situated east of the Cordillera Blanca between it and the valley of the Marañón River. The Marañón separates the Western Cordillera from the Eastern Cordillera.

7.1.3 Geology

Between the Antamina area and the Pacific Ocean lies the Coastal Zone and the Cordillera Negra. Together these form a magmatic arc that was active from the Late Jurassic to the Tertiary. The main components of this arc are the Casma Volcanics (Albian, ca 105 to 95 Ma), the Coastal Batholith (ca 100 to 50 Ma) and the Calipuy Group Volcanics (Late Cretaceous to Paleogene, ca 95 to 30 Ma). The latter formed the Cordillera Negra. The arc was deformed during the mid-Cretaceous (Mochica Phase) and Late Cretaceous (Peruvian Phase).

To the east of the magmatic arc thick sediments were deposited in a deep, extensional, ensialic marine back-arc basin called the Western Trough (or Western Peruvian Geosyncline), also active from Late Jurassic to Late Cretaceous times. The sediments consist of slates and quartzites (Chicama Formation, Late Jurassic, ca 152 to 144 Ma) followed by thick deltaic sandstones, shales and coal with a marine limestone (Goyllarisquisga Group, Early Cretaceous, ca 144 to 114 Ma). Next came a marine transgression and deposition of thick marine carbonates (Mid Cretaceous, ca 113 to 88 Ma, Pariahuanca, Chulec, Pariatambo and Jumasha Formations), followed by marine shales (Celendin Formation) in the Late Cretaceous (ca 88 to 84 Ma). Following marine regression and basin uplift, there was deposition on continental red bed sediments (Casapalca Formation) in the Late Cretaceous and Paleocene. Jumasha Formation limestones in the eastern part of the Western Trough host the Antamina deposit.

This basin was bounded to the east by a basement high (the Marañón High, Axial Threshold or Marañón Geoanticline) formed of Late Precambrian schists, phyllites and slates (Marañón Complex), which now forms the Eastern Cordillera. Mesozoic sediments are much thinner on this basement high. To the east, a sequence of Mesozoic sandstone and carbonates was deposited in an external foreland basin (the Eastern Basin or Eastern Peruvian Geosyncline) overlapping the Brazilian Shield. This sequence is thinner than that of the Western Trough and now forms the Sub Andean Zone fold and thrust belt.

The Western Trough was deformed by the Inca 2 fold phase in the Late Eocene (ca 41 – 40 Ma). This resulted in an extensive folding and reverse faulting throughout the basin; and formation of a fold-thrust belt in the eastern part along the boundary with the Marañón High (Marañón Fold-Thrust Belt). Antamina is located in this fold-thrust belt. During the Miocene there were three short compressive periods (Quechua 1 to 3) at ca 19 Ma, 12 Ma and 6 Ma, separated by neutral or extensional periods.

In the Middle to Late Miocene, the Cordillera Blanca batholith was intruded in the eastern part of the Western Trough sequence (Chicama Formation) to form the Cordillera Blanca (ca 16.5 to 5 Ma). There are coeval ignimbrites (Yungay Formation). At the same time there was widespread magmatism (middle- to high- K calcalkaline) throughout the Western and Eastern Cordilleras. The Antamina stock is a part of this event.

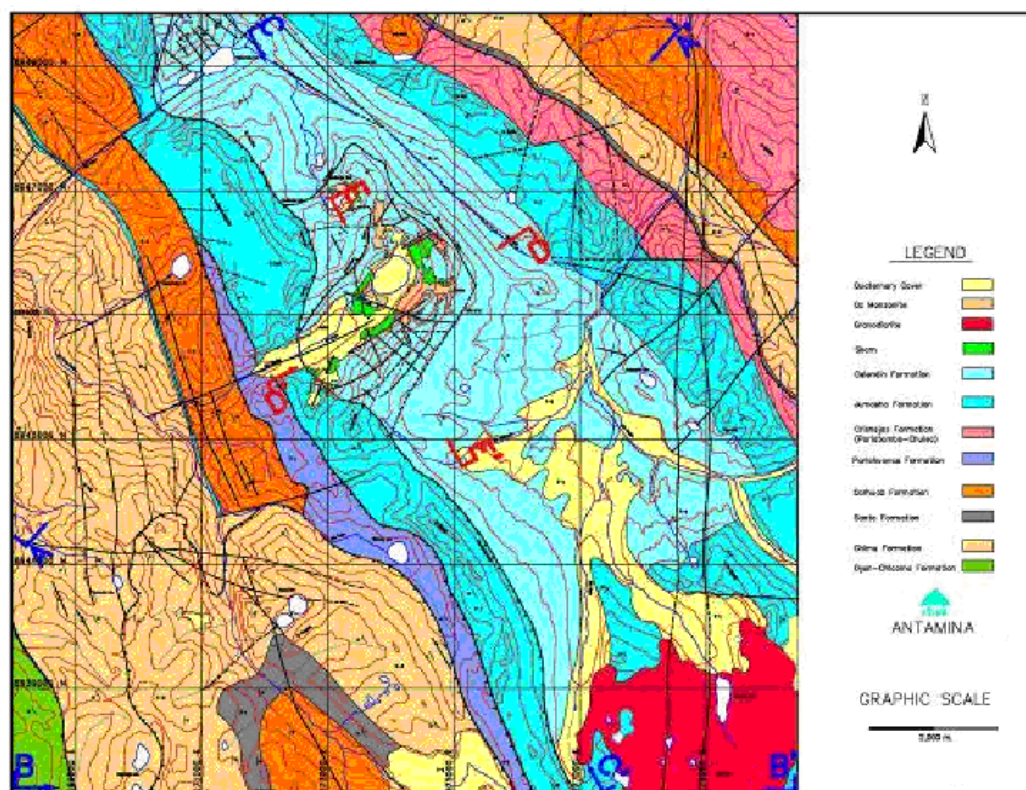
7.1.4 Metallogeny

Antamina lies in the eastern part of the polymetallic belt of Central Perú. The polymetallic belt is located in the Western Cordillera between about 6° South (Huancabamba Deflection) and 14° South (Abancay Deflection) and is defined at either end by transverse, arc-normal structural features. Mineralization in the belt shows a Zn-Pb-Ag-Cu-Au association, mainly in hydrothermal deposits related to Middle to Upper Miocene calc-alkaline subvolcanic and high level intrusions. The belt was traditionally known for the major Zn-Pb-Ag mines such as Cerro de Pasco, Milpo, Casapalca-Morococha and others. Porphyry Cu and Cu-Au deposits also occur, and since the early 1990s the belt has become the major gold producer in South America with the discovery of epithermal gold deposits such as Yanacocha, Pierina, Quicay and Alto Chicama. The deposits of this belt are characterized by high amounts of other metals, some of which may be produced as by-products at different mines. These metals include As, Bi, Cd, Ge, Hg, In, Mo, Sb, Se, Sn, Te, and W.

7.2 Local Geology

The local geologic setting can be seen in Figure 7.2.

Figure 7.2: Local Geology





7.2.1 Geomorphology

The oldest landform in the eastern part of the Western Cordillera, where Antamina is situated, is the puna Surface. This is the remnant of a peneplain of regional extent now shown by concordant mountain peaks. Around the Antamina the peaks are generally at 4,500 to 4,800 meter altitude.

The puna surface is cut by the valley phase and canyon phase, which formed the major river valleys in the region such as the Marañón and Santa. The high parts of valleys formed in the valley stage are broad and open and usually populated and cultivated. The lower parts are narrow canyons and formed as a result of rapid uplift and erosion. They can be up to 2,000 meters deep. The puna surface has been dated as younger than 14.5 Ma, the valley phase as post 14.5 Ma to pre 6 Ma, and the canyon phase as post 6 Ma.

The youngest features are from the Pleistocene glaciation with an ice-limit down to about 3,500 meters altitude. There were at least three phases of glaciation. The main phase formed U-shaped valleys a few kilometers long, such as Antamina (4,100 – 4,200 meters) and Contonga (Lake Pajoscchocha at 4,110 meters). A younger glaciation formed corrie (cirque) basins such as Lake Antamina (4,337 meters), Lake Contonga (4,380 meters) and the lower Condorcocha valley (4,360 meters). The last phase formed small corrie basins at the lakes Condorcocha (4,510 meters) and Contonga (4,620 meters). The Antamina valley feeds into a series of deep valleys formed during the earlier valley and canyon stages. It is probable that a small valley of this stage existed at Antamina before glaciation and allowed ice to accumulate.

7.2.2 Stratigraphy and Structure

The stratigraphy of the Antamina district is shown in Figure 7.1. Antamina is located within a part of the Marañón Fold-Thrust belt which, at Antamina, has a width of about 40 kilometers. This is one of the widest parts of the belt, which can be as narrow as 10 kilometers. The structures and stratigraphy trend northwest-southeast and the thrusts are east verging. The age of thrusting is Incaic 2 (Late Eocene).

The Antamina deposit is hosted by Jumasha Formation in a thrust tongue over the younger Celendin Formation (Figure 7.2 and Figure 7.3). West of Antamina the Jumasha Formation forms a steep thrust ramp over the Jumasha thrust tongue (i.e. over itself) and the Celendin Formation. The Jumasha forms prominent steep mountains of well-bedded, light gray limestone. Continuing west, successive thrusts bring in the Parihuanca Formation over the Jumasha (the Chulec and Pariatambo Formations are missing), the Carhuaz and then the Chimu Formation. The latter forms a synclinorium with the Santa and Carhuaz Formations outcropping to the south. Further west the Oyón Formation is thrust over the Chimu Formation.

The Celendin Formation outcrops to the east of Antamina and is soft with little exposure. It forms the core of a regional synclinorium. The axis plunges gently to the southeast and runs along Quebrada Huincush to Rosita de Oro. The axis must continue beneath the Antamina thrust tongue since Quebrada Tucush is on the north limb of the syncline.

The actual trace of the fault separating Celendin from Jumasha northeast of Antamina Lake appears to be quite steep. It is not clear whether this is merely a steep ramp of the thrust or a later normal fault offsetting the thrust.

The syncline closure in the northwest around Contonga has very complicated minor folding in the Jumasha Formation. Going east, the Celendin is in stratigraphic contact with the Jumasha and Crisnejas Formations (the latter is the eastern facies of the Chulec and Pariatambo Formations). This is thrust eastwards over an anticline of the Chimu, Santa, Carhuaz and Crisnejas Formations.



7.2.3 Intrusions and Mineralization

The Antamina intrusion (9.8 Ma²) is a multiple phase, quartz monzonite porphyry and has the Antamina Cu-Zn skarn developed around it. There is a nearby intrusion of similar composition with narrow, weakly mineralized skarn 1.5 kilometers northwest at Condorcocha.

The Contonga and Taully stocks lie 4 kilometers north of Antamina and intrude Jumasha Formation limestone. The composition is quartz monzonite and the texture varies from porphyritic to equigranular, with phyllic alteration. The stocks are small (300 meters and 650 meters in diameter respectively) and form subvertical cylinders with a narrow ring of garnet skarn (averaging 3.2 meters wide) with Zn-Ag-Pb-low Cu mineralization, which has been mined at Contonga.

The Lucia pluton (Estrella del Norte property) is located about 7 kilometers southeast of Antamina. It is a relatively large stock (4 kilometers x 2.5 kilometers) of quartz monzonite and granodiorite with an equigranular texture in the main body and porphyritic in the northwest part. It intrudes Jumasha Formation limestone and has narrow garnet skarns developed at the contact with widths of 0.5 to 2.5 meters and lengths of hundreds of meters, as well as veins in the limestone. The skarns have Zn-Pb-Ag mineralization with low Cu. There has been no mining apart from some small prospect workings.

7.3 Deposit Geology

The Antamina and Usu Pallares deposits comprise a very large copper skarn with zinc, silver, molybdenum, arsenic, and bismuth formed by the intrusion of quartz monzonite into a sequence of limestone and marls. Currently two areas of mineable mineralization have been identified. These are called Antamina Main Deposit and Usu Pallares. The two deposits are very similar in geology and lithological zoning; however the style of mineralization may vary in size and extension. Limited information suggests that both deposits were formed contemporaneously. It is believed that both deposits are actually physically connected and are part of the same deposit and mineralizing system. However, at this time there is insufficient data to demonstrate a physical connection between the two deposits.

[Due to document format reasons, please find the graphic of this section in the next page]

² McKee, et al., *Age of porphyry, potassic alteration and related skarn mineralization, Antamina District*, 1979.

Figure 7.3

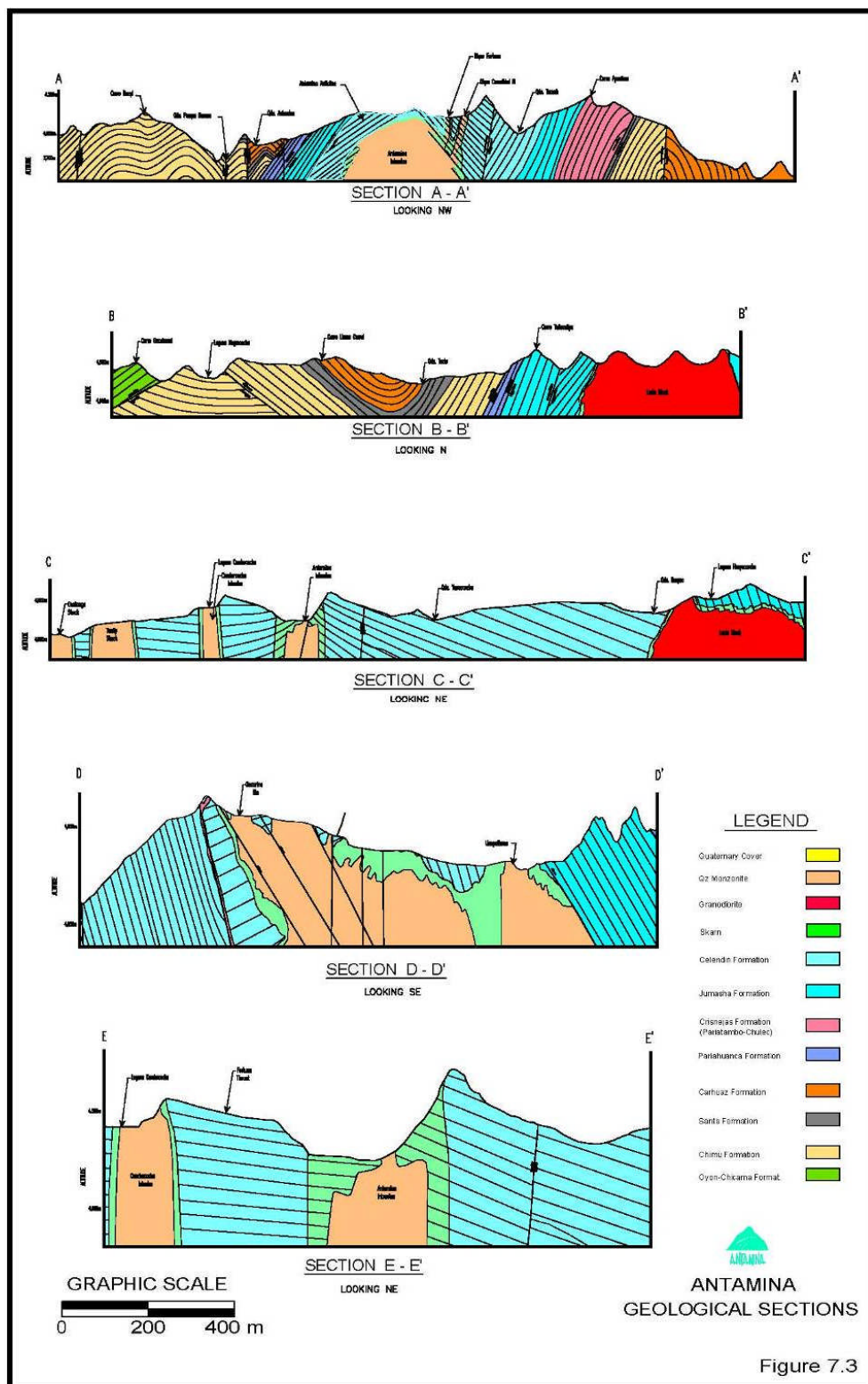


Figure 7.3



7.3.1 Geomorphology

The pre-mine Antamina valley was a 4 kilometers long glacial U-shaped valley with steep sides, a flat floor and corrie lake in an upper valley at the head. The lake surface was at an elevation of 4,337 meters (August 1996) and the lake was 760 meters long by 510 meters wide and up to 51 meters deep. It was separated from the main valley by a rock ridge, known as the Taco (elevation 4,375 meters), which was 38 meters above the lake level and 155 meters above the main valley. The northwest part of the ridge was an ice-smoothed rock outcrop with roche moutonnées, while the southeast half had moraine deposits and the lake spillway. This was an outlet from the corrie glacier as an ice tongue or stream.

The main Antamina valley altitude varies from 4,200 meters to about 3,800 meters and has a gentle gradient. There are two hanging valleys, Usu Pallares and Vallecito.

The head of Antamina valley (northeast) had a sharp ridge with an altitude of 4,640 meters to 4,717 meters. Ridges and peaks up to 5,073 meters (Cerro Tornillo) form the northwest side of the valley. The ridge on the southeast side of the valley varies from 4,683 meters up to 4,924 meters (Cerro Buque Punta).

The pre-mine Antamina valley thus had a depth of 500 to 600 meters below the enclosing ridges and up to almost 1,000 meters below the highest peak. The pre-glacial topography is interpreted to have been a puna surface at around 4,700 meters with peaks over 5,000 meters. There was probably a small and shallow river valley in Antamina formed as a headwater during the valley and canyon stages. Ice accumulated in this valley during the main Pleistocene glaciation, which formed the main valley and hanging valleys.

Moraines from both glacial events outcropped on the southeast side of the Taco rock barrier below the former Lake Antamina. The older moraine was pyritic and the younger was limonitic and in places there was a ferricrete layer between the two.

Lake Antamina had a U-shaped bathymetric profile with steep sides and a flat bottom. Vertical drill holes completed from barge drilling in 1999-2000 from the lake surface showed soft bedded silt-clay lake sediments. The sediments varied from less than a meter to approximately 45 meters in thickness along the lake bottom.

7.3.2 Classification of Rock-Types

7.3.2.1 Introduction

Several critical lithological distinctions have been made, including the discrimination of exoskarn from endoskarn, and the recognition and systematic identification of the degree of brecciation in the orebody. These distinctions are considered important from a resource modeling as well as an ore-genetic point of view because of differences in the grade and style of mineralization in the various rock-types as described below. There are currently 170 rock types identified and logged within the Antamina and Usu Pallares deposits. Procedural control and logging consistency is enforced through rigorous application of the Compañía Minera Antamina S.A. Core Logging Manual.³ The same logging system is used for both deposits as the rock types present in both deposits are the same.

The general skarn zonation from the intrusive core outward is as follows: pink garnet endoskarn, brown garnet endoskarn, mixed brown and green garnet transitional skarn, mixed brown and green garnet exoskarn, green garnet exoskarn, diopside exoskarn, wollastonite exoskarn, hornfels, marble, limestone. Enriched brecciation zones composed

³ Compañía Minera Antamina S.A., *Antamina Mine Geology Logging Manual and Coding Instructions Ver. 1.54*, March 18, 2003.



of all skarn types can occur in any lithology type but more frequently within the intrusive and endoskarn units, see Figure 9.1 and Figure 9.2.

7.3.2.2 Intrusives – Geology, Age and Structure

For the Antamina Main deposit an empirical classification of the intrusive rocks based on their petrography is used. Four main types of intrusive rock are recognized by distinguishing on the basis of phenocryst type and abundance. They are: crowded plagioclase porphyry, crowded plagioclase K-feldspar porphyry, crowded plagioclase K-feldspar-megacryst porphyry, and sparsely porphyritic plagioclase K-feldspar porphyry. A minor intrusive rock-type, sparse hornblende plagioclase porphyry, forms a dyke in the Poderosa area, at the northeast end of the Antamina valley. These varieties are classified as quartz monzonite porphyry.

The main mass of the unskarned porphyry is predominantly crowded plagioclase porphyry with common quartz, biotite and relict hornblende phenocrysts and rare K-feldspar phenocrysts. The K-feldspar phenocrysts are locally more common and thus another subunit of crowded porphyry has been distinguished. Sharp contacts between these two types of crowded porphyry are very rarely evident. Apparent magma commingling textures between a minor phenocryst-poor darker-gray phase and a paler phenocryst-rich phase are locally displayed.

Although the contact relationships among the different porphyry phases are not clear everywhere, in general the crowded plagioclase porphyry is the earliest phase, and the sparse porphyry, the latest. Locally, crowded-porphyry contains quartz vein stockwork and associated biotitic, i.e., potassic alteration. This altered but unskarned porphyry locally hosts disseminated and vein molybdenite mineralization and minor disseminated chalcopyrite ($\leq 0.2\%$ Cu and 0.05% Mo). Sparse porphyritic dykes cut crowded porphyry, skarn and breccia, and locally contain skarn xenoliths.

Multiphase intrusions are typical of porphyry deposits and the different phases were probably intruded in a period of less than one million years (they are statistically identical in radiometric dating). The relative ages of intrusion are recognized by textures such as cross-cutting relationships, xenoliths, chilled margins, truncated veinlets, degree of quartz veining and hydrothermal alteration, igneous texture and overall rock quality.

A USGS study in the 1970s did address the question of the age of the Antamina intrusions. It was dated by K-Ar at 9.8 Ma (mean of 9.1 ± 0.4 Ma to 10.4 ± 0.4 Ma, 5 samples) with no significant differences between primary biotite, primary K-feldspar and K-feldspar megacrysts, nor between different intrusive phases (quartz monzonite porphyry in Taco – Laberinto; late mineral quartz monzonite with K-feldspar megacrysts in Usu Pallares; and minor intrusions with no quartz phenocrysts north of the lake).⁴ Age dating in recent years by researchers utilizing various methods, has produced ages similar to that stated above.

The intrusives form a complicated anastomizing shape in the upper part of the deposit, which is believed to be the result of a complex structural arrangement and multiple phases of intrusion. The main trend of the intrusive bodies is northeast, and there are also east-west and northwest trends. The intrusive bodies form broad dikes that converge into a stock at depth (below approximately the 4,200 meter level).

It is known from drilling that at the 3,900 meter level the intrusion forms an almost circular stock approximately 750 to 800 meters in diameter. There is a 350 meter long, northeast-trending spur (150 meters wide) at the northeast end of the Lake Zone, and a 500 meter

⁴ McKee et al., *Age of porphyry, potassic alteration and related skarn mineralization, Antamina District*, 1979.



long, southwest-trending spur (50 – 170 meters wide) below the valley at the southwest end. In total the intrusions occur over a length of 1,700 meters at this level. Skarn forms a shell of 300 to 400 meters in width around the intrusions. There are assumed separate intrusions in the Valley and Usu Pallares Zone, however; at this time there is insufficient evidence to determine if the intrusive bodies are connected with the larger intrusive body of the Antamina deposit. There is little variation in the shape of the intrusion down to the 3,850 meter level based on the limited drill data available.

At high levels (e.g. the 4,260 meter level) the intrusion is comprised of several discrete broad dikes with a dominant northeast trend, and also southeast and east-west local trends and edges. These dips vary from vertical to moderately inclined (ca 45°). The skarn around and above these has widths of 70 to 800 meters at this level.

The same classification scheme is applied to the intrusive rocks at Usu Pallares where these units occur similarly as described for the Antamina Main deposit. In the Usu Pallares area the predominant intrusive rocks are the K-feldspar-megacryst porphyry and the crowded plagioclase porphyry in the form of narrow dykes. In Usu Pallares the sparsely porphyritic plagioclase K-feldspar porphyry is rarely observed. The emplacement of this intrusive is thought to occur latter than the before mentioned intrusives due to the lack of alteration and mineralization. The intrusive rocks occur as narrow dykes coincident to the north-west oriented thrust faults. A major intrusive body is localized between sections 19100N and 19200N.

7.3.2.3 Pink Garnet Endoskarn

Within the Antamina Main deposit the coarse-grained pink garnet endoskarn is the other widespread type of endoskarn. It consists of milky-white plagioclase-rich matrix (distinguishable from the pale-gray, translucent matrix of unskarned porphyry) enclosing large pink garnets and more sparse maroon garnets, and displaying relict porphyritic texture. Significant mineralization is not associated paragenetically with the development of this coarse-grained endoskarn, which, although commonly containing disseminated molybdenite, can eventually host blebs and/or veinlets of chalcopyrite of ore grade.

Narrow intervals of the rarely occurring plagioclase endoskarn are located between porphyry and coarse-grained pink-garnet endoskarn. Plagioclase endoskarn rarely contains copper ore-grade, and entirely lacks of zinc, but it is a useful indicator of proximity to ore. Due to the irregularity and its narrow width this unit is included as part of the pink garnet endoskarn domain.

Porphyry-style Mo mineralization, as disseminations and veins, is overprinted by the pink garnet endoskarn, and veins are rendered indistinct because quartz was consumed in the skarn formation. The high grade portion of the copper mineralization in endoskarn is later than the low grade copper and molybdenum mineralization associated with the porphyry intrusives.

In Usu Pallares the pink garnet endoskarn is formed from weakly altered intrusive where hydrothermal activity took place and formed veinlets of chalcopyrite and minor molybdenite. This unit can be recognized by relicts of the porphyritic texture in the intrusive and a mass of fine-grained pink garnet around the plagioclase crystals. The pink garnet endoskarn in Usu Pallares is usually mixed with diopside and forms a wide unit around the intrusive core in the north and west part of the deposit. It also forms finger-like bodies in the south part of the deposit. This unit hosts both low-grade and high-grade copper ore but is relatively low in molybdenum.



7.3.2.4 Brown Garnet Endoskarn

Brown garnet endoskarn is the dominant endoskarn unit and also the dominant skarn unit of both endoskarns and exoskarns major domains. It is the most important ore hosting lithology within the Antamina deposit for its copper content. Generally it hosts the copper only (chalcopyrite), low bismuth ore type (type 1), often containing molybdenum. Rarely will this unit host zinc or high bismuth ore types. The texture of this unit can vary from fine grained < 1mm to coarse grained > 10mm, occurring as a compact garnet skarn, but often disaggregated in character. The percentage of disaggregated brown garnet endoskarn can be significant in the SAG mill feed stream, the chalcopyrite tends to be coarse grained and easily liberated with grinding. Zinc tends to be low typically about 0.2-0.3% and therefore not recoverable. Silver typically is in the 6-12 gram/tonne range and occurs in solid solution within the chalcopyrite.

This unit is also the main source of copper ore in the Usu Pallares area. The brown garnet endoskarn is a wide unit around the intrusive core between sections 19150N and 19250N. Mineralization is hosted in narrow veins, but there is also some dissemination. The rock type is predominantly a fine-grained garnet with less open spaces for interstitial mineralization. Unlike the Antamina Main deposit, this unit constitutes a dense and competent rock that may affect grinding and milling during processing. An explanation for this fact is that the retrograde alteration, fracturing and disaggregation events are not significantly spread in Usu Pallares.

7.3.2.5 Transitional Skarn

There is a recognized skarn type within the Antamina Main deposit whose origin is uncertain but shows mixed characteristics of both endoskarn and exoskarn and therefore is called transitional. This is a frequent occurrence at the margins of proto intrusive (now endoskarn) with country rock, formerly limestone. It is hypothesized that where the skarn process was very intense at these contacts, both endoskarn and exoskarn approached the same composition and mineralogy and the two lithologies became indistinguishable.

This lithology is commonly brown, medium-grained and granular, but overall is variable in texture, grain-size and color. Granular, medium-grained, brown garnet skarn with intergranular chalcopyrite would be classified as brown garnet exoskarn where it constitutes the end-member of a continuous gradation from green garnet exoskarn. Fine-grained, dark-pink garnet skarn having veins with chloritic selvages but lacking relict porphyritic texture would be classified as endoskarn if it were in gradational contact with porphyry. Locally the sparse maroon garnets in coarse-grained pink garnet exoskarn are larger and more abundant and they coalesce to form granular, medium-grained intervals of skarn, which appear quite similar to exoskarn. Fine-grained, pale-brown skarn would be classified as exoskarn if it constituted part of gradation from limestone or hornfels. However, in rare locations, fine-grained, pale-brown skarn also occurs in gradational contact with porphyry and containing relict porphyritic texture. Every gradational variation amongst these skarn types occurs, and if the critical characteristics and/or gradational variation relationships for interpreting the skarn as endoskarn or exoskarn were absent or ambiguous then the unit was classified as transitional skarn.

Transitional brown and green skarn usually contains pale-brown or beige, coarse- to very-coarse-grained garnets with intergranular medium- to dark-green garnets. Under the north part of the former Lake Antamina, and locally elsewhere in proximity to endoskarn, this transitional brown and green skarn contains diopside, in part as radial clusters. In the Pantano area this facies is common, although there it does not contain diopside, but locally contains wollastonite.



The occurrence of this unit in Usu Pallares is restricted to a few drilling intercepts that represents 0.7% of the total intercepted lithologies. The characteristics are similar to that described for the Antamina Main deposit.

7.3.2.6 Brown-Green Garnet Exoskarn

For the Antamina Main deposit the mixed green and brown type of exoskarn, brown garnet commonly occurs as veinlets cutting green garnets. In places brown garnet preferentially replaces some layers in green garnet exoskarn at a centimeter-scale, producing banded brown and green garnets. This facies may contain sphalerite $\pm$ chalcopyrite with the same style of mineralization and grades as green garnet exoskarn. In the field the brown-green exoskarn can be noted as an irregular band next to the green exoskarn with a varied width of a few meters to tens of meters. This unit can be considered one of the main sources of copper-zinc ores at Antamina.

In the Usu Pallares area the occurrence of the brown-green exoskarn is in the form of a narrow band of less than 30 meters wide with similar textures as described for the Antamina Main deposit. It contains zinc and copper mineralization averaging 1.39% zinc and 1.07% copper. For the Usu Pallares area this unit is an important host of copper and copper-zinc ores.

For modeling purposes and due to the limited data the brown garnet exoskarn was included into this modeling domain.

7.3.2.7 Green Garnet Exoskarn

In much of the deposits, the skarn facies adjacent to marble or hornfels is green garnet skarn. In much of this facies, garnet appears to be replacing calcite directly, i.e., there is no evidence that garnet replaced wollastonite. It is believed that green garnet exoskarn in different parts of the deposits formed by two different reaction paths, one, mentioned above, via wollastonite skarn, and the other, directly from marble. The two types are texturally indistinguishable.

Green garnet skarn contains either chalcopyrite–sphalerite ore or sphalerite alone and rarely barren or low grade: the sulphides range from disseminated to massive and interbanded with green garnet. Sphalerite commonly averages 3 to 5 percent in green-garnet exoskarn. However, it is erratically distributed, commonly occurring as rich bands separated by relatively barren sections.

The green garnet exoskarn in Usu Pallares is characterized by the presence of disseminated sphalerite and chalcopyrite. Unlike the Antamina Main deposit the presence of massive bodies of sphalerite is uncommon. The green garnet exoskarn in Usu Pallares is predominantly fine to medium grained and commonly contains minor proportions of diopside and wollastonite.

7.3.2.8 Diopside Exoskarn

In the Antamina Main deposit the diopside exoskarn occurs on the north east flank of the Antamina valley next to the Oscarina dike complex. Apparently this is a more reactive front related to former sequences of silty limestones because it occurs in the vicinity of hornfels. Texturally shows more than 60% of fine-grained diopside with minor portions of wollastonite and garnets. This unit may form a wide zone with little to no copper and zinc mineralization in the upper levels to increasing copper and restricted zinc. Lead veins are associated to this unit in contact with dykes. In general it is a good host for high values of bismuth and arsenic.

In Usu Pallares the diopside exoskarn is a 30 meter wide unit present on both flanks of the skarn margins in the north part of the deposit, but the current sparse drillhole spacing may



not reflect the true width of the unit. It is considered a host for low grade copper and zinc mineralization. It is generally adjacent to hornfels.

7.3.2.9 Wollastonite Exoskarn

Within the Antamina Main deposit the wollastonite exoskarn comprises an inner zone (contiguous with green garnet skarn) of bornite ore and an outer zone (closer to marble) of bornite – sphalerite ore. The contact between wollastonite skarn and green garnet skarn is a broad gradational replacement zone where green garnet replaces wollastonite, in a zone classified as wollastonite – green garnet skarn. Bornite ore occurs throughout this intermediate zone as well as in both green garnet exoskarn and wollastonite exoskarn. The contact between bornite ore and chalcopyrite ore is a broad, gradational zone of coexisting chalcopyrite – bornite, generally within green garnet skarn near wollastonite skarn. Two sub-units are differentiated based on the type of garnet present. A brown garnet wollastonite exoskarn contains low bornite and high molybdenum. A second unit is formed by green garnet wollastonite exoskarn with significant amounts of bornite and sphalerite. In general the wollastonite exoskarn contains high arsenic and bismuth.

There is a second variety of breccia, wollastonite breccia, in which comminuted wollastonite constitutes the matrix. Because of the textural similarity of this breccia type to the more common garnetiferous enriched brecciation zone, this could also be phreatic hydrothermal breccia. This type of breccia is distinctive because it does not contain magnetite and is not significantly elevated in Cu grade.

In Usu Pallares only the green garnet wollastonite exoskarn has been identified to date. This unit is restricted to the north-west portion of the deposit where it forms irregular and discontinuous zones next to the marble contact. Intercepts with a bornite intensity of one were reported in the drilling logs.

7.3.2.10 Hornfels

In the Antamina Main deposit the fine-grained hornfels can be pale-brown, pale-green, khaki, or yellowish-gray, and varies from fine-grained to aphanitic. It ranges from massive to laminated, with fine, wavy, compositional banding, and generally consists of a very fine-grained aggregate of garnet and diopside with minor wollastonite. It has been identified locally at or near the margins of the deposit. This rock has no apparent porosity or permeability and rarely contains minor sulphides. Where these layers occur on the margins of the intrusion, they appear to limit development of the ore and skarn units. One can suspect that the process of skarnoid formation rendered these units impermeable to further fluid flow and/or unreactive and thus limited ore/skarn formation in them. They are thought not to have additional elements added by hydrothermal fluids. Hence they were formed only by thermal metamorphism. The exception to this is diopside skarn or diopside hornfels which are thought to have additional elements added to the rock by hydrothermal process. This rock often contains minor sulphides and often can exhibit levels of economic sulphides to become marginal or low grade ore, but rarely high grade ore.

In Usu Pallares hornfels are rare and localized in the northeast flank of the deposit. They are predominantly brown and grey in color and occur as thin layers within the Jumasha formation. There is no mineralization associated with the hornfels in Usu Pallares.

7.3.2.11 Limestone-Marble

Most limestone occurring at Antamina Main and Usu Pallares deposits cut by drilling at the margins of the skarn is light gray, very fine grained and micritic with parallel bedding on a scale of several centimeters but no fossils, shell fragments or other biogenic or sedimentary structures. In outcrop in the upper valley slopes, these limestones are thickly bedded (1 to 3 meters), light gray and in cliff faces weather to a white or creamy color. These limestones are classified as micritic. These are interpreted as belonging to the



Jumasha Formation that is anomalously thick at Antamina as a result of repetition of the sequence by thrust faulting.

At the head of the Antamina valley there are interbedded (2 to 3 meter beds) micritic limestone and stromatolitic limestone.

The limestones exposed in the anticline on the east side of the valley has a light gray core, followed by a unit of black limestone with thin gray beds, then a light gray micritic limestone. This limestone is distinct from the overlying (overthrust) beds typical of the Jumasha Formation.

The Jumasha Formation limestone is susceptible to karst weathering. There are no karst features at surface in the Antamina deposit area but there are karst features in surrounding areas. However, underground cavities with high water flow (karst or fault zones) were encountered in diamond drilling in the Laberinto and Valley – South Taco areas.

In the Antamina Main deposit marble is the recrystallized and altered front of the sediments in contact with the skarn units. It is mainly composed of calcite and diopside with thin veinlets of sphalerite, pyrite, lead and minor chalcopryrite. The marble unit can be as wide as 100 to 200 meters and it can be recognized by a typical bleaching (whitening) of the dark grayish colored Jumasha and Celendin formations. The diopside marble a subtype which is in contact with the green exoskarn unit may be as wide as 50 meters and contains traces of sulfides in veins and as dissemination. The diopside marble is not modeled in a single unit due to the lack of drilling in the waste material.

In Usu Pallares the same characteristics are recognized as described for the Antamina Main deposit.

7.3.2.12 Enriched Brecciation Zone

The occurrence of hydrothermal breccias in the Antamina Main deposit, lacking juvenile components and therefore probably classifiable as phreatic, is widespread and is an important ore host in what is termed the brown garnet endoskarn zone. It can now be concluded that virtually all of the breccia lacks a magmatic or juvenile component. The reader should note that hydrothermal breccia is rarely been observed within the Usu Pallares deposit, and hence the lower copper grades in the deposit when compared to the Antamina Main deposit.

The enriched brecciation zone is formed by a wider portion of the hydrothermal breccias and the surrounding area of a few meters in which the mineralization is higher than the host rock. This unit cuts all types of skarn, including wollastonite skarn, but is particularly common at the endoskarn – exoskarn contact, i.e., the original margins of the former stock. They are also considered as intramineral because they cut and contain clasts of mineralized skarn, yet themselves host replacement-style and vein mineralization of pyrite – chalcopryrite – magnetite. Breccias and veins in endoskarn contain a similar association of metallic minerals, although magnetite is a very minor component of veins in endoskarn.

The core of the unit is the heterolithic breccia which is commonly matrix-dominated containing angular to subrounded fragments of all of the rocks and minerals it cuts, as well as clasts of sulphides, magnetite, and quartz, which appear to be derived from veins and replacement selvages restricted to the breccias. The breccia matrix ranges from massive to laminated, and the clasts range from randomly oriented to locally shingled or imbricated.



The enriched brecciation zone is generally above average ore-grade in Cu and Ag, and locally contains ore-grades of Mo and/or Zn inherited from the wall rocks. Breccias may locally contain galena. Often post skarn mineralization veins containing sulfosalts cross cut the breccias.

The lack of hydrothermal breccias at Usu Pallares suggests a lesser dynamic event during the mineralization of this area. In this area only 60 meters of drilling intercepted breccias representing 0.4% of the total intercepted lithologies. Its texture and geometry are similar to those described for the Antamina Main deposit.

7.3.3 Retrograde Alteration of Skarn

Within the Antamina Main deposit two main variants of retrograde skarn alteration are recognized: chlorite-rich vein selvages in endoskarn; and alteration of the intergranular matrix of transitional skarn to pale-green clay. The matrix of hydrothermal breccia had been considered to be altered to olive-brown-green clay, but no clay minerals were detected in it by preliminary laboratory analyses.

Retrograde alteration of endoskarn is complex, and ranges from epidote associated with chalcopyrite – pyrite blebs and veinlets that have white selvages, to chlorite associated with chalcopyrite – pyrite blebs and veinlets that have white selvages, to chlorite associated with pyrite – chalcopyrite ($\pm$ magnetite) veins with crackle and mosaic breccia. Worthy of note is that retrograde alteration does not affect exoskarn. Retrograde alteration occurs in much of the endoskarn and in the transitional skarn.

In Usu Pallares the retrograde alteration is commonly in the form of intergranular pale-green clay pyrite-rich matrix with minor epidote. This event appears to be localized in the center of the deposit between sections 19150 and 19250N.

7.3.4 Structure

The Antamina deposit as shown in

Figure 7.4 is within a fold-thrust belt and structure is the main control on intrusion and skarn formation (Figure 7.5). The interpreted structural history of the deposit is described as follows:

A northeast to southwest striking longitudinal fault is the oldest structure. This controlled part of the intrusion and the Antamina valley. There is one exposure of this fault at the head of the valley below a later thrust.

A northeast verging thrust sequence developed during the Late Eocene, Incaic 2 phase (ca 40– 41 Ma). The Antamina deposit is located within a localized thrust tongue formed by at least six flat-lying thrust sheets. The tongue is 3 kilometers wide and 3 kilometers long, although it may have had a greater original extent. This thrust sequence is an imbricate stack, which has resulted in a super thickening of the favorable host rocks (Jumasha) in the area.

Very localized extension occurred on the southeast side of the present day Antamina valley. Extension was accommodated by listric faulting and by strike-slip fault movement along the main northeast-southwest longitudinal fault. This minor extensional phase may be correlated with the Quechua 2 phase regionally. The Antamina intrusions are interpreted to have been controlled by the listric faults as they are seen occupying fault planes in the limestone above the deposit. Within the deposit no obvious sign of these faults exists since their loci are now completely obliterated by intrusion and skarn.

Post-mineral high angle fault movement appears to have occurred on the southeast contact of the skarn on the southeast side of the valley. There is neither significant displacement nor cut-off of the skarn. However, there is considered to be some slip due to ductility contrast at the skarn/marble contact.

Within the skarn and intrusions there are zones of brittle breakage and slickensided surfaces, however no significant post-mineral fault displacements have been identified. At the margins of the deposit in Usu Pallares and Fortuna, thin intrusive sheets are apparently controlled by thrust ramps.

Figure 7.4

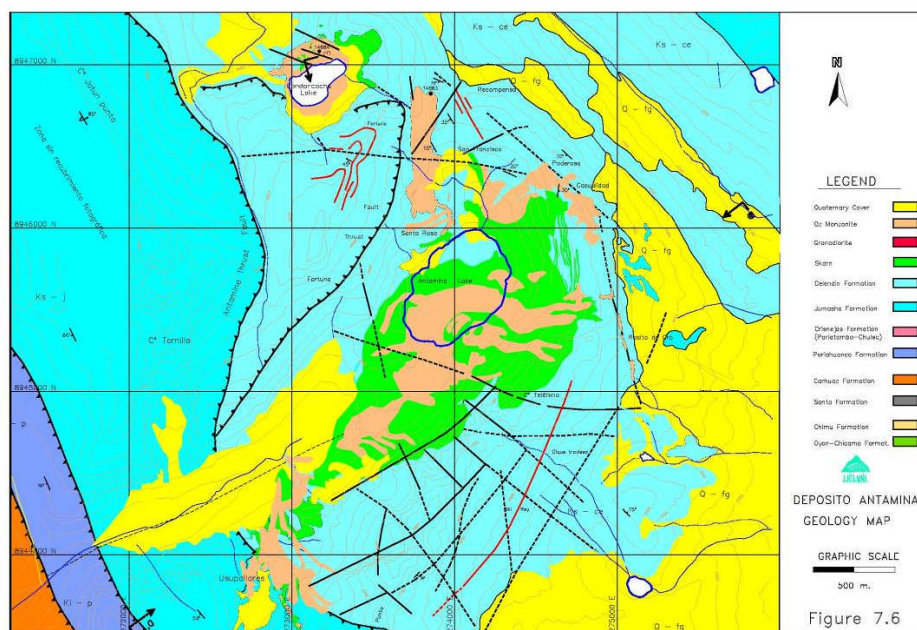
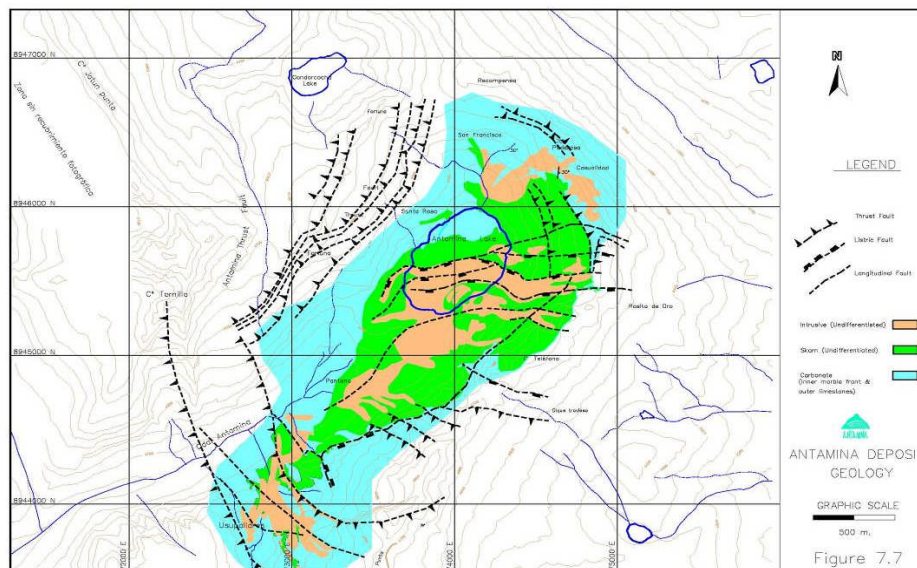


Figure 7.5



8 Deposit Type

The Antamina and Usu Pallares deposits comprise a very large copper skarn with zinc, silver, molybdenum, lead, bismuth and arsenic formed by the intrusion of quartz monzonite body into limestones. Characteristics of the skarn zone depend upon the nature of both the intruded rock and the emanations from the intrusive body. Two types of alteration are recognized; recrystallization, or rearrangement of the constituents already in the rocks, and addition of chemical elements. Most skarn deposits show both features. The most striking skarn deposits are developed when the host rock is a sedimentary rock of carbonate composition. Mineral deposition in skarn deposits typically displays structural and stratigraphic controls. Skarn lithology is usually zoned from the fluid source both inward and outward. A typical sequence is; intrusive altered to garnet rich endoskarn, garnet rich exoskarn, diopside exoskarn, wollastonite exoskarn, hornfels, marble, and unaltered limestone.

9 Mineralization

The Antamina and Usu Pallares orebodies have proven to be consistently well mineralized and predictable, both in terms of grade and metal zoning over large areas, however short distance variability is quite high as is typical in most skarn deposits. Very little of the skarn lithology comprising the deposits is unmineralized.

As with the skarn silicate mineralogy, Antamina and Usu Pallares are horizontally and vertically zoned with respect to major metal components. This lateral zoning is clearly related to the orientation of the intrusive and limestone contacts and continues throughout the nearly one kilometer of vertical range of the deposit explored to date.

Metal zonation is quite distinctive within the deposit (see Figure 9.1 and Figure 9.2). Copper occurs relatively evenly distributed from endoskarn to the limestone contact. Zinc and bismuth tend to occur within 70 meters of the contact of green garnet exoskarn with limestone/marble/hornfels. Molybdenite is generally located within the intrusive core and surrounding endoskarn. Silver is present in any of the exoskarn lithologies. Lead is generally located in green garnet and diopside exoskarn and hornfels. Cobalt is generally associated with sphalerite mineralization. However veins and blebs of tennantite and other minerals can be found as rare occurrences in any rock type at Antamina. Arsenic occurs in solid solution within the copper minerals and in a variety of arsenic bearing minerals, contained within a valley parallel structural corridor and within a zone at the intersection of this structural trend with the Oscarina Dike structural trend.

Figure 9.1: Lithology and metal zoning

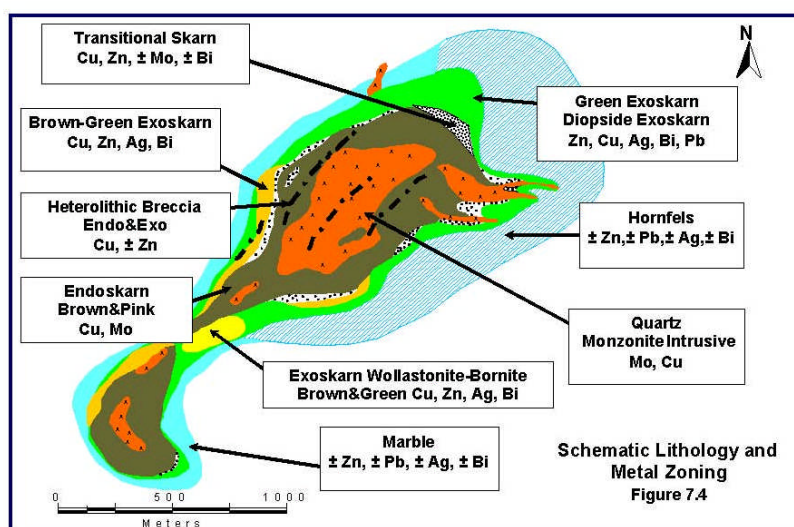
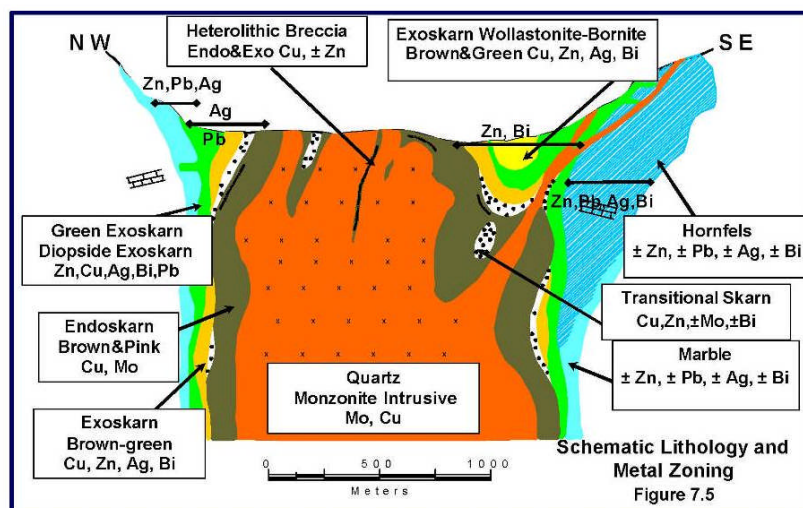


Figure 9.2: Lithology and metal zoning



Chalcopyrite is the predominant copper sulfide mineral. Approximately five percent of the copper mineralization is in the form of the copper sulfide mineral bornite. Zinc occurs primarily as the sulfide mineral sphalerite. Silver is normally associated in solid solution with chalcopyrite. However, it is also associated with galena, bismuth sulphosalts and tennantite. Molybdenum exists as the sulfide mineral molybdenite.

The most common bismuth minerals are bismuthinite, cosalite, whittichenite, cuprobismutite, aikinite, and kobellite, although various other Bi minerals as well.

There was little exposure of the deposit to oxidation and the evidence is in the way of a very limited oxidation cap and minimum supergene enrichment. This is due to the recent glaciation of the deposit, and the cold climate. However, there were zones of acetate soluble copper mineralization in Phase 2, resulting in material unsuitable for the production of salable quality concentrate. This oxidized portion has been almost completely mined to date.

10 Exploration

The Antamina project progressed beyond the exploration phase during the EME work. As the project is now in production, it is the author's opinion that not a lot of detail is required on early exploration work.

The Antamina deposit has been drilled and underground sampled to the level that a resource and reserve model was constructed and a feasibility study undertaken in 1998. Project construction was commenced in 1999 with initial production realized in June 2001.

Antamina Geology staff, supported by technical personnel from sponsor companies, performed all work since 1996; with the exception of outside consultants used to assist or supervise various aspects of the project. Antamina has the highest level of confidence in the exploration, development, and resource estimation work.

During 2002 and 2003 the Mineral Research Deposit Unit (MRDU) of the University of British Columbia visited Antamina as part of the study "Sources and Exhausts in Polymetallic Carbonate rock-hosted ore deposits: Miocene magmatism and alteration in Central Perú". As a result of this



study the satellite areas of Fortuna, Usu Pallares and Condorcocha were associated by geochronology, isotopic and alteration patterns to the main intrusive and Antamina skarn. These distal exposures require mapping and sampling to identify economic targets.

In 2005 Larry Smith from AMEC conducted a review of the historical exploratory information within the district. A series of potential areas for grass-roots exploration were identified based on stratigraphic, structural and morphological patterns. No exploration efforts have been performed in those areas but detailed information can be obtained in the report *District and Regional Exploration Program, Antamina, Ancash Department, Peru by AMEC to Compañía Minera Antamina, February 20, 2006*, which is available upon request.

11 Drilling

The Antamina data set contains core drill data, reverse circulation drill data, and data from underground drifts. The data set is described below.

Cerro de Pasco Corporation drilled 32 drill holes, 18 from surface and 14 from underground for a total of 3,200 meters. They also drove 4,300 meters of the 6,000 meters of underground exploration drifts and 220 meters of exploration raises. The data that survived is of dubious quality as there is no quality control procedures documented for sampling and assaying. This data is not used in resource modeling.

Empresa Minera Especial (EME) drilled 174 core holes for 19,885 meters and extended the underground drifts to the present total of 6,000 meters. In addition the entire 6,000 meters of underground drifts were channeled sampled. All sample rejects and pulps are stored on site in an organized fashion and are available for inspection and re-sampling.

The Antamina 1996 – 1997 core drill program consisted of 294 holes for a total of 106,446 meters. The holes were mostly NQ in diameter, but some were HQ in diameter. Most of the sample rejects, pulps and remaining core are stored on site.

The Antamina 1999 – 2000 core drill program consisted of 93 holes for a total of 12,482 meters. 59 of the holes were drilled on Lake Antamina using a barge. The holes were mostly HQ in diameter, but some were NQ in diameter. The sample rejects, pulps and remaining core are stored on site.

The Antamina 2002-2003 core drill program consisted of 162 holes for a total of 32,431 meters. In addition 32 reverse circulation holes for a total of 5,688 meters were drilled as part of the waste definition program.

The Antamina 2003-2004 core drilling program consisted of 493 holes for a total of 111,736 meters. This extensive program included 245 holes for a total of 72,697 meters focused on upgrading inferred resources into the indicated category. A second program included 246 holes for a total of 38,542 meters focused on reducing uncertainty for ore scheduling during the period 2005-2007 by upgrading resources into the measured category. A third component was the waste definition program with two holes for a total of 492 meters.

The Antamina Main 2005 drilling program consisted of 112 holes for a total of 20,152 meters. This program included 24 holes for a total of 7,689 meters focused on upgrading inferred resources into the indicated category. A second program included 80 holes for a total of 10,721 meters focused on reducing uncertainty for ore scheduling during the period 2007-2010 by upgrading resources into the measured category. A third component was the waste definition program with 8 holes for a total of 1,742 meters.

The 2005-2006 Usu Pallares drilling program consisted of 88 holes for a total of 22,819 meters. The objective of this program was to convert the resources into reserves for mine planning purposes. Previously, in 1997 eleven holes were drilled over this area and the potential was identified for skarn



mineralization similar to the Antamina deposit but with lower grades for copper and zinc. Based on this evidence and recent surface mapping a drilling program was designed to define the extent of the economic mineralization.

The 2007-2008 drilling programs were focused on the conversion of inferred resources into the indicated and measured categories with the consequent addition of reserves for mine planning. The program completed 574 holes for a total of 264,899 meters drilled mainly in the Antamina Main area.

Table 11.1 summarizes the various drill and underground programs conducted at Antamina to date. Additional reverse circulation condemnation and geotechnical drill programs have been conducted; however, this information is not used in resource modeling.

The drill hole and underground data has been entered into acQuire database software for validation and storage. This information has been audited by Antamina staff and independent consultants in various occasions in order to guarantee the integrity and suitability for resource modeling purposes. This will be explained in detail in section 14 Data Verification.

The drill hole and underground data is then transferred using comma delimited files into MineSight v3.0 modeling software and a three-dimensional model for the geology has been constructed using this information.

Table 11.1: Summary of Antamina Drilling and Underground Data

Company	Year(s)	Sample Type	Num. Of Holes	Num. of Assayed Samples	Total Meters	Assay Lab	QAQC	Blanks	Dups	Stds	Umpire Lab
EMEA (Centromin)	1970-1975	Core	109	6,296	13,486	Centromin	Unk	Unk	Unk	Unk	Unk
EMEA (Centromin)	1970-1975	Channel Sample	65 (128 tunnel walls)	10,507	12,399	Centromin	Unk	Unk	Unk	Unk	Unk
Antamina	1996-1997	Core	294	38,253	106,446	SGS and Chemex	Yes (10%)	2%	5%	4%	Yes
Antamina	1997 (assayed 2003)	RC	58 (2 assayed)	384	3,407	ALS-Chemex	Yes (10%)	2%	4%	5%	None
Antamina	1999- 2000	Core	93	3,588	12,482	SGS and Chemex	Yes (11%)	2%	4%	4%	Yes
Antamina	2002- 2003	Core	162	15,000	32,431	ALS-Chemex	Yes (10%)	2%	4%	5%	None
Antamina	2002- 2003	RC	31	3,685	5,688	ALS-Chemex	Yes (10%)	2%	4%	5%	None
Antamina	Late 2003	Core	25	2,705	4,686	ALS-Chemex	Yes (13%)	3%	4%	7%	None
Antamina	2004	Core	468	54,961	107,050	ALS-Chemex	Yes (19%)	3%	5%	11%	SGS
Antamina	2005	Core	112	10,216	20,152	ALS-Chemex	Yes (17%)	3%	3%	11%	None
Usu Pallares	2005-2006	Core	88	12,476	22,819	ALS-Chemex	Yes (17%)	3%	3%	11%	None
Antamina	2006-2007	Core	200	49,097	105,057	ALS-Chemex	Yes (20%)	5%	7%	8%	None
Antamina	2007-2008	Core	374	77,343	159,842	ALS-Chemex	Yes (20%)	5%	7%	7%	None

12 Sampling Method and Approach

Assay data used for resource modeling is predominately from drill core samples (95%), with lesser amount of underground channel (4%) and reverse circulation (1%) samples (see Table 11.1). Drill core sampling methodology for all Antamina drilling campaigns was the same: the drill core was cut in half, and sample intervals were set at 3 meters in length; however, the length of samples were adjusted to start and end at major geologic alteration and lithology contacts, all efforts were made to collect samples with a minimum length not less than 1 meter, except for density samples which average approximately 0.15 meter in length. The whole length of the drill hole was sampled, whether mineralized or not, except where low recovery prevented the collection of a representative sample. There are 269,935 drill core assay samples. Underground tunnels were channel sampled by EMEA (Centromin) in a textbook fashion. This level of detail is not usually seen in the industry. The left and



right walls of the drift were sampled separately, with the average sample length being 1 meter. There are 10,507 underground assay samples. Reverse circulation (RC) drilling was used for condemnation and waste drilling. Pre-2002 RC holes were sampled every 3 meters; RC holes drilled during the 2002-2003 campaign were sampled every 1.5 meters. There are approximately 4,069 RC assay samples.

MRDI performed an analysis of grade and drill core recovery during the Compañía Minera Antamina S.A. August 2000 Resource Model interpolation study. No bias was noted for Cu, Zn, Mo, Ag, Bi and Co with respect to core recovery.⁵ Therefore the samples used for the resource estimation can be considered representative.

It is the geologic section author's opinion that the assay dataset has a very good spatial coverage (both aerially and vertically), and contains significant number of measurements for each of the rock types used in the resource estimation.

Density data have been collected with varying degrees of frequency and methodology throughout Antamina's project history:

Centromin: 6,582 samples, unknown method, data not used in resource model
Antamina 1996-1997: 2,153 whole core samples, caliper method
Antamina 1996-1997: 473 split-core samples, wax coat method
Antamina 1999: 76 split-core samples, wax coat method
Antamina 2002-2003: 1,418 whole core samples, wax coat method
Antamina 2002-2003: 254 whole core samples, caliper and wax coat methods
Antamina 2004: 8,268 whole core samples, wax coat method
Antamina 2005-2006: 14,236 whole core samples, wax coat method of which 2,285 belong to Usu Pallares⁶
Antamina 2007-2008: 16,943 whole core samples, wax coat method, large size density samples ("triple tube") 3,224 samples

As can be seen in the above list, most of the data has been collected using two methods: caliper and wax-coat water immersion. Both of those methods have been applied to approx. 15 centimeter-long whole core samples, although some half-core samples have also been used. A study performed for the 2003 Interim Resource Model determined that the Antamina wax and caliper datasets are equivalent⁷. Thus, both datasets were employed in the 2009 Resource Model.

Additionally, as part of the 2004, 2005 and partial 2006 drilling campaigns, studies were performed to determine correction factors for the bias between the whole core density and the in-situ density⁸. In 2004 the program consisted in the collection of a set of 1,014 3-meter long undisturbed core samples (large-scale density samples, or LSDS), as a proxy for in-situ density. In the 2005-2006 drilling campaign the 3 meter long samples were increased by an additional 741 samples that were incorporated into the database to validate density determinations for resource estimation. In the 2007-2008 drilling campaign the 3 meter long samples were substantially increased by 3,224 determinations, providing higher confidence on the adjustment factors applied in the 2009 Resource Model⁹.

The density of the LSDS samples was determined by measuring the weight of the 3 meter long sample (after drying it in the oven to constant weight), and its volume (measuring the true length of the sample, and at least two measures of the core diameter using a caliper). After determining the density of the LSDS sample, one or more 15 centimeter long whole core samples were collected from

⁵ Parker, H., *MRDI Memorandum No. 1, Drill Core Recovery Analysis*, July 2000.

⁶ AMEC, *Compañía Minera Antamina S.A., 2006 Resource Model, Huaraz Peru*, chap. 15 Density, November 2006.

⁷ AMEC, *Compañía Minera Antamina S.A., 2003 Interim Resource Model Antamina Mine, Huaraz Peru*, chap. 20 Density, December 2003.

⁸ AMEC, *Compañía Minera Antamina S.A., 2005 Resource Model, Huaraz Peru*, chap. 16 Density, August 2005.

⁹ AMEC, *Compañía Minera Antamina S.A., 2009 Resource Model, Huaraz Peru*, chap. 16 Density, November 2009



within each of the LSDS samples (for a total of 1,152 whole core samples). The density of those whole core samples was determined using the standard wax coat water-immersion method. Finally, a comparison was made of the results obtained by both methods. The results of that study validated the application of density correction factors based on the degree of fracturing and disaggregation of the rock. Those factors were subsequently used during the resource estimation procedures.

It is the geologic section author's opinion that the density dataset has an excellent spatial coverage, and contains significant number of measurements for each of the rock types used in resource estimation.

A bulk metallurgical sample was taken for pilot plant testing to determine grinding and bulk flotation characteristics of the ores. Due to oxidation of the existing drifts and the location of the drifts, a drift location for the bulk sample was selected in the likely starter pit area. A drift measuring 2.5 X 2.5 meters was driven for a total length of 225 meters, over which 135 meters were skarn, 40 meters were intrusive and the remaining 50 meters were limestone-marble. Compañía Minera Antamina S.A. collected material for the bulk sample by slashing from the adit walls taking care to keep fragment size as large as possible (for SAG mill testing). The locations of the slash were determined from channel sample assay results and the ore type required. Approximately 400 tonnes of material was shipped for testing.¹⁰

13 Sample Preparation, Analyses and Security

No details are known of the Centromin sample preparation, assaying, and QAQC protocols. Antamina checked a number of Centromin Holes in 1997 with a twin hole program and check assays of pulps. The results of those checks endorsed the Centromin Cu and Zn data for resource modeling, but the Mo and Ag data were rejected.

For all drilling programs run by Compañía Minera Antamina S.A., sample preparation, assaying, analytical, and quality control procedures have always followed acceptable industry standard practices. The procedures followed included the use of independent assay labs for all sample preparation and assays, and the use of QAQC protocols in all drilling campaigns. Additionally, most of the drilling programs included an independent audit of the QAQC program and results.

During the first Antamina drilling program (1996-1997), the following sample preparation procedure was followed¹¹:

- Crush half-core sample to ~75% -10 mesh
- Split -10 mesh fraction to ~250 grams
- Pulverize the 250 grams split to ~95% -150 mesh
- Analyze at SGS-Lima (Cu, Zn, Mo, and Ag by AA) and XRAL Labs in Canada (ICP suite).

Core cutting and QAQC sample insertion was performed by Antamina personnel at the mine site. Crushing and splitting of the -10 mesh fraction was performed by SGS personnel at the mine site. SGS personnel at their labs offsite performed the remaining sample preparation.

The 1996-1997 QAQC program comprised the use of standards, blanks, two types of duplicates (core and coarse-reject duplicates), and check assays (see proportions used in Table 11.1). Sieve screen tests were performed regularly to check performance of both crusher and pulverizer. Check assays (pulp re-assays at an umpire lab) were performed at Cone Geochemical Inc. (Colorado) or at Chemex Labs in Vancouver. The sample preparation, assay and QAQC programs were designed, supervised, and audited by an independent geochemical consultant (J. S. Zuker of Lakewood, Colorado).¹²

¹⁰ Bechtel International Inc., *Antamina Project Feasibility Study*, March 1998, vol. 2, app. 1, chap. 5.

¹¹ Bechtel International Inc., *Antamina Project Feasibility Study*, March 1998, vol. 2, pp 50-56.

¹² Bechtel International Inc., *Antamina Project Feasibility Study*, March 1998, vol. 2, p. 50.



The sample protocol for the 1999-2000 drilling program was slightly modified from the previous sample protocol¹³:

- Crush half-core sample to ~75% -10 mesh
- Split the -10 mesh fraction to ~5 kilograms
- Pulverize the 5 kilograms split to ~95% -150 mesh
- Analyze at SGS-Lima (Cu, Zn, Mo, and Ag by AA) and XRAL Labs in Canada (ICP suite)

All other procedures, including Quality Control procedures, were identical to those used in the 1996-1997 drilling program.

The sampling and assay protocols for the 2002-2003, 2004 and 2005-2006 drilling campaigns^{14,15} underwent further, but slight, modifications, and are as follows:

- Crush half-core sample to ~90% -10 mesh
- Split the -10 mesh fraction to ~2.5 kilograms
- Pulverize the 2.5 kilograms split to ~95% -150 mesh
- Analyze at ALS-Chemex Labs in Lima (Cu, Zn, Mo, and Ag by AA) and Vancouver (ICP suite and Bi by ICP-MS).

Core cutting and QAQC sample insertion was performed by Antamina personnel at the mine site. ALS-Chemex personnel at their Labs in Lima performed the remaining sample preparation.

The 2002-2003 QAQC program¹⁶ comprised the use of standards, blanks, and three types of duplicates (core, pulp, and coarse-reject duplicates; see Table 11.1). No check assays were performed, and sieve screen tests were performed regularly only for the -150 mesh fraction.

The QAQC program for the 2004 drilling campaign was designed and supervised by an independent geochemical consultant (S. Long from AMEC, USA and Armando Simon of AMEC, Lima Perú).¹⁷ This QAQC program comprised the use of standards, blanks, three types of duplicates (core, pulp, and coarse-reject duplicates), and check assays (see proportions used in Table 11.1).¹⁸ Sieve screen tests were performed regularly to check performance of both crusher and pulverizer. Check assays were performed at SGS Labs in Lima.

The QAQC program for the 2005-2006 and 2007-2008 drilling campaigns was conducted by Antamina personnel and it was audited by the independent consultant PhD Jeff Sullivan and Silvia Satchwell from Consultores de Recursos Minerales S. A. The auditor stated that the Antamina QAQC procedures and protocol were best practice in the world mining industry¹⁹.

In all Compañía Minera Antamina S.A. drilling programs, samples have been assayed for Cu, Zn, Mo and Ag by AA with aqua regia digestion, and As and Co (together with a suite of other 30 to 40 elements) by ICP-AES with aqua regia digestion. Lead has been assayed by either ICP or AA methods, depending on the Pb content of the sample. Bismuth has been assayed by various methods which are described below. Prior to the 2002-2003 drilling campaign, five methods were used for Bi analyses. It was determined that some of these methods did not accurately report the Bi value at levels below 70 ppm. Additionally, for some analytical methods, an increase in copper values resulted in a slightly increased Bi value being reported. An extensive investigation was undertaken as part of the Antamina August 2000 Resource Model work performed by MRDI and Eric Lipten. Correction equations were developed and a priority scheme was developed to select the Bi

¹³ Espinoza, J., *Drill Core Simple Protocol 1999-2000 Drill Program*, Antamina Internal Memorandum, January 15, 2000.

¹⁴ ALS-Chemex, *Quotation LI-068/02R prepared for Cia Minera Antamina S.A.*, August 1, 2002.

¹⁵ Pareja, G., *2004 Drilling Program Assay Protocol*, Antamina Internal Memorandum, December 3, 2003.

¹⁶ Pareja, G., *2002 Drilling Program QAQC Protocol*, Antamina Internal Memorandum, November 25, 2002.

¹⁷ Long, S., *QA/QC Report for the Antamina Data Entry Project*, AMEC, May 25, 2005.

¹⁸ Pareja, G., *2004 Drilling Program QAQC Protocol*, Antamina Internal Memorandum, October 15, 2003.

¹⁹ Sullivan, J., Satchwell, S., *QA/QC Audit, 2005 Drilling Program, July to September, 2005*, October 18, 2005.



assay used for block estimation for all pre-2002 Bi data²⁰. On all holes drilled after 2000, bismuth has been analyzed using a multi-acid digestion ($\text{HF-HNO}_3\text{-HClO}_4$ digestion + HCl leach) and a mass spectrometer coupled to an ICP (ICP-MS method). A study performed by AMEC in 2004²¹ showed that the ICP-MS method is without significant copper interference and without significant bias relative to Bi assayed by hydride-generation (considered the best method for bismuth assaying).

Starting with the 2002 drilling campaign, samples have been assayed for partially-soluble Cu and Zn using acetate-soluble Cu and Zn, and cyanide-soluble Cu. The soluble copper/zinc datasets are less than one-tenth of their respective total metals' datasets. Acetate-soluble copper and zinc are indicators of copper and zinc oxidized by weathering. Cyanide-soluble copper is an indicator of cyanide consumption during ore processing at the mill. The partial-soluble methods are important in the resource/reserve estimation process since they help forecast metallurgical performance for the copper and zinc concentrates. The partially-soluble assays have been performed using methods developed for ore control by the Compañía Minera Antamina S.A. concentrator laboratory.

The certifications for the various laboratories used by Minera Antamina S.A are as follows

Acme Analytical Laboratories – BS EN ISO 9001:2000
ACTLABS Skyline Peru S.A.C. - ISO 9001:2000 BVQI February 2005
ALS Peru – ISO 9001 (2000) ANSI/ASQ Q 9001:2000 BVQI October 2003
Genalysis Laboratory Services Pty Ltd – ISO/IEC 17025 (1999), NATA Accreditation No 3244
SGS Del Peru S.A.C – ISO9001:2000, AOAC, IFIA
Ultra Trace Pty Ltd – ISO/IEC 17025 (2001) NATA Accreditation No 14492

All original data (geological logs, field forms, printed assay certificates, core photographs, and all other types of paper forms) have been archived at the mine site inventoried and filed by hole. All the paper forms have been scanned and electronically stored, together with all the information that was originally received in an electronic form. A backup copy of all electronically-stored data (including backups of the SQL-based resource database) is stored in the Compañía Minera Antamina S.A. vault in Lima.

Industry accepted procedures were utilized for sampling, sample preparation, security and analytical procedures. The author has reviewed the sampling, analytical assaying and QAQC procedures and is confident on their adequacy for resource estimation at Antamina. The sample checks have demonstrated that the samples are representative of the mineralization and that there is no bias in the sampling.

14 Data Verification

Data verification has been an integral part of all Compañía Minera Antamina S.A. drilling campaigns and resource estimation programs. The details of the data verification procedures used in each drilling campaign are explained in the following paragraphs.

As part of the initial Compañía Minera Antamina S.A. exploration work (1996-1997) a rigorous data handling and data checking protocol was followed for the treatment of geologic, analytical and QAQC data. During the same time period, a comprehensive effort was undertaken to validate assay data from earlier drilling campaigns. This work included twinning of some drill holes, re-analysis of a selection of old sample pulps and rejects, and comparison of overall results within a defined volume (test block) of the deposit. The conclusion drawn from this work was that the Centromin copper and zinc assay data was suitable for inclusion in the sample database, but that the silver and molybdenum assay data was not of sufficient surety for inclusion.²² Additional audits of geologic

²⁰ Parker, H., Lipten, E., *MRDI Memorandum No 2, Adjustment of Bismuth Assays*, July 5, 2000.

²¹ Long, S., *Re-assay for Bi by ICP/MS of samples previously assayed by hydride method*, AMEC, November 16, 2004.

²² Bechtel International Inc., *Antamina Project Feasibility Study*, March 1998, vol. 2, pp 77-84.



sample collection and data management were performed by an independent geologic consultant²³ and by bankers' representatives as part of the feasibility and financing exercises.

Several audits, data checks, and data additions were undertaken in 2000 as part of the construction of the August 2000 Resource Model:

- A 10 percent random audit of the Cu, Zn, Ag, Mo, and multi-element ICP assay database was performed.
- It was found that some Bi assay data had not been entered for the previous resource model. A 100 percent audit of Bi assay data was undertaken. The missing Bi assay data was entered using the double entry method.
- The core relogging data was entered and checked by multiple geologists.
- Down-hole acid survey tubes were read (by senior members of the Antamina Geology Department) then entered and compared using a double entry method.
- A 10 percent random audit of down hole gyroscopic data and core recovery data was performed.
- A 20 percent random audit of density data and acid-base accounting assay data was performed.
- All drill hole collars available in the field were surveyed or re-surveyed (by the Antamina Mine Engineering Department) using a high precision Garmin GPS instrument.
- Topographic correction was applied to holes/tunnels that were not re-surveyed.

All the audited datasets were found to be below the industry accepted error rate of one percent.

Prior to the 2002-2003 drilling campaign, it was decided to compile all pre-existing datasets into one single, Microsoft SQL-based database. One of the advantages of the SQL-based database, besides better data management and data security, is that it provides automatic data validation for most fields on data entry and import. This minimizes the chances of loading erroneous data into the database.

In 2002, as part of database migration, several additions, modifications, and corrections were done to the pre-existing data. The main modifications were as follows:

- Addition of Chemex Bi data
- Completion of gyroscopic downhole survey data
- Addition of Centromin recovery and density data
- Addition of Antamina condemnation drillhole data
- Revision and modification of Centromin tunnel data
- Change of drillhole names (name length ≤ 6 characters long)
- Addition of final assay fields

After all the pre-2002 data had been migrated into the SQL-based database, Antamina Geology Department personnel audited the data. The error rate was found to be below the industry accepted level of one percent. All errors noted during the audits were corrected and verified by Antamina Geology Department staff. The audits that were undertaken are as follows:

- Collar survey data – 100% audit
- Downhole surveys – 100% audit
- Assay data – 11.9% random audit
- Centromin assay data – 11.2% random audit
- Centromin assay data blank Cu and Zn – all blank intervals for Cu and Zn checked
- High-grade Cu and Zn assay data (Centromin and Antamina) – 100% audit for Cu $\geq 3\%$ and Zn $\geq 4\%$

²³ Bechtel International Inc., *Antamina Project Feasibility Study*, March 1998, vol. 1a, Comments by Strathcona Mineral Services Limited. pp. 4-53 to 4-55.



- Antamina pre-2002 Bi assays – 100% audit of all methods and data except for Chemex Bi data which was blind double-entered, compared and corrected
- Geology lithology data (Centromin and Antamina) – 12.9% random audit
- Geology mineral data (Centromin and Antamina) – 13.3% random audit
- Geology vein data (Centromin and Antamina) – 12.3% random audit
- Geology alteration data (Centromin and Antamina) – 13.2% random audit
- QAQC data – 100% audit
- Recovery data pre-2002 Antamina – 10% random audit
- Density data Antamina – 15.75% random audit, recalculation of density value; problems in original hand calculations were resolved.

Data entry for the 2002-2003 drilling campaign was not performed using a blind double-data entry. In order to avoid errors creeping into the database due to the lack of blind double-data entry, Antamina Geology Department staff performed a data checking procedure, similar to an audit. All errors noted during data checking were corrected and verified by Antamina staff. Listed below is the level of data checking performed for each data item in the 2002-2003 drill program:

- Collar survey – 100% check of entered data
- Downhole survey – 100% check of entered data
- Assay intervals – 100% check of entered data
- Assay data – 14.3% check of entered data
- Geology lithology data – 100% check of entered data
- Geology mineral data – 100% check of entered data
- Geology vein data – 100% check of entered data
- Geology alteration data – 100% check of entered data
- Density data – 100% check of entered data
- Recovery data – 100% check of entered data
- Point load data – 100% check of entered data

Consultores de Recursos Minerales S.A. of Chile (CRM) performed an independent audit of the Antamina geology database²⁴. CRM's database review included both Pre-2002 and 2002-2003 drilling data. CRM also performed a variety of tests on Centromin tunnel data. No compelling evidence was found to remove the Centromin tunnel data from the resource model database. All errors noted by CRM were corrected before initiation of resource estimation.

AMEC (Perú) S.A. performed all data entry for the 2004 drilling program in Lima²⁵. Antamina provided data to AMEC in a set of paper forms and digital files (a copy of all paper forms was retained at the mine site for security purposes). AMEC personnel subsequently entered data, using a blind double-data entry procedure, in which four personnel entered the data into two separate databases (two people per database). The two databases were systematically compared, and all discrepancies resolved. Data confirmation memos were sent regularly by AMEC to Antamina reporting errors that required further investigation. Those errors were fixed based on the corrected data provided by Antamina. AMEC prepared weekly and monthly reports documenting the data entry status, including a list of issues and a summary QAQC report.

CRM performed an independent audit of the 2004 drilling data²⁶. No significant errors were found, although inconsistencies between downhole survey methods (for a given hole) were detected, particularly for Maxibor® results. Antamina addressed this problem, by performing a study to prioritize the downhole survey data to be subsequently used in resource modeling²⁷. CRM concluded that the database was acceptable for use in resource estimation.

²⁴ Sullivan, J., *Antamina Resource Audit Results, Phase 1*, CRM, June 6, 2003.

²⁵ AMEC, *Data Entry Project, Ancash, Perú*, March 2005.

²⁶ AMEC, *Data Entry Project, Ancash, Perú*, March 2005, app. A.

²⁷ Gomez, P., *Down hole Survey Priority Update*, Antamina Internal Memorandum, October 27, 2004.



During the 2005-2006 drilling campaign in Antamina Main and Usu Pallares deposits the lithological unit indeterminate skarn was renamed to transitional skarn. Additionally, this unit was relogged to determine the accuracy of the logged intervals as it was found to be geologically inconsistent when modeling. This resulted in a better resolution of the indeterminate skarn and brown garnet endoskarn. The drilling data from the 2005-2006 drilling programs of the Antamina Main and Usu Pallares deposits was audited by CRM in the following exercises: Holes A1048-A1128 September 10th 2005, holes A1129 to A1159 November 22, 2005, holes A001-A1247 March 21, 2006 and A1160-A1247 March 30, 2006. The QAQC data and procedures were also audited by CRM²⁸ with no issues found to address. In detail, the database with this drilling data was verified, audited and validated by Antamina staff and CRM auditors as follows:

- Collar survey – 100% check of entered data
- Downhole survey – 100% check of entered data
- Assay intervals – 100% check of entered data
- Assay data – 100% check of entered data
- Geology lithology data – 100% check of entered data
- Geology mineral data – 100% check of entered data
- Geology vein data – 100% check of entered data
- Geology alteration data – 100% check of entered data
- Density data – 100% check of entered data
- Recovery data – 100% check of entered data
- Point load data – 100% check of entered data

During the 2007-2008 drilling campaign in Antamina Main the drilling information was audited by CRM in the following exercises:

Audit of Holes A1248 to A1368 (9 April 2007)
Audit of Holes A001 to A1467 (31 Aug 2007)
Audit of Holes A1468 to A1511 (12 Nov 2008)²⁹

AMEC (Peru) generated an annual QAQC report including all batches from June 2007 to December 2007. In this report AMEC assesses accuracy, precision and contamination of the Antamina sampling protocol and states that assays for all metals (except arsenic) fall within acceptable ranges. A consistent overestimation of the Arsenic values was established and reported to Antamina. Antamina requested to ALS-Chemex to conduct a study to verify values and to identify sources for bias. ALS-Chemex conducted a series of tests determining that the use of “hot-blocks” instead of hot-plates for the acid digestion prior to the analytical determination resulted in a more reliable measurement of Arsenic. The “hot-plate” digestion was found a more aggressive digestion provoking the sublimation of arsenic and therefore values tend to be lower. Around 20% bias was detected in the range of 600 ppm, this order of magnitude do not have a significant impact in the ore characterization because 80 ppm is set as the dividing value for low and high arsenic ores. In future resource estimations Antamina will assess the need to uniform this type of digestion in the current dataset.

The drilling database was verified, audited and validated by the Antamina staff and the external auditor CRM as follows:

- Collar survey – 100% check of entered data
- Down hole survey – 100% check of entered data
- Assay intervals – 100% check of entered data

²⁸ Sullivan J, Satchwell S., *Review of July to September 2005 QAQC reports*, Ancash, Peru, 2005

²⁹ Satchwell, S, and Sullivan, J, *Antamina Database Audit, Audit of holes A001 to A1888*, CRM memorandum dated 31 January, 2009, 139 pp.



- Assay data – 100% check of entered data
- Geology lithology data – 100% check of entered data
- Geology mineral data – 100% check of entered data
- Geology vein data – 100% check of entered data
- Geology alteration data – 100% check of entered data
- Density data – 100% check of entered data
- Recovery data – 100% check of entered data
- Point load data – 100% check of entered data

All the internal data audits performed since the year 2005 have been carried out under the direct supervision of Jhon Espinoza.

15 Adjacent Properties

No information from adjacent properties has been used in the exploration program or in the estimation of the mineral resource.

16 Mineral Processing and Metallurgical Testing

Extensive metallurgical testing was conducted on the Antamina ore body from November 1996 to December 1997. The work was carried out by reputable metallurgical laboratories under the direction of a committee of experienced metallurgical experts from within the ownership group supported by independent consultants. Much of the critical initial work was carried out in duplicate at different laboratories.

The metallurgical test results were further reviewed by independent engineers reporting to the banking consortium as part of the financing process. Metallurgical parameters derived from the test work were applied during the feasibility work after review by these independent engineers.

The following sources of pre-operational metallurgical test data are available:

- Over 400 bench scale flotation tests on 11 drill core composite (Metcom) samples at Lakefield Research Limited (Appendix 15H) and at G and T Metallurgical Services Ltd. (Antamina Project Feasibility Study, March 1998, Appendix 16)
- More than 75 small scale continuous grinding tests on 11 drill core samples by A. R. MacPherson Consultants Ltd. (Antamina Project Feasibility Study, March 1998, Appendix 17)
- Greater than 85 pilot plant grinding tests at Lakefield Research on 17 bulk samples (Antamina Project Feasibility Study, March 1998, Appendix 15B)
- Mineralogical investigations on 138 metallurgical test samples at Lakefield Research (Antamina Project Feasibility Study, March 1998, Appendices 15C, 15D, and 15E)
- 37 pilot plant and bench scale flotation tests at Lakefield Research (Antamina Project Feasibility Study, March 1998, Appendix 15A)
- 17 bench scale grinding and greater than 200 flotation tests at Lakefield Research (Antamina Project Feasibility Study, March 1998, Appendix 15G) on Metcom drill core samples selected for initial metallurgical test work.
- Mineralogical studies on 16 samples by AMTEL of Bi, Zn, Cu, and Ag deportment in flotation products (Antamina Project Feasibility Study, March 1998, Appendix 22)
- Determination of grinding requirements from 37 flotation tests at Lakefield Research (Antamina Project Feasibility Study, March 1998, Appendix 15H)
- Lakefield Research proposed grinding systems based on 168 samples from 17 pilot plant tests (Antamina Project Feasibility Study, March 1998, Appendix 15F)
- 23 bench scales dewatering tests on pilot plant concentrate samples by Larox, Inc. (Larox) (Antamina Project Feasibility Study, March 1998, Appendix 18), Pocock Industrial Inc. (Pocock) (Antamina Project Feasibility Study, March 1998, Appendix 19), and Lakefield Research (Antamina Project Feasibility Study, March 1998, Appendix 15)



- 25 tests for coarse ore and concentrate flow characteristics by Jenike and Johanson Ltd. (Antamina Project Feasibility Study, March 1998, Appendices 21A and 21B) and concentrate transportable moisture limits by SGS Canada Inc. (SGS) (Antamina Project Feasibility Study, March 1998, Appendix 15A)
- 25 bench scale tests on 5 samples by Fuller Company (Fuller) (Antamina Project Feasibility Study, March 1998, Appendix 20) on lime production from mine-site limestone samples
- 168 bench scale grinding and greater than 200 flotation tests at Lakefield Research (Antamina Project Feasibility Study, March 1998, Appendix 15I) on Valmet drill core samples from the initial 5 year mine plan

Concentrator operations were started in May 2001. Over ten years of operational and metallurgical data have been realized since the original feasibility test work. The operation data and metallurgical relationships developed subsequent to the feasibility study have been applied to the current resource model and used as part of the determination of Reserves and Resources reported herein.

17 Mineral Resource and Mineral Reserve Estimates

Antamina Geology conducted an exercise to update the 2009 Resource Model from February through August 2010 producing the 2010 Resource Model which is the basis for the current Compañía Minera Antamina S.A. reserve estimate. The motivation was to enhance the confidence on the estimation and classification of the waste material surrounding the skarn. For the mineralized portion of the deposit this version of the resource model is a replication of the 2009 Resource Model except for the update of the top 5 benches below August 20, 2010 topography that was used in the estimation of the Mid Term Resource Model. The Limestone/marble and Hornfels barren units were divided into 10 rock types in the updated geological model. On the other hand, the 2009 Resource Model is an incremental update to the 2008 Resource Model and included over 150,000 m of drilling completed since the database closure for the 2008 Resource model, in September 2007, and the end of the drill campaign in December 2008. All estimates in this report are referred to the 2010 Resource Model although the methodology is referred to the exercise for the construction of the 2009 Resource Model.

The geologic and resource models were constructed using MineSight 3D Version 3.30-05³⁰. The geologic model was built incorporating all available information including lithology identified in drill holes, blast holes and a structural model interpretation. All software used during the exploratory data analysis and interpretation was verified. This included the compositing and back tagging of drill holes and block model, histograms, contact plots, soft/firm/hard (SFH) composite selection, scatter plots, variography and kriging software programs.

Grade estimation was conducted within grade envelopes and/or domains (defined on rock groups, variogram, indicator and/or distance classes) adopting SFH boundaries as determined using the contact plots. Bismuth and zinc used low, medium and high grade envelopes. Molybdenum used low and high grade envelopes. Metals without a deterministic domain were copper, silver, arsenic, lead, iron, cobalt and the solubles (ammonium acetate soluble copper, cyanide soluble copper and ammonium acetate soluble zinc). The estimation technique for all metals will be described in Section 17.6 .

A series of validation procedures of estimated block grades was made against exploration drill holes and production blast holes. This included comparison of block values and composites in swath plots of 60x60x45 meters, histograms and contact plots of block values to compare to original composite values, Herco validation to detect the smoothing effect, blast holes reblocked to 20x20x15 meters for comparison to resource model blocks, and finally visual inspection.

³⁰ AMEC, *Compañía Minera Antamina S.A., 2009 Resource Model - Huaraz Peru*, chap. 3 Deterministic Models, November 2009.



The following summary description includes all relevant information with regard to the construction of the geologic model and the mineral resource estimate:

17.1 Model Data and Geologic Model

All data from drill holes, blast holes, and tunnels used for the geologic modeling is stored in acQuire databases³¹. Drill hole and blast hole data included all available information as of October 15, 2007. A structural interpretation model updated in year 2006 was also used to guide lithological/alteration contacts at the deposit limits. The geologic interpretation consisted of closed polygons following drillhole intercepts, blast hole logging and surface mapping. Antamina geology staff completed geologic interpretations in plan view every 15 meters at mid-bench elevation and using two sets of vertical sections spaced at 50 meters oriented N45W/S45E and N45E/S45W. The following lithology domains were interpreted: limestone-marble, hornfels, intrusive, pink garnet endoskarn, brown garnet endoskarn, transitional skarn, brown and green exoskarn, green exoskarn, diopside exoskarn, brown wollastonite exoskarn, green wollastonite exoskarn, and heterolithic breccia. Antamina geology staff reconciled the plan view and section interpretations in all three orientations.

Similarly to the 2006 geological model highwall and surface mapping was used to interpret the lithological units in a three dimensional model. For the Antamina Main deposit blast holes were also correlated within benches to outline thin breccia bodies and projected to deeper benches. In Usu Pallares blast holes were not available as there is no current mining activity in the area. In addition, the indeterminate skarn (called transitional in previous models) was relogged to better define its nature. These factors along with the increased amount of drilling at closer grid spacing in Antamina Main had a positive impact in obtaining a high confidence geologic model and hence increased resource model resolution.

The lithological rock code was stored in the block model using the block centroid designation with its proportion as percentage of the 10 x 10 x 15m primary block model. The block modeling was completed on a whole block basis. This means that each 10 x 10 x 15 m block has only one rock type or grade-domain codes. Each block has only one grade for each element. The final grade and density in the block was obtained by weight averaging by density and partial volume. Finally, the primary model values were reblocked into a 20 x 20 x 15 m block model which was imported into the final model.

The 15 meter and 7.5 meter from the drill hole composites and model blocks were tagged using the 15 meter mid-bench plan polygons. Drill hole assays were composited down-hole with a length of 7.5 meters to be used in interpolation.

The block model was rotated 45 degrees clockwise and coordinates were reduced by subtracting 8,940,000 in the north coordinate and 270,000 in the east coordinate to facilitate software processing. The model origin in the new reduced/rotated system is: Easting 1,255.57, Northing 4,967.17, and elevation 5003.00 meters. The block dimensions are 20 x 20 meters horizontal and 15 meters vertical. Within the model there are 160 columns by 185 rows by 105 levels.

A comparison of the drilled (as-logged) to the interpreted (back tagged) composites reveals an overall matching of 89%³², with the majority units brown garnet endoskarn and intrusive at 94% and 92%, 83% in the pink endoskarn, brown-green exoskarn is at 90% and green exoskarn is at 89% matching. This is evidence of the high resolution of the geologic model.

³¹ , *Compañía Minera Antamina S.A., 2009 Resource Model - Huaraz Peru*, chap. 2, Database, November 2009.

³² AMEC, *Compañía Minera Antamina S.A., 2009 Resource Model - Huaraz Peru*, chap. 3, Deterministic Models, November.



17.2 Deterministic Models

Deterministic interpretation models were built for zinc, bismuth, bornite and molybdenum³³ in both areas Antamina Main and Usu Pallares. For zinc and bismuth three grade envelopes were defined based on the grade value of the 7.5 meter composite for low, medium and high grade. The cut-offs for bismuth were 25 and 150 ppm and for zinc the cut-offs were 0.25 and 2.50 percent. The bornite deterministic model encompassed all bornite occurrences to constrain the bornite ore. The bornite ore is defined by a relationship between bornite and chalcopyrite intensities that produces a saleable Bornite Concentrate. For molybdenum a high-grade domain was developed corresponding to molybdenum grades greater than or equal to 0.02% Mo; everything outside the high-grade domain was assigned the default 'low-grade' domain. This threshold was established based on combined criteria considering the molybdenum grade distribution and metallurgical implications.

17.3 Contact Analysis

Lithology plays a critical role in grade distribution in Antamina. An extensive contact analysis is required to guarantee that appropriate interpolation rules are applied. This involves an examination of the behavior of assays across each type of lithology contact grouping for each metal³⁴. Based on this work the interpolation plan will either code a contact as hard, soft or firm. This controls the behavior of composite utilization at contact boundaries when block interpolation is performed. A hard boundary does not allow the use of composites from one side of the contact to interpolate blocks on the other side of the contact. A soft boundary places no restriction on the use of composites to interpolate blocks on either other side of the contact. A firm boundary is coded such that composites are used to interpolate blocks across the contact but there is a limited distance and composite paring factor with introduces a gradient at the contact.

17.4 Risk Adjustment

Capping or top-cutting has been traditionally applied to limit the influence on the resource estimate of very high-grade assays, which demonstrate no spatial continuity. If these assay grades are not reduced or otherwise restricted in the interpolation process, it is possible for large areas surrounding the high-grade assays to be assigned unreasonably high grades ('grade smearing'), particularly where a weighted average linear interpolation is used (e.g., kriging). In the worst case, waste will be assigned ore grades. Sometimes, using downhole grade plots and or drift-blast-hole samples, it is possible to establish the spatial influence of a very high-grade zone. In these cases, the actual high grade of a sample may be used in the interpolation process over a local area. Outside the local area it may be cut, but not eliminated, so as to have a reduced influence on grade estimates. In other places the very high grades may occur totally at random, with no indication in adjacent assays that the grade is adjacent to a high-grade zone. This is common where occasional veins cut disseminated deposits.³⁵

Of the two scenarios, the first is probably more applicable to skarn at Antamina, and indeed most porphyry-related deposits. The highest-grade composites will be used at the actual composite grade in a restricted neighborhood. Outside this neighborhood, the composites will be capped prior to interpolation. The second scenario would be more applicable to minor mineralization in the Limestone/Marble, Hornfels and Intrusive (in the core of the deposit area).

Outlier restriction was used to control the range to which high grade composites were used in estimation in the copper, zinc molybdenum, silver, and lead, and the results are detailed in the individual report chapters. A formal assessment of metal-at-risk was not carried out as part of the

³³ AMEC, *Compañía Minera Antamina S.A., 2009 Resource Model - Huaraz Peru*, chap. 3, Deterministic Models, November 2009.

³⁴ AMEC, *Compañía Minera Antamina S.A., 2009 Resource Model –Huaraz Peru*, chap. 1 Summary, p. 2-4, November 2009.

³⁵ AMEC, *Compañía Minera Antamina S.A., 2009 Resource Model – Huaraz Peru*, chap. 1 Summary, pp. 4-1 to 4-2, November 2009.



2009 modeling efforts, but, the outlier restriction applied to the metals of principal economic importance served to reduce risk.

Outlier thresholds were determined in two different ways:

- By visual observation of the probability plots
- Review of plans and sections to check for high grade “blowouts” in parts of the model where there is little geological support for large volumes of high-grade mineralization.

The first method gives a minimum threshold that usually pulls CVs down to an acceptable level when too high and this method has been used for copper, zinc molybdenum and silver. The second method was implemented during the review of plans and sections of each of the grade models near the end of the modeling process.

The effective amount of metal removed by the use of outlier restriction was checked with a separate kriged model, generated using uncapped, un-restricted composites, as part of the validation process.

17.5 Variography

Multidirectional variograms were computed using the Sage2001 program for Cu, Zn, Ag, Mo, Bi, As, Pb, Fe, Co, CuCN, CuAC and ZnAC³⁶. Variogram directions in the program were set to change every 30 degrees in azimuth and 15 degrees in dip giving a total of 61 correlogram figures. The down hole correlograms were used to establish the nugget effect for the directional variogram domains. Variograms were run within geographic domains for each metal. These domains were established by examining contoured grades on bench plots and the domains have been drawn to enclose areas where the perceived strike of the mineralization is relatively constant. In variogram domains where the amount of pairs was insufficient for variography the composites from the adjacent domain with similarities of anisotropy were included.

17.6 Grade Interpolation

As previously discussed in Mineralization (Section 9) the metal zonation in Antamina is quite complex within the deposit. Copper occurs relatively evenly distributed from endoskarn to the limestone contact. Zinc and bismuth tend to occur within 70 meters of the contact of green garnet skarn with limestone/marble/hornfels. Molybdenite is generally located within the intrusive core and surrounding pink garnet endoskarn. Silver is present in any of the exoskarn lithologies. Lead is generally located in green garnet and diopside exoskarn and hornfels. This zonation is related to the lithological zonation, structural features and stratigraphic controls during or prior to the mineralization events. This fact controls the geometry of the different variogram domains for each metal and also how the rock units can be grouped. A selection of the most appropriate interpolation method for each metal was performed in the 2003 Modeling Methodology Test Cases. The 2003 Modeling Methodology Test Cases was a comparison study of various estimation techniques for the Antamina Deposit. The techniques applied were: deterministic (grade envelopes), probabilistic (high, medium and low indicators), multiple indicator kriging and distance class grouping from the marble contact. Results were compared, validated and reconciled against the blast hole model from production.

The interpolation methods for the 2010 Resource Model for each metal will be discussed in the following sections. A summary is presented in Table 17.1.

³⁶ AMEC, *Compañía Minera Antamina S.A., 2009 Resource Model – Huaraz Peru*, chap. 1 Summary, pp. 3-1 to 3-2, November 2009.



Table 17.1: Element Interpolation Method Summary

	Cu	Zn	Bi	Ag	Mo	As	Pb	Fe	Co	CuCN	CuAC	ZnAC	Bornite
Deterministic Grade domains	No	High-Medium-Low	High-Medium-Low	No	High-Low	No	No	No	No	No	No	No	Yes
Probabilistic Domains (indicators)	No	No	No	No	No	Low-Med-High	Low(Low-High)-High	No	No	Yes	Yes	Yes	No
Indicator threshold	No	0.25, 2.5%	25, 150 ppm	No	0.02	85 ppm	0.10%	No	No	Ratio CuCN/CuTOT >= 20%	Ratio CuAC/CuTOT >= 5%	Ratio CuAC/CuTOT >= 7%	No
Distance class	No	No	No	No	No	No	Yes	No	No	No	Yes	Yes	No
Rock groups	24	3	0	11	2	5	4	6	28	4	7	4	No
Variogram Domains	100-111	200-213	400-410	800-811	300-309	600-609	500-508	700-710	900-908	NA	NA	NA	NA
Kriging passes	4	4	4	4	4	Indicator: 4 LAK Grade: 4	4	4	4	4	4	4	IDW2 *
Ordinary kriging	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	Pass 1, 2	NA
Simple kriging	Pass 3	Pass 3	Pass 3	Pass 3	Pass 3	Pass 3	Pass 3	Pass 3	Pass 3	Pass 3	Pass 3	Pass 3	NA
Assignment of local mean	Pass 4	Pass 4	Pass 4	Pass 4	Pass 4	Pass 4	Pass 4	Pass 4	Pass 4	Pass 4	Pass 4	Pass 4	NA
Outlier restriction	Varies by rock type	Low: 0.25%, 20m Med: 2.0%, 35m	Low: 100ppm, 50m Med: 150-3500ppm, 50m	Varies by rock type	Varies by rock group	Not applied	Not applied	Varies by rock type	Varies by rock type	Not applied	Not applied	Not applied	NA

* Bornite and chalcopyrite values were interpolated using Inverse Distance Squared for the bornite model.

17.6.1 Copper Model

Copper is the principal product metal for Antamina as the major contributor of the calculated profit for most ores. Rock groups were defined to simplify variogram analysis and make the block grade estimation process more efficient. Grouping was performed respecting Antamina's geology and mineralization trends (e.g. Oscarina Zone), and was supplemented by population statistics derived from histograms and box plots. A total of 24 rock groups were defined for the 2008 model update.

Copper was estimated using four pass ordinary and simple kriging with outlier restriction. Rock types were grouped based on geological observations, and statistical criteria, into 28 rock groups. In general copper grade seems to be uniform with increasing depth but in some areas grades increase slightly. Soft-firm-hard (SFH) lithological contact constraints were applied on selection of composites used for kriging.. The interpolation was performed within the 10 x 10 x 15m block and this primary model was reblocked into a 20 x 20 x 15 m block model which was imported into the final model.

17.6.2 Zinc Model

A combination of ordinary and simple kriging has been used to interpolate block grades within deterministic grade shells. This technique was tested against four other methods during the test modeling program in 2003, and was found to out-perform the others in terms of constraining high grade zinc composites at the contact with the limestone. Three grade domains were interpreted manually from drillholes and blast holes. The grade domains are called the high-grade domain, the medium-grade domain and the low-grade domain.

Zinc interpolation was constrained by grade domains and rock types. Independent soft-firm-hard (SFH) contact relationships, and outlier restriction were applied in the low-, medium- and high-grade domains for the Capped and Uncapped Models. Outlier restriction with a high-grade threshold was applied in the low- and medium-grade domains and outlier restriction with a low-grade threshold was applied in the high-grade domain.

The high nugget effects and limited spatial grade continuity, even within the deterministic domains, result in a relatively high smoothing-effect. The model's local predictability is expected to be poor, although on a large-scale, it should perform well. High-grade areas next to or in contact with Limestone/Marble or Hornfels and surrounded by thick medium-grade zones appear to be narrower and erratic at depth. This could be an artifact



of wider-spaced drilling; historically, close-drilling results in increased volumes of high-grade areas.

17.6.3 Bismuth Model

Due to metallurgical issues, bismuth may report to the copper concentrate, causing penalties that reduce the estimated profit for the Antamina ores. Its content in the ore is a very important factor that determines the nature and classification of ore types at Antamina. In the Antamina Main area bismuth was modeled using deterministic shapes for three grade domains: high grade (≥ 150 ppm), medium grade (25 to 150 ppm) and low grade (< 25 ppm). The high grade bismuth domain is typically present next to the contact between exoskarn and marble-hornfels. The envelopes can be as wide as 180 meters in the Oscarina area, but typically are 25 to 40 meters wide. The medium grade envelopes are wider and incorporate all skarn units. The low grade is typically comprised of intrusive and marble-hornfels lithology units.

Bismuth interpolation was constrained by grade and variogram domains; utilizing contact relationships varying from hard, soft or firm. As the deterministic interpretation was only related to drill holes/blast holes each 10 x 10 x 15m block could host one grade domain. The interpolation was performed independently for the grade envelope assigned by block centroid. The primary model was reblocked into a 20 x 20 x 15 m block model which was imported into the final model.

17.6.4 Silver Model

Silver is a by-product metal in Antamina concentrates. Silver is broadly controlled by lithology and can be grouped into lithology domains within the deposit. Low grade silver is predominantly present in marble-hornfels and intrusive in concentrations averaging less than 5 grams/tonne. Brown Endoskarn is also a relatively low-grade unit, averaging 9.1 g/t Ag. Brown-Green, and Green Exoskarn constitute intermediate units, with average grades of 15.0 g/t Ag, and 21.5 g/t Ag respectively. Brown Wollastonite and Green Wollastonite Exoskarn contain high silver concentrations, averaging 29.2 g/t Ag and 43.6 g/t Ag, respectively. Diopside Exoskarn, averaging 11.7 g/t Ag, and Enriched Brecciation Zone (13.5 g/t Ag in Enriched Brecciation Zone in Brown Endoskarn and 23.5 g/t Ag in Enriched Brecciation Zone in Exoskarn) are minor units showing low to intermediate grades.

Contact analysis was performed for selection of soft, firm and hard boundary control. Silver was interpolated using four-pass ordinary and simple kriging approach with outlier restriction; constrained by rock group and variogram domain. The interpolation was performed according to the rock group in the 10 x 10 x 15m block. The primary model was reblocked into a 20 x 20 x 15 m block model which was imported into the final model.

17.6.5 Molybdenum Model

Molybdenum is a by-product metal for the Antamina mine, and its contribution to revenue is reflected in the ore value per hour (OVPHR) block valuation metric used for planning purposes and Ore Control. Molybdenum is found in relatively high concentrations in some parts of the ore body. Molybdenum concentrations are controlled by structures and rock type and vary throughout the deposit. Economic grade concentrations ($> 0.02\%$ Mo) are found in the Intrusive, Endoskarn, and Brown Wollastonite Exoskarn rock types. Relatively very high grades ($> 0.1\%$ Mo) are found in the Main Pit area but not in the Usu Pallares area. The Exoskarn and Enriched Brecciation zones have lower grades, while Limestone-Marble and Hornfels are barren for molybdenum ($< 0.001\%$ Mo).

The 2008 drill-campaign has considerably increased the resolution of the molybdenum resource model, and helped to identify a deterministic shape using grades higher than 0.02% Mo following rock type or structural controls in North, East and Plan sections. The use of the



deterministic model as a control during grade estimation does an essential job in controlling the overestimation or smearing of high grades into the core of the Intrusive.

Molybdenum is one of the metals with the best continuity at Antamina; as is demonstrated by vertical continuity of the high-grade zone over 1,000 m. This continuity is confirmed by deep drilling. Deep drilling shows expansion of corridors into zones approximately 100 m wide in intrusive on both sides of the deposit. This material typically has copper grades less than 0.3% Cu. In particular Pink Endoskarn is favorable for molybdenum mineralization greater than 0.1% Mo. As the mine deepens it will gradually convert from a copper-zinc operation to a copper-molybdenum operation. Pit geometry could depend on molybdenum prices and/or concentrator recovery for molybdenum.

Contact analysis was performed within lithologies and grade envelopes obtaining soft, firm and hard relationships. Interpolation was performed within variogram domains, respecting rock type and grade code; using a four-pass ordinary and simple kriging approach with outlier restriction. The interpolation was performed according to the rock type and grade domain in the 10 x 10 x 15m block. The primary model was reblocked into a 20 x 20 x 15 meter block model which was imported into the final model.

17.6.6 Arsenic Model

Arsenic is deleterious in smelting and is therefore undesirable in feed to the concentrator. A head grade of 85 ppm could result in a concentrate with approximately 2,000 ppm which is the value where penalties are typically applied. High arsenic is localized mainly in two areas: one is at the intersection of the valley trend structural corridor with the Oscarina dyke structural trend and the second is coincident with the bornite-wollastonite area. Five rock groups were defined based on similarities of mean values. The value of 85 ppm was defined as the threshold between low and high grade domains.

Although arsenic was thought to be localized in structural corridors that parallel the NW-SE Oscarina trend and NE-SW Valley trend, additional drilling has shown these corridors are not well defined. Blasthole maps show localized areas of high-grade arsenic, but even at this scale of sampling, the detailed lithological or structural corridor controls are not discernable. Given the high density of exploration drillhole samples on the upper levels, where arsenic is more concentrated, the 2009 modeling method for arsenic has been simplified from previous years. A probabilistic method is no longer necessary given the relatively high density of exploration drillholes. Rather a more simple deterministic method has been employed. An arsenic nearest-neighbor (NN) model constrained by rock type forms the fundamental building block of the final arsenic model. The NN extrapolation of high arsenic grades (> 85 ppm As) was constrained with an anisotropic distance of 50 m. Exploratory data analysis (EDA) resulted in the classification of the NN model into two groups by grade (G1 < 85 ppm As and G2 ≥ 85 ppm As). Subsequent EDA resulted in the division of group G1 into 2 subgroups defined by rock type, and the division of group G2 into 3 subgroups defined by rock type. Contact plots were generated between all possible groups, subgroups, and rock types. These provided the basis for the definition of Soft-Firm-Hard (SFH) contact boundaries and distance from contacts (DClass) used in the kriging plan for the estimation of arsenic.

Validations indicate that the modeling approach is reasonable, considering the very high-“nugget” nature of arsenic, whose maxima can be more than a hundred times the average grades. The relatively high density of drillhole data is sufficient for the deterministic definition of stationary domains out of non-stationary data. At depth, the drillhole density becomes noticeably less. However, given the interpretation that the concentration of arsenic decreases with depth, and that the deterministic patterns defined at higher elevations are likely appropriate at depth, the deterministic method is also deemed appropriate at depth.

Capping was applied in order to avoid smearing of locally very high grades by using an



outlier restriction methodology. This resulted in the removal of approximately 2% of the total arsenic metal content from the model.

17.6.7 Lead Model

Lead is considered a by-product metal at Antamina mainly restricted to the upper elevations in the deposit. When grades higher than 0.1% or 0.3% (depending on ore type) are processed, lead overloads the flotation circuits causing concentrate quality issues. Some occurrences of veins with very high-grade lead ranging 1% to 2% are associated with structural features, some are impossible to extend and correlate to the lower areas in the deposit. In general, lead appears to be controlled by the skarn-marble contact and grade tends to diminish away from the contact.

Distances from the contact between the metasediment rocks and the skarn-intrusive rocks were categorized by distance-classes (DCLAS), and a DCLAS code was assigned to the composites and blocks. Groupings for indicators, lead grade estimation, and local means were obtained using DCLAS.

A threshold of 0.1% Pb was used to define the low-high indicator. Low-grade (PBLO) and high-grade (PBHI) lead grades were interpolated independently.

The final lead estimate (PBKR) was obtained with the following expression:

$$PBKR = (1 - PBIND) * (PBLO) + (PBIND) * (PBHI)$$

Where PBIND is the kriged indicator value.

Lead grades show little spatial continuity (high nugget-effect and short range variograms), which diminish the model's local predictability. This was reflected on bench and section plots by erratic patches of high grades. Validations proved that the modeling approach for lead was globally reasonable, considering the lack of continuity of lead grades. On a local scale, the model's predictability is poor.

17.6.8 Iron Model

Iron is a deleterious element that affects the processes at the concentrator when the iron head-grade exceeds 20%; at this level, iron causes a severe increase in reagent consumption and increases the tailings density.

The iron model does not require the deterministic or probabilistic zones that were deemed necessary for other metals at Antamina. Iron was modeled using a four-pass ordinary and simple kriging approach with outlier restrictions applied. One-pass nearest-neighbor estimation was also implemented using 10 m x 10 m x 7.5 m blocks for validation purposes. Five rock-groups were created by grouping rock types based on iron statistics and genetic grounds. Soft firm hard (SFH) lithological contact constraints were applied to the selection of composites used for kriging.

During the 2008 modeling work, a study comparing iron assays from ALS Chemex, the primary laboratory in the resource diamond drilling database, with the Antamina mine laboratory was completed. The study revealed a complex conditional bias between the resource model grade (ALS Chemex) and the mill grade (Antamina laboratory). These differences are caused by the different assay methods used in the two laboratories and negatively impacts the use of the resource model iron grade in long-term mine planning. A set of polynomial correction equations are used to correct the resource model iron grade to the final iron grade or mill iron grade prior to block valuation. The results obtained from the validation techniques applied on the Estimation and Final Models proved the models to be acceptable.



17.6.9 Cobalt Model

Cobalt is a deleterious element in zinc concentrates, and penalties are assessed if cobalt concentrations in zinc concentrates exceed 300 ppm. There are no penalties associated with cobalt in the copper concentrates. Cobalt concentrations are broadly controlled by rock type. Concentrations range from below detection limits of 1 ppm Co in distal Limestone to over 300 ppm Co in Exoskarn rock types. Modeling of cobalt was straightforward, and the strategy used was similar to that used for silver, copper, and iron. Seventeen rock groups were created based on the geology of the fourteen rock types and variogram domains modeled at Antamina and their cobalt distribution. Grade trends across the rock type contacts were examined, and each contact was assigned a soft, firm or hard classification to allow all, some, or no composites to be shared between the rock types during interpolation. The rock groups were further subdivided based on geographic domains to allow the development and application of more precise variogram models. The interpolation was performed according to the rock group in the 10 x 10 x 15m block. The primary model was reblocked into a 20 x 20 x 15 m block model which was imported into the final model.

17.6.10 Cyanide-Soluble Copper and Acetate-Soluble Copper and Zinc

Cyanide soluble copper (CUCN) and acetate soluble copper (CUAC) grades are controlled by surficial weathering of primary copper mineralization bornite occurrences within the copper mineralization. Acetate soluble zinc grades are controlled by surficial weathering of primary zinc mineralization. The final available copper (CUAVL) and available zinc (ZNAVL) grades used for block valuation are calculated from the total copper and zinc grades with a deduction for acetate soluble copper and acetate soluble zinc to account for metal that is not recoverable at the concentrator.

As a result of a study completed during the 2009 resource modeling campaign investigating the impact of new drilling on the definition of new mineralized zones with appreciable soluble grades, the block models for CUCN, CUAC and ZNAC were not updated in 2009. Grades for solubles in the final 2009 block models have been imported from the previous resource models.

CUCN, CUAC, and ZNAC models were estimated in 2008 using an indicator probability kriging approach to define domains containing appreciable quantities of soluble copper and zinc, followed by a series of passes of ordinary kriging, simple kriging and direct local mean assignment to estimate the soluble grades.

Exploratory data analysis (EDA) was carried out on 7.5 m composites including box plots, contact plots, summary statistics, variography of the indicators for soluble metal ratios, and the grade variables. The indicators were estimated from 7.5 m composites into 10 m x 10 m x 15 m blocks, referred to as the Kriged Models. Validation of both the indicator and the estimated grades was carried out using box plots, histograms, swath plots and merged contact plots. The Kriged Models are re-blocked into the 20 m x 20 m x 15 m blocks, referred to as the Final Models, using density weighting. Validation of the Final Models is carried out using box plots, histograms and swath plots.

The need to incorporate CUCN in metallurgical recovery equations, in order to address shortcomings with the current system, is being discussed by Antamina process staff, and a project to develop new equations for use in ore valuation is anticipated for early 2010. The use of CUCN in the new block valuation will place enhanced importance on the CUCN grade model, and an update of the CUCN model, taking into account the recommendations included in this report, is strongly recommended.

17.6.11 Bornite Model

Bornite is the second mineral source of copper mineralization at Antamina and represents approximately five percent of total copper ores. One dominant characteristic of the bornite



mineralization is the presence of bismuth sulfosalts in microscopic solid solution within the bornite minerals. The metallurgical treatment also requires additional cyanide consumption, hence the economic importance of assessing bornite mineralization. Bornite ores may only include a restricted amount of chalcopyrite when processed in order to produce a marketable bornite concentrate, see Table 17.2. If bornite is present in the ore but chalcopyrite is significantly greater in percentage, then the material is classified as transitional ore, which will produce a transitional concentrate product.

For modeling, a bornite envelope was interpreted based on any visual identification of bornite. Blocks were considered fully inside the bornite envelope if the percentage of the block within the bornite envelope was greater than zero (even a block only one percent within the bornite envelope is considered as a bornite block); otherwise it was considered outside the bornite envelope. Inside this envelope the intensities of chalcopyrite and bornite were interpolated by inverse distance at the power of 2. If the interpolated values of chalcopyrite and bornite within the block, met the criteria to be classified as bornite ore (value of 2 in Table 17.2) then blocks were flagged as bornite ore for ore type classification. If the values of chalcopyrite and bornite within the block were such that the block met the criteria of transitional ore, the block was flagged as transitional ore. Otherwise the blocks are tagged as chalcopyrite ore blocks.

Table 17.2 Bornite Flag coding system, 0=chalcopyrite ore, 1= transitional bornite-chalcopyrite ore, and 2= bornite ore

Chalcopyrite Intensity	Bornite Intensity					
	0	1	2	3	4	5
0	0	2	2	2	2	2
1	0	2	2	2	2	2
2	0	1	2	2	2	2
3	0	1	2	2	2	2
4	0	1	1	2	2	2
5	0	1	1	1	2	2

17.7 Validation of Block Model

The block model has been validated in several ways. An extensive review of block model grade plans and sections was performed jointly by the Antamina Geology Department and AMEC resource modeling teams. This was to ensure that interpolation was honoring the data and interpolation envelopes. Comparing an uncapped kriged model against the capped kriged model provided a check of the capping levels to ensure that the targeted reduction of metal at risk was achieved in practice. The average grades of capped kriged, and nearest neighbor model were compared by 60 meter wide slices through the block model parallel to the X, Y and Z axes of the model. The average grades of composites were also compared to those of the capped kriged and nearest neighbor models. It was found that in general there is reasonable agreement among the various validation methods.

Dr. Jeff Sullivan and Ms. Sylvia Satchwell of Consultores de Recursos Minerales S.A. (CRM) were the third-party auditors retained by Antamina for the 2009 Resource Model³⁷. They performed the same services for the 2003, 2005, 2006 and 2009 Resource Models, and are thoroughly familiar with the Antamina deposit, and the resource estimation and classification methodology employed by Antamina and AMEC staff.

Auditing began in early 2009 with a check of the database and a review of the geologic model, and has continued to October 2009 even during the construction of the Resource Model. The

³⁷ Sullivan, J, and Satchwell, S, Resource estimation audit Antamina Resource Model, 2009, report dated October 26, 2009, 216 pp



CRM audit has therefore focused on intermediate work products, handouts from presentations, and verbal information received from discussions with project personnel.

17.8 Resource Classification

Resource model blocks were classified in accordance with the CIM Standards on Mineral Resources and Reserves Definitions and Guidelines adopted by the CIM Council on August 20, 2000 and updated on December 11, 2005 to meet the new requirements of National Instrument 43-101 effective at the end of 2005. Additionally the resource model blocks were classified in accordance with the Joint Ore Reserve Committee (JORC) of the Australasian Institute of Mining and Metallurgy (AusIMM requirements).

The current methodology apply at Antamina to classify resources was tested and validated in the 2003, 2005 and 2008 resource models, therefore, for the 2009 Resource Model the same classification scheme was utilized for the entire deposit. The block classification criteria was modified for the 2009 Resource Model to account for situations where molybdenum occurs in sub-economic copper grades and where the principal economic metal is therefore molybdenum.

Resources were classified as Measured, Indicated or Inferred by considering the following:

Geology, sampling and grade estimation aspects of the model.
Confidence limits on grade based on variography.
Confidence limits on tonnage, grade and contained metal for different drill hole spacings.
Geometric configurations of ore in various grade ranges on maps and sections.
Reliability of data, and in particular Centromin drill holes and tunnels.
Conditional bias in local resource estimates.
Anisotropy in horizontal and vertical ore projections.
Ore/waste ratio in geologic units by vertical position in the deposit.

In the description below, composites have a nominal length of 7.5 meters (the same as used for kriging). A "hole" is considered as a drill hole or tunnel. Four kriging passes with increasing search radius were used to assign grade into blocks. Passes 1 and 2 are suitable for resource classification as the search ranges are within 150 meters³⁸.

Classification is done separately by grade class group:

- Grade class 0, 0.0 to 0.499% Cu or 0.0 to 0.499% Zn
- Grade class 1, 0.5 to 0.999% Cu or 0.5 to 0.999% Zn
- Grade class 2, 1.0 to 1.999% Cu or 1.0 to 2.499% Zn
- Grade class 3, $\geq 2.0\%$ Cu or $\geq 2.5\%$ Zn

17.8.1 Measured Resources

The Centromin data (drill hole + tunnel) indicator must be less than 0.8 and the Centromin tunnel indicator must be less than 0.7.³⁹

Classification is done separately by grade zones:

- For a block in grade zone 3, there must be three composites of the same grade domain from different holes within 30 m of the block center, with the closest being within 20 m of the block center.

³⁸ AMEC, *Compañía Minera Antamina S.A., 2009 Resource Model – Huaraz Peru*, chap. 24 Resource Classification, November 2009.

³⁹ As discussed in Section 11 (Drilling) the sampling and assaying quality of the Centromin underground tunnels and drillholes are not at current industry best practice levels. This fact compromises the confidence of resources estimated predominately with Centromin data to achieve the category of measured. Kriged indicators were assigned to blocks using the same kriging plans and search ranges used for Cu and Zn interpolation. The resulted value varied from 0 to 1 indicating no influence to total influence of Centromin data during grade interpolation. The block classification was downgraded as appropriate if the estimation of block grade was affected by Centromin data.



- For a block in grade zone 0, 1 or 2, there must be three composites of the same grade domain from different holes within 35 m of the block center, with the closest being within 25 m of the block center.

The first criterion approximates a 25 meter hole spacing; the second criterion approximates a 30 to 35 meter hole spacing. A special Measured Resource zone occurs within 30 meters of current pit topography if:

- The block was kriged for zinc in pass 1 or 2.
- There are composites within 80 meters of the block center.
- The zinc grade zone is equal or higher than the grade zone for copper.

17.8.2 Indicated Resources

Classification is done separately by grade class (zone) groups:

- Grade Zone 0
- Grade Zone 1
- Grade Zones 2+3

A block is considered Indicated if:

- There are three composites of the same grade group from different holes within 55 meters of the block center, with the closest being within 40 meters of the block center, or
- There are two composites of the same grade group from different holes within 65 meters of the block center, with the closest being within 30 meters of the block center, or
- There are at least 50 composites within 75 meters of a block center with at least 90% having the same grade group as that of the block. This is called second pass Indicated (long class 29).

The first two criteria approximate a 50 meter hole spacing; the last criterion approximates a 75 meter hole spacing in a homogeneous area.

17.8.3 Inferred Resources

Remaining blocks kriged in passes 1 or 2 that have not been classified as Indicated or Measured, and blocks kriged in pass 3 are considered Inferred. Around 50% of the reported resources at Antamina are classified as Inferred. Because of this, and because the resources at Antamina are reported under the JORC Code, it was considered appropriate to subdivide the Inferred resource at Antamina into Interpolated and Extrapolated categories. The methodology for this consisted of a series of polygonal interpretation carried out on mine bench plans at 15 m intervals with drill hole data extrapolated vertically 45 m on each side of the bench. The interpolation outlines were drawn around the drillhole data joining drill holes up to 300 m apart. The interpretation was extended a few meters above the pre-mine topographic surface ensuring all blocks at surface were coded properly. The bench interpretation resulted in a set of closed polygons representing the area within which data have been interpolated (i.e., where no extrapolation had occurred).

17.9 Dilution of Resource Block Model

The 2009 Resource Model differs from earlier Antamina resource models in that a 10 x 10 x 15m block size was used for interpretation of lithological units including barren rock types. A later reblock process included this primary block model into the Antamina SMU 20 x 20 x 15m block size. Therefore, it is possible that within the same SMU block some primary blocks can grade above cut-off and some primary blocks can grade below cut-off. The final resource block model grades are weighted averages of the primary block grades. If the overall block grade is above



cut-off, it may contain sub-blocks of pure ore and dilution. If the overall block grade is below cut-off, it may contain primary blocks of ore loss⁴⁰ and other material below the cut-off grade.

No additional dilution is considered necessary for mine planning purposes.

17.10 Mineral Reserves and Resources

The pit design and the life-of-mine plan product of the mine planning process described in Section 23.3 was used to establish the spatial limits for the Mineral Reserve Estimate.

The current life-of-mine plan was developed taking into account mine and concentrator capacities according to the current expansion project (Compañía Minera Antamina - Expansion Program Feasibility Report, October 2009). The waste dump capacity was updated with the latest operational, geotechnical, hydrology, legal and environmental criteria, and the remaining storage capacity of the tailings impoundment was calculated at the 4165 dam crest elevation.

The life-of-mine plan also includes a variable cut-off policy for the first two years, beyond that period a flat cut-off was selected; the cut-off is based on the net value before taxes that the material will generate per hour of concentrator operation.

An economic analysis based on the production schedule for Proven and Probable Mineral Reserves using the Antamina price/cost protocol demonstrates a positive Net Present Value (NPV) for Antamina is describes in Section 23.9

In-pit material for reserves reporting purposes has been defined as that material which is between the December 2010 month-end pit surface topography and the designed limit of the final phase scheduled in the current life-of-mine plan.

Proven Reserves include in-pit measured blocks with a net value greater than or equal to the cut-off grade and existing high-grade stockpiles, excluding MP (a high-lead copper-zinc ore) and Oxidized ore stockpiles.

Probable Reserves include in-pit indicated blocks with a net value greater than or equal to the cut-off grade.

Material for resources reporting includes existing low-grade ore stockpiles, inferred material in life-of-mine plan above cut-off grade and the material located between the limit used in the life-of-mine plan and the optimized pit shell 2010 MII pit.

Measured Resources include measured blocks contained in the 2010 MII pit with a positive net value, existing low-grade ore stockpiles and measured material sent to the low-grade ore stockpile in the life-of-mine plan. Bornite ores (M5 and M6 ores) are not included and are treated as waste as there is not an area available to reasonably stockpile these ores for potential future processing. These ore types have also shown significant deterioration even in short-term stockpiles used for blending purposes and therefore long-term stockpiling of lower grades cannot be viable based on current ore processing methods.

Indicated Resources include indicated blocks contained in the 2010 MII pit with a positive net value and indicated material sent to the low-grade ore stockpile in the life-of-mine plan. Bornite ores (M5 and M6 ores) are not included and are treated as waste.

Inferred Resources include inferred blocks in the life-of-mine plan above the cut-off, inferred material sent to the low-grade ore stockpile in the life-of-mine plan and inferred blocks contained in the 2010 MII pit with a positive net value.

⁴⁰ Ore loss: sub-blocks above cut-off in blocks below cut-off. Dilution: sub-blocks below cut-off in blocks meeting cut-off.



The January 1, 2011 formal statement of Mineral Reserves and Mineral Resources for the Antamina Deposit is as follows:

Table 17.3: Statement of Mineral Reserve Estimate as of January 1st 2011.

Classification	Ore (Mtonnes)	Cu (%)	Zn (%)	Ag (g/t)	Mo (%)
Proven Copper Ores	100	1.07	0.16	8.3	0.034
Proven Copper Zinc Ores	47	0.84	1.83	15.0	0.007
Probable Copper Ores	492	0.95	0.14	8.9	0.027
Probable Copper Zinc Ores	183	0.83	2.00	14.5	0.006
Total Proven & Probable Reserves	822	0.93	0.66	10.4	0.022

Table 17.4: Statement of Mineral Resource Estimate (incremental)* as of January 1st 2011.

Classification	Ore (Mtonnes)	Cu (%)	Zn (%)	Ag (g/t)	Mo (%)
Measured Copper Ores	35	0.46	0.12	4.6	0.037
Measured Copper Zinc Ores	15	0.46	0.85	9.1	0.024
Indicated Copper Ores	276	0.93	0.12	9.2	0.023
Indicated Copper Zinc Ores	78	0.87	1.90	13.6	0.006
Meas. + Indicated Resources Total	404	0.86	0.49	9.6	0.021
Inferred Copper Ores	531	0.79	0.11	9.2	0.018
Inferred Copper Zinc Ores	177	0.54	1.33	9.7	0.003
Inferred Resources Total	708	0.73	0.42	9.3	0.014

*None of the Material identified as Mineral Reserve in Table 17.3 is included in the quantities stated as Mineral Resource in Table 17.4.

To the authors' knowledge, there are no known environmental, permitting, legal, title, taxation, socioeconomic, marketing, political, or other relevant issues which may materially affect the estimate of mineral resources and mineral reserves stated above.

18 Other Relevant Data and Information

The authors believe that the information presented within this reports meets all requirements for filling of the 43-101 report and that no additional explanations are required.

19 Interpretation and Conclusions

The Antamina ore body represents a large polymetallic skarn deposit with substantial resources. The resource estimate overall is adequate for quarterly, or for Life-of-Mine planning. Drill data density is adequate to enable a good product estimate to be realized on a monthly basis for a rolling 12 month period. The associated accuracy of the work done in support of the reserve and resource estimate meets or exceeds the accuracy expected at an operating mine for this type of deposit.

20 Recommendations

As mining progresses the deposit will require in-fill drilling to bring the confidence limits of future yearly production to a quarterly level. Grade control and blast hole sampling procedures have been developed and refined over ten years of production experience. Additional refinement of ore control and operational procedures will continue as operational experience increases. Additional drilling of resources for extending mine life will be scheduled as required to support future production requirements and studies for waste and tailings disposal will be developed.

At depth molybdenum will likely become an increasing contributor to block valuation. A metallurgical test program is required to confirm the level of amenability to processing of molybdenum resources at depth.



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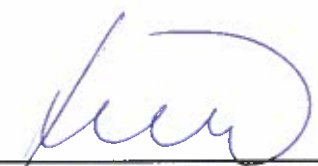
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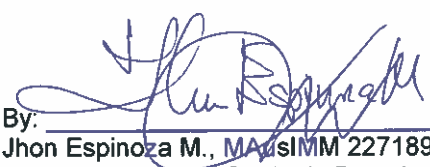
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TECHNICAL REPORT MINERAL RESERVES AND RESOURCES, ANTAMINA DEPOSIT, PERÚ 2010

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Qualified Person – Geologic Description and Resource Estimation

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23 Additional Requirements for Technical Reports on Development Properties and Production Properties

23.1 Introduction

A decision by the ownership group to proceed with the development of the Antamina Project (conditional on successful financing) was reached in the third quarter of 1998 concurrent with the final agreement of the Peruvian Government as to the development plan. Finalization of financing arrangements was obtained in 1999. The project development plan and financing agreement was based on the project scope and budget as finally detailed in *Antamina Project Updated Feasibility Study*, January 1999.

Pre-stripping mining operations began in late 1998. The onset of stripping coincided with the start of construction of fundamental infrastructure elements for the project, such as the access road and the construction camp. Concentrator construction began in the second quarter of 1999 followed by the start of port and pipeline construction in 2000.

Antamina is currently a producing property, with the mine and concentrator operating at the designed capacity.

23.2 Operations

Antamina has been a producing property for over ten years. Operations are described as currently conducted.

23.2.1 Mining

Antamina is a large open pit mining operation using standard mining equipment and methods. Drilling is done with large rotary drills and blasting uses bulk explosives. Electric cable shovels and haul trucks do the principal material movement mining in 15 meters benches. The fleet sizes for primary mine production are planned to be increased according with the current expansion project; fleet sizes are shown in Table 23.1.

Table 23.1: Primary Mine Production Equipment

Description	Nominal Size	# Units		
		2011	2013	2020
Bucyrus 49RIII Drill	351 mm	7	10	10
Ingersoll-Rand DMM2 Drill	211 mm	2	2	2
P&H 4100XPC Shovel	52 - 59 m ³	1	3	3
Bucyrus 495B Shovel	42 m ³	4	4	4
Letorneau L2350	40 m ³	1	3	3
Cat 994D/F F.E. Loader	16 m ³	4	3	1
Cat 793C/D/F Haul Truck	236 tonnes	71	118	138

Under the current life-of-mine plan, mine production is schedule to increase from approximately 150 million tonnes per year to over 240 million tonnes per year by 2013. Appropriate auxiliary equipment and light-duty vehicles provide support for the production operations.

Waste is hauled to final deposition on large waste dumps in areas outside the ultimate pit. Ore is either delivered directly to the Primary Crusher (located south of the pit in the Antamina valley) or to a stockpile for later feeding to the crusher. The long-term operational strategy is currently based on the use of a variable cut-off grade over time to improve the Net Present



Value of the project. As a consequence of this strategy, large ore stockpiles are created and then reclaimed through the life of the operation. This strategy is reviewed annually.

23.2.2 Processing

The Antamina Concentrator processes multiple ore types on a campaign basis. These campaigns range from a number of days to an entire month or longer depending upon ore development and concentrate marketing requirements. The Concentrator Facilities were designed to process an average of 70,000 tonnes per day of ore at an assumed 90% total plant availability. Optimizations of the various internal processes and additional capital investment in the mine to mill project, pebble crushing plant and option 5 project have resulted in a substantial increase in the average processing rate to approximately 106,000 tonnes per day at a realized 94% total plant availability. Although production through the concentrator is currently restrained due to failures in the SAG Mill and Ball Mill motor windings, plans have been implemented to ensure the realization of the higher throughput rate.

Currently run-of-mine (ROM) ore is delivered to the 1.524 meter x 2.261 meter primary gyratory crusher via 240 tonnes haulage trucks. The ore is reduced in size from 1.5 meters to a nominal -178 millimeter product. The primary crusher product is transported via a 3.0 kilometer conveyor through a 2.7 kilometer tunnel which runs southeast of the crusher location and into the Yanacancha valley where the Concentrator process facilities are located. The coarse ore conveyor passes the material onto a radial-arm stacker that deposits the crushed ore onto one of two 50,000 tonne live-capacity coarse ore stockpiles. Coarse ore is reclaimed by apron feeders located underneath the coarse ore stockpiles in a reclaim tunnel that runs longitudinally to the stockpiles. The SAG Mill feed conveyor discharges into a single, open circuit SAG Mill (11.6 meter x 6.4 meter EGL). Pulp discharged from the SAG Mill is distributed to three parallel Ball Mills (7.3 meter x 10.8 meter). The Ball Mills are operated in closed circuit with hydrocyclones and serve as the secondary stage of size reduction prior to the flotation circuits.

According with the current Expansion Project, the grinding circuit will be updated adding a second SAG Mill and a fourth ball mill to provide the required grinding power for 130,000 tpd and mitigate the production continuity risk associated with a single line operation (*Expansion Program Feasibility Report, October 2009*).

The initial flotation step following grinding is the Bulk Copper Rougher Circuit. The flotation product from the Bulk Copper Roughers is fed to the Bulk Copper Cleaner/Regrind Circuit. The cleaned flotation product from the Bulk Copper Cleaner/Regrind is then fed to the Bulk Copper Thickener.

From the Bulk Copper Thickener, the Bulk Copper concentrate is fed to the Bi/Mo Circuit. This flotation circuit operates in a bimodal fashion, depending on the ore type being campaigned. The circuit is used to either clean the Bulk Copper Concentrate of bismuth (a penalty element) by producing a Pb-Bi-Ag concentrate, or to separate a saleable Molybdenum Concentrate from the Bulk Copper feed material. The flotation product (either Pb-Bi-Ag concentrate or Mo concentrate) is fed to a small dewatering and product bagging plant. The 'tailings' product from the Bi/Mo Circuit is a final Copper Concentrate, which is fed to the Copper Concentrate Thickener.

The tailings from the Bulk Copper Rougher and Cleaner Circuits are fed to the Zinc Rougher Circuit. Zinc Rougher flotation product is processed through the Zinc Cleaner/Regrind Circuit whose product is the final Zinc concentrate, which is fed to the Zinc Concentrate Thickener. Tailings from the Zinc Circuits constitute the final tailings material, which is disposed of in the tailings impoundment.



Additionally the primary copper and zinc flotation circuits will be upgraded as required to maintain separation performance; concentrate dewatering will be also upgraded for bulk, zinc and final copper concentrates (*Expansion Program Feasibility Report, October 2009*).

The Antamina flotation flowsheet is designed to provide the flexibility to treat the different ore types as they are campaigned through the plant. Consequently, the plant is operated differently for different ore types as shown in Table 23.2.

Table 23.2: Mill Flotation Circuit Configuration by Ore Type

Ore Type		Cu Circuits	Bi/Mo Mode	Zn Circuits
M1	Copper Ore – Low Bismuth	On	Mo	Off
M2	Copper Ore – High Bismuth	On	Mo	Off
M2A	Copper Ore – Very High Bismuth	On	Mo	Off
M3	Copper Zinc Ore – Low Bismuth	On	Bi	On
M4	Copper Zinc Ore – High Bismuth	On	Bi	On
M4A	Copper Zinc Ore – Very High Bismuth	On	Bi	On
M5	Copper Bornite Ore	On	Off	Off
M6	Copper Zinc Bornite Ore	On	Off	On

23.2.3 Metallurgy

Because of the wide variability of potential ores, the Antamina deposit has undergone an exhaustive amount of metallurgical test work⁴¹ as listed above in Section 16.

The most significant conclusions from the Antamina Metallurgical test work are:

- Separation of the ore into distinct ore types will optimize the recovery of payable metals and provide the best opportunity to manage product quality.
- The ores are amenable to traditional differential flotation techniques using standard reagent schemes.
- The minerals of interest are quite coarse grained.
- Satisfactory metallurgical results are obtained at a P₈₀ secondary grind size of 150µm - 220µm for Copper-only ores (types 1, 2, and 2A), at a P₈₀ secondary grind size of 150µm - 220µm for Copper-Zinc ores (types 3, 4, and 4A) and at a P₈₀ secondary grind size of 140µm - 180µm for Bornite ores (types 5 and 6)
- There are a wide variety of grinding characteristics, both from the standpoint of total grind power required and from the standpoint of the relative power requirement split between primary (SAG) grinding and secondary (Ball Mill) grinding.
- Consequently, a great deal of operating flexibility and control was required from the design of the grinding circuits. This flexibility was further augmented with the incorporation of a pebble crushing circuit in 2008.
- For the high bismuth chalcopyrite ores (types 2A and 4A), bismuth minerals can be differentially floated from the Bulk Copper Concentrate by employing standard Lead-Copper separation techniques, albeit with some resultant copper and silver recovery losses.
- For Bornite ores, the primary bismuth mineral is wittichenite, which is present as very fine inclusions in bornite crystals.
- Consequently, Copper Concentrates from bornite ores are not practically separated by flotation for Bi and thus, have elevated bismuth contents.

⁴¹ Bechtel International Inc., *Antamina Project Feasibility Study*, March 1998, vol. 1a, sec. 6 and vols. 24-28.



- Silver generally reports to the final Copper Concentrate and Pb-Bi-Ag Concentrate.

Copper recoveries and grades vary by ore type. For chalcopyrite type ores, the copper recovery typically ranges from 70% to 95% and concentrate grade typically ranges from 27% to 30% Cu. In the case of the bornite ores, copper recovery typically ranges from 75-83% and the concentrate grade typically ranges from 38-42% Cu. Zinc recovery typically ranges from 0% (for copper only ores) to 90% and the concentrate grade typically ranges from 53-57% Zn. Molybdenum recovery typically ranges from 0% to 70% and concentrate grades of approximately 52% Mo. Silver recovery (to Copper Concentrate) typically ranges from 65% to 90%.⁴²

As part of the expansion program, metallurgical testing of core was carried out using samples from the 2007-2008 drilling program, each sample was prepared for flotation response testing with G & T Metallurgical Services and those results show that the expected metallurgical performance of the ore body will remain consistent with the performance obtained historically (*Expansion Program Feasibility Report, October 2009*).

23.2.4 Concentrate Handling

A concentrate pipeline transports final Copper and Zinc Concentrates from the mine site to the port site over 302 kilometers away at the coastal town of Huarney. A single pipeline is used to transport both concentrates (when appropriate, depending on the ore type being campaigned) on a batch-wise basis using an intervening “water plug” to separate the products. The minor quantities of Pb-Bi-Ag and Mo Concentrates produced are dewatered at the mine site, packaged in drums or bags and transported to various ports or clients using highway trucks.

Concentrates are drawn from the final thickeners into one of five large storage tanks ahead of the pipeline to manage the batching process. At the port site, the concentrate is received into three large storage tanks that provide surge capacity between the pipeline and the dewatering plant. Copper and Zinc Concentrates are dewatered to final shipping moisture content by large pressure filters before being deposited in a dry concentrate storage shed with a nominal capacity of 180,000 tonnes. Finally, concentrates are loaded onto ocean going vessels using a sophisticated ship loading facility at the port site.

23.2.5 Infrastructure Elements

In addition to the primary production steps outlined above, the Antamina Project has had to incorporate a number of significant infrastructure elements. The most important elements are: Access Road, Power lines, Tailings Impoundment, Freshwater Supply and Storage Dam, Truck Shop, Man Camp, and an Employee town site.^{43,44}

23.3 Mine Planning

Based on the Antamina 2010 Resource Model, Antamina staff continued technical evaluation. An optimization process was performed using Whittle 4X software; this model was updated with the latest information from the 2010 Resource block model, metallurgical recoveries, throughput, long term metal prices, costs, geotechnical domains and market conditions; only blocks classified as measured or indicated were used to define the pit limit. Final pit and phase design were performed using QPit software including operative and geotechnical constraints.

Long term metal prices used for the optimization process come from the company internal price protocols updated in April 2010.

The current life-of-mine plan was developed taking into account mine and concentrator capacities according to the current expansion project capacities. The waste dump capacity was updated with the latest operational, geotechnical, hydrology, legal and environmental criteria, and the

⁴² Bechtel International Inc., *Antamina Project Feasibility Study*, March 1998, vol. 1a, sec. 6.

⁴³ Bechtel International Inc., *Antamina Project Feasibility Study*, vols. 1a and 1b, March 1998.

⁴⁴ Bechtel International Inc., *Antamina Project Updated Feasibility Study*, vols. 1 and 2, January 1999.



remaining storage capacity of the tailings impoundment was calculated at the 4165 dam crest elevation.

23.3.1 Mine Planning Parameters

Key assumptions used to develop the current life-of-mine plan were updated using more recent information available.

The metal price protocols are based on independent and observable information and which result in impartial price estimates that can be verified by parties external to the company.

The protocols take the following principal considerations:

- US Dollars denomination in real terms as of January 1st of current year
- Use of market forward prices: LME for copper, zinc and lead, Comex for silver.
- Use of views from independent sources for each applicable year and long term projection. Use of interpolation in the case of molybdenum and bismuth if required.
- High and Low cases developed on the basis of correlation of historical prices to stock levels and/or historical price variance.

The detailed methodology for the April 2010 update is shown in Figure 23.1; the long term metal prices resulted are shown in Table 23.3.

Figure 23.1: Summary of April 2010 Price Protocol Considerations

5 Yr Plan >>>> Calendar Years >>>>		Yr1	Yr2	Yr3	Yr4	Yr5	Long Term
		2010	2011	2012	2013	2014	
S o f o r c e a t i o n	Copper US\$/lb	Average LME Fwd & Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the LT
	Zinc US\$/lb	Average LME Fwd & Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the LT
	Lead US\$/lb	Average LME Fwd & Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the LT
	Silver US\$/oz	Average Comex Fwd & Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the LT
	Molybdenum US\$/lb	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the year	Average of Views for the LT
	Bismuth US\$/lb	Recent Average	Interpolation to LT Price	Interpolation to LT Price	Interpolation to LT Price	Interpolation to LT Price	Historical LT Average
	Peruvian Sol PEN:USD	Average of Views for the year	Average of Views for the year	Medium term Projection	Medium term Projection	Medium term Projection	LT Projection

Table 23.3: Long Term Metal Prices

Metal	Price Protocol April 2010
Copper (US\$/lb)	1.99
Zinc (US\$/lb)	0.87
Molybdenum (US\$/lb)	10.19
Silver (US\$/oz)	10.18

Concentrate grades and metal recoveries are based on comprehensive models updated annually by Antamina metallurgical staff with actual mill performance, concentrate grades and metal recoveries are estimate for each significant metal in each concentrate by the individual ore types. For further information on metallurgical parameters see Section 23.2.3



Mine equipment productivities and mill throughput have been updated with actual information and projections base on the current expansion project (Compañía Minera Antamina - Expansion Program Feasibility Report, October 2009).

Geotechnical criteria are based on recommendations prepared by Piteau Associates, issued at the end of 2008 and revised by Antamina staff.

23.3.2 Block Value Calculation

Due to the polymetallic nature of the Antamina deposit, the presence of multiple ore types and a widely variation in concentrator throughput rates per ore type a profitability value was used. The profitability value is defined as the net profit before depreciation, financing expenses and taxes to be expected from a given unit of ore material per calendar hour after all relevant long-term costs (mining, concentrating, transportation, smelting, refining) are subtracted from the gross revenues derived from said unit. The profitability calculation incorporates expected metallurgical results and concentrator instantaneous throughput rates adjusted for expected concentrator availability

As a post-processing step, a profitability value (US \$/hour) was assigned to each interpolated block in the Antamina 2010 Resource Model. For validation, the profitability value for a random subset of the blocks (incorporating all ore types and a wide range of interpolated grade values) was calculated by hand and the results compared to ensure the two calculations produced similar results.

23.3.3 Pit Optimization

Using the given technical and economic parameters and the Antamina 2010 Resource Model, pit optimization was performed using the Whittle 4X software to produce a “family” of potential pit shells generated using a range of metal prices which start at a low value and are incrementally increased to the specified long term metal prices.

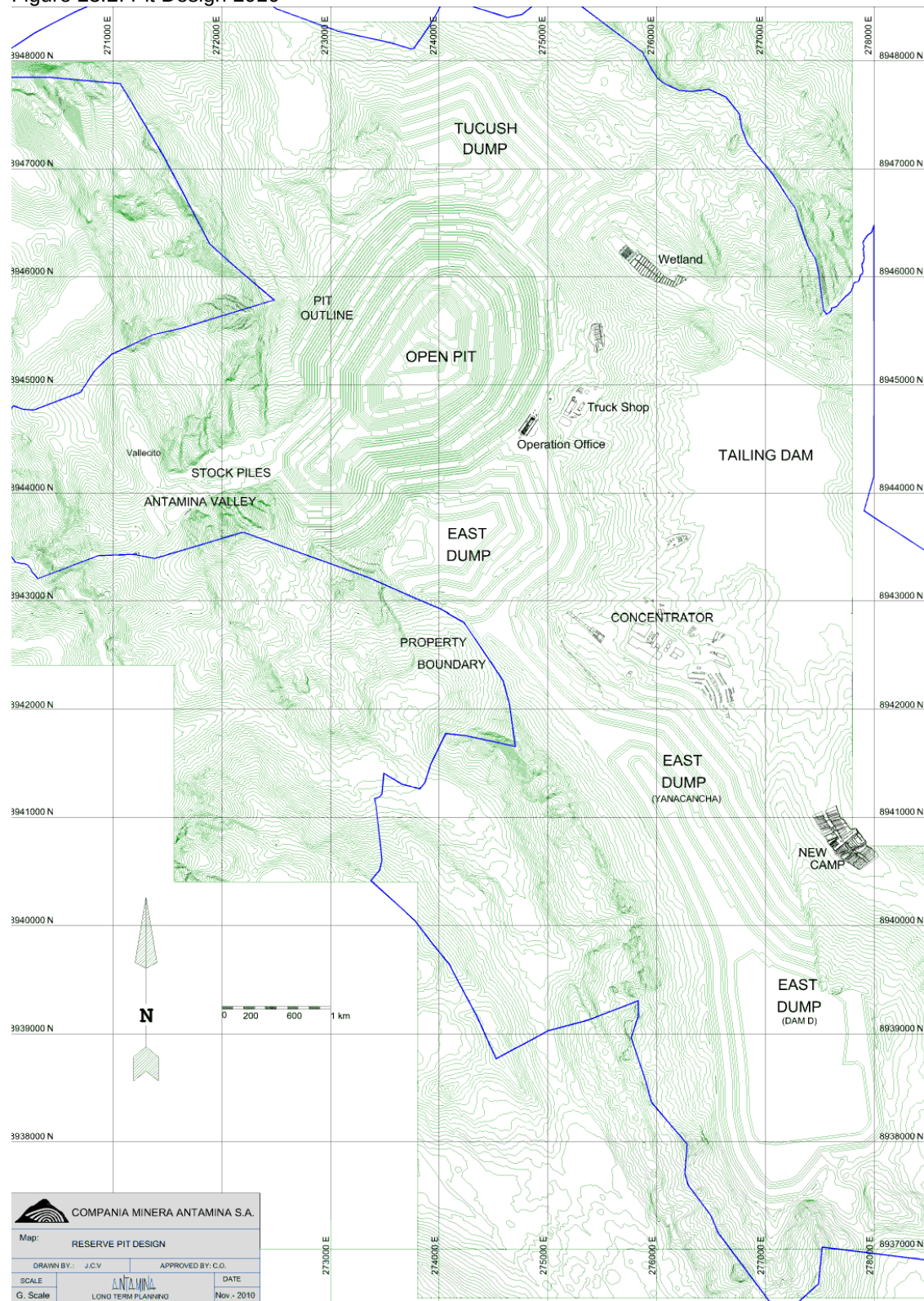
It is important to note that the Whittle 4X algorithm considered only measured and indicated blocks as revenue generators within the 2010 Antamina Resource Model while executing its optimization algorithm. Inferred blocks were automatically considered as “waste” for the purposes of this evaluation.

23.3.4 Pit Design

Antamina mine engineers used the Q’Pit software to produce a smoothed ultimate pit design. This design was based on the Pit Shell generated during the pit optimization work. This design hereafter referred to as Pit 2010, is shown in Figure 23.2. This smoothed design is an operative design that honors geotechnical wall angle constraints and Antamina mine design standards.

Within the ultimate pit, a series of fourteen successive pit phase designs have been established. Phase 1 and 2 has already been completed and mining is currently taking place in Phases 3, 4, 5, 6 and 7. The pit phasing roughly follows a sequence of intermediate pit shells from the pit optimization exercise while attempting to respect the mine design constraint of a 140 meter minimum mining width for phases.

Figure 23.2: Pit Design 2010





23.3.5 Cut-off Grades

Antamina continues with the use of cut-off grade strategy as developed using Comet software provided by Strategy Optimisation Systems Ltd. The current life-of-mine plan includes a variable cut-off policy for the first two years, beyond that period a flat cut-off was selected; the cut-off is based on the net value before taxes that the material will generate per hour of concentrator operation. The cut-off grades for the reserve life-of-mine plan are shown in Table 23.4. The cut-off grade is expressed in terms of dollars per hour pre-tax profit through the concentrator. This takes into account the variable cost and metallurgical performance of the various ore types processed.

Table 23.4: Cut-off Grades by Year

Year	2011	2012	2013	2014	2015	2016-2028
000 \$/hr	60	50	6	6	6	6

23.3.6 Mine Planning

The long range planning group within the Antamina Mine Engineering Department developed a Life-of-Mine plan and a mill feed schedule utilizing the 2010 Pit, the 2010 Antamina Resource Model and all known operating and production constraints. Only measured and indicated blocks within the ultimate pit were used in developing this mine plan. The LOM plan is technically and operationally viable. This LOM mine plan indicates that the reserves reported provide a positive net value. Mine production per year is shown in Table 23.5.

Table 23.5: Life-of-Mine Plan (million of tonnes).

LOM	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Total Mined	153	210	241	242	243	239	237	233	232	210
Total Ore Feed	39	49	48	48	48	48	47	48	48	48

LOM	2021	2022	2023	2024	2025	2026	2027	2028	Total
Total Mined	198	182	175	135	111	91	73	18	3,222
Total Ore Feed	49	48	48	48	48	48	49	16	822

23.4 Markets

Compañía Minera Antamina S.A. has long-term concentrate commitments of 800,000 tonnes per year of copper concentrate and 350,000 tonnes per year of zinc concentrate. These long-term contracts will expire at the end of 2013. Presently, all Copper and Zinc Concentrates produced in excess of the long-term commitments will be taken by the shareholder companies and shareholder companies will take all copper and zinc concentrates produced after the expiration of such long-term contracts.

Compañía Minera Antamina S.A. also has two contracts for Molybdenum Concentrate for 2011. One contract calls for delivery of 600 tonnes per month and the other one for a minimum of 420 tonnes per month and all excess production is also allocated to this second contract during 2011.

Concentrates from Compañía Minera Antamina S.A. are sent to smelters around the world including South America, Asia, North America, and Europe.

23.5 Contracts

All contracts for mine site services, smelting and refining, and transportation entered into by Compañía Minera Antamina S.A. are negotiated to be within prevailing market parameters and terms.



23.6 Environmental Considerations

A discussion of the Environmental Aspects of the Antamina project and long term liabilities is contained in Section 4.8 Environmental Liabilities. From an economic standpoint it should be noted that financial guarantees for closure of mining operations in Peru is required.

Compañía Minera Antamina S.A. is planning ongoing reclamation activities to be carried out throughout the life of the operation. Significant reclamation and revegetation work has already been carried out during the project construction phase to remediate disturbed areas and top soils are being stockpiled for future. These stockpiles are stabilized with the growth of ichu grass.

Compañía Minera Antamina S.A. has developed a Closure Plan document which lays out the long-term remediation strategy. This was the first closure plan accepted by the Ministry of Energy and Mines in Peru.

23.7 Taxes

As part of the investment agreement entered into in September 16th 1998 between Compañía Minera Antamina S.A. and the Peruvian Government, Compañía Minera Antamina S.A. has a guaranteed Tax Stability Regime for a period of 15 years for all significant taxation policies. The 15-year started in January 2001 and will be due on December 31st 2015.

In summary:

- There is an 8-year carry forward of Tax Losses which can take advantage of the depreciation of the original capital investment.
- There is a legislated "employee profit sharing" levy of 8% of Net Income before taxes that is distributed to all workers after year-end. This distribution to the employees has a set maximum and if the distribution exceeds that maximum to the employees, the remainder is returned to the Peruvian Government.
- There is a prevailing Income Tax of 30%.
- There is an 18% Value Added Tax, known in Perú as IGV, against the value of goods and services purchased and then balanced against the value of goods sold. For Exported Goods, the IGV paid to produce is refunded. Consequently, the net effect for Compañía Minera Antamina S.A. is effectively zero, except for the working capital involved to administer the rolling collection versus refund cycle. Additionally, Compañía Minera Antamina S.A. benefited from a special law during the construction period, which saw early recovery of IGV paid on construction purchases, even though no products were being produced to offset.
- There is a 12% import duty on machinery, equipment and supplies coming into Perú. Compañía Minera Antamina S.A. has the right to spread the payment of any of these duties incurred during the Construction Period over a 6.5-year period.
- Antamina is provided a benefit of Reinvestment of Profits (ROP) for all capital investments that increase production by 10% or more. The capital investment can be deducted against pre-tax profits. This program does not impede the normal use of depreciation for all installations and equipment.
- As at December 2009, Compañía Minera Antamina received approval from the government of Peru for the use of Reinvestment of Profits (ROP) for an amount of \$900M for the expansion of the operation. The expansion project is slated to cost \$1.3 billion. Through this mechanism the Company obtains an income tax shield equivalent to 30% of the ROP approved amount, for the execution period of September 2009 – March 2013.



However, despite the tax stability agreement the new government resulting from the election of Alan Garcia as president of Peru has resulted in Antamina's participation in a "voluntary contribution" program beginning for the full calendar year 2006 (despite the agreement being finalized on December 22, 2006). The program life extends to December 31, 2010. This program mandates an additional 3.75 percent of after tax profits to go into a development fund for use in the impacted areas of Ancash province. This fund will be administered by Antamina personnel to ensure that it is used effectively. The agreement is in effect for Antamina for each year (January through December) when the LME (London Metal Exchange) copper price is over \$1.79 per pound. If the average price is lower, then no contribution is made. The basis price is to be adjusted annually based on the United States of America's Producer Price Index.

23.8 Capital and Operating Cost Estimates

Capital and operating cost estimates have been prepared in conjunction with the economic analysis done to verify the Life-of-Mine Plan prepared to support the new reserves. The average long term operating costs are shown in Table 23.6.

Table 23.6: Long Term Operating Cost

Parameter	Operating Cost
Mining	US\$ 2.09/t mined
Processing Plant	US\$ 4.06/t milled

23.9 Economic Analysis

A summary cash flow analysis based on the production schedule for the Proven and Probable Mineral Reserve was produced and is shown in Table 23.7. The analysis was done using the Antamina price/cost protocol and is consistent with the stated mineral reserves, and demonstrates a positive Net Present Value (NPV) for Antamina. The capital for the initial investment has already been repaid. Assumed economic parameters are as per Table 23.3. This positive value demonstrates that the Proven and Probable Mineral Reserve as stated, does in fact constitute ore and meets the criteria laid out in the Canadian Securities Administrators' System for Electronic Document Analysis and Retrieval (SEDAR) section 43-101 to classify it as a reportable mineral reserve.

Table 23.7: Project Net Cash Flows (US\$ million, undiscounted)

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
Cash Flow (US\$ M)	1,397	1,576	1,507	1,009	1,312	916	675	1,013	623

Year	2020	2021	2022	2023	2024	2025	2026	2027	2028
Cash Flow (US\$ M)	165	729	599	716	523	368	777	1,242	-13

23.10 Sensitivities

Table 23.8 shows Antamina's cash flows sensitivity to the copper and zinc prices, and the operating cost. Variations in the copper price will have the more significant impact, but even under a 10% drop in the copper price, cash flows are still significantly value accretive.



Table 23.8: Sensitivities

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
+10% Cu Price	1,632	1,792	1,674	1,116	1,475	1,017	791	1,140	719
-10% Cu Price	1,162	1,359	1,341	902	1,154	810	589	888	525
+10% Zn Price	1,429	1,588	1,524	1,028	1,324	946	705	1,029	651
-10% Zn Price	1,367	1,563	1,490	989	1,300	885	645	997	596
+5% Operating Cost	1,369	1,560	1,484	987	1,293	894	652	995	599
-5% Operating Cost	1,425	1,591	1,530	1,030	1,332	938	698	1,032	648

Year	2020	2021	2022	2023	2024	2025	2026	2027	2028
+10% Cu Price	239	824	688	823	601	430	873	1,395	7
-10% Cu Price	91	635	510	608	445	306	681	1,088	-33
+10% Zn Price	184	733	624	728	538	385	787	1,253	-4
-10% Zn Price	146	726	573	703	507	352	768	1,230	-22
+5% Operating Cost	141	710	576	693	501	348	763	1,222	-30
-5% Operating Cost	189	749	622	739	545	389	792	1,261	4

23.11 Payback

As the main investment has already been made, the payback period is not relevant.

23.12 Mine Life

The projected life for the schedule prepared in support of the Mineral Reserve statement is approximately 18 years beginning in 2011 and finishing in 2028. A conceptual study oriented to identify additional areas for waste and tailings disposal is ongoing, this project is supported by the additional resources identified beyond the limit of the current life-of-mine.

An intensive program of drilling is scheduled to be executed during the next five years in order to expand both resources and reserves.



Appendix A - Certificate of Authors



CERTIFICATE OF AUTHOR

Luis Lozada, Member AusIMM
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llozada@antamina.com

I, Luis Lozada, Member AUSIMM, am the Mine Engineering Superintendent for Compañía Minera Antamina S.A., corporate office located at Av. El Derby 055, Torre 1 – Oficina 801, Santiago de Surco, Lima 33 - Perú.

I graduated from the Pontificia Universidad Catolica Del Perú with a Bachelor of Science degree in Mining Engineering in 2001. I am a member of the Australasian Institute of Mining & Metallurgy (305930).

Since 1997 I have been continually working in the mining industry, I am employee of Compañía Minera Antamina S.A. and have worked at the Antamina mine site since January 2000.

I have read the definition of "qualified person" set out in National Instrument 43-101 ("NI 43-101") and certify that by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101 for this report.

I visit periodically the property due to the nature of my employment; the most recent personal inspection to the mine site was conducted during the week of November 29, 2010.

I have overall responsibility for the technical report titled MINERAL RESERVES AND RESOURCES, ANTAMINA DEPOSIT, PERÚ 2010 with an effective date of January 1st, 2011, relating to the Antamina Property.

I am not independent of the issuer as described in Section 1.4 of NI 43-101. I am employee of Compañía Minera Antamina S.A. since January 2000.

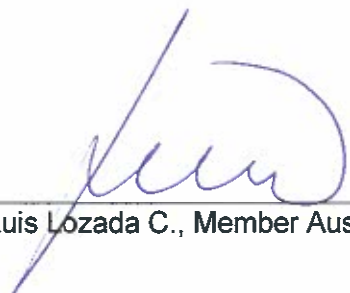
I have read NI 43-101 and Form 43-101F1, and this report has been prepared in compliance with that instrument and form.

As of the date of this certificate, to the best of my knowledge, information and belief, the technical report contains all scientific and technical information that is required to be disclosed to make the technical report not misleading.

I consent to the filing of the Technical Report with any Canadian stock exchange and other Canadian regulatory authorities and publication by them for regulatory purposes, including electronic publication in the public company files on their websites accessible by the public, of the Technical Report.

Dated the 31st day of January, 2011.

"Signed and sealed"



Luis Lozada C., Member AusIMM.



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CERTIFICATE OF AUTHOR

Jhon Espinoza, Member AusIMM
Av. El Derby 055, Torre 1 – Oficina 801
Santiago de Surco - Lima 33, Perú
Tel 511 217-3180; Fax 511 217-3055
jespinoza@antamina.com

I, Jhon Espinoza, Member AUSIMM, am the Development Geology Superintendent for Compañía Minera Antamina S.A., corporate office located at Av. El Derby 055, Torre 1 – Oficina 801, Santiago de Surco, Lima 33 - Perú.

I graduated from the Universidad Nacional de Ingenieria Peru with a Bachelor of Science degree in Geological Engineering in 1989. I am a member of the Australasian Institute of Mining & Metallurgy (227189).

Since 1990 I have been continually working in the mining industry, I am employee of Compañía Minera Antamina S.A. and have worked at the Antamina mine site since November 1999.

I have read the definition of "qualified person" set out in National Instrument 43-101 ("NI 43-101") and certify that by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101 for this report.

My endeavors as Geology Superintendent were conducted at the mine site as of September 2010 and therefore the most recent personal inspection to the mine site was conducted during the week of September 30, 2010.

I have responsibility on the Geology and Mineral Resource chapters for the technical report titled MINERAL RESERVES AND RESOURCES, ANTAMINA DEPOSIT, PERÚ 2010 with an effective date of January 1st, 2011, relating to the Antamina Property.

I am not independent of the issuer as described in Section 1.4 of NI 43-101. I am employee of Compañía Minera Antamina S.A. since November 1999.

I have read NI 43-101 and Form 43-101F1, and this report has been prepared in compliance with that instrument and form.

As of the date of this certificate, to the best of my knowledge, information and belief, the technical report contains all scientific and technical information that is required to be disclosed to make the technical report not misleading.

I consent to the filing of the Technical Report with any Canadian stock exchange and other Canadian regulatory authorities and publication by them for regulatory purposes, including electronic publication in the public company files on their websites accessible by the public, of the Technical Report.

Dated this 31st day of January, 2011.

"signed and sealed"

A handwritten signature in blue ink, appearing to read "Jhon Espinoza", written over a horizontal line.

Jhon Espinoza M., Member AusIMM.



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Appendix B – Main Permits Required for Operations

ENVIRONMENTAL IMPACT ASSESSMENT

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
1	Modification of the Environmental Impact Assessment - Variant of the Section P"0" - S.E. Antamina of the Auxiliary Power Line 220 kV. S.E. Vizcarra - S.E. Antamina	DGAAM - MEM	Directorial Resolution No. 222-2010-MEM-AAM	June 25, 2010	Permanent	Non applicable
2	Modification of the Environmental Impact Assessment - Auxiliary Power Line 220 kV. S.E. Vizcarra - S.E. Antamina	DGAAM - MEM	Directorial Resolution No. 280-2009-MEM-AAM	September 11, 2009	Permanent	Non applicable
3	Environmental Impact Assessment - Extension of the Open Pit and Process Optimization	DGAAM - MEM	Directorial Resolution No. 091-2008-MEM/AAM	April 22, 2008	Permanent	Non applicable
4	Modification of the Environmental Impact Assessment - Environmental Management Plan Update	DGAAM - MEM	Directorial Resolution No. 381-2006-MEM-AAM	September 4, 2006	Permanent	Non applicable
5	Modification of the Environmental Impact Assessment - Complementary Report of the Port	DGAAM - MEM	Directorial Resolution No. 420-2003-EM-DGAA	October 21, 2003	Permanent	Non applicable
6	Modification of the Environmental Impact Assessment - Relocation of the Sector V2 - V3 of the Power Line 220 kV. Huallanca - Antamina	DGAAM - MEM	Directorial Resolution No. 274-2002-EM-DGAA	September 13, 2002	Permanent	Non applicable
7	Modification of the Environmental Impact Assessment - Project Design, Social Component and Environmental Plan of the Canrash and Conococha Lakes	DGAAM - MEM	Directorial Resolution No. 184-2002-EM/DGAA	June 26, 2002	Permanent	Non applicable
8	Addendum 3 of the Environmental Impact Assessment	DGM - MEM	Directorial Resolution No. 065-99-EM/DGM	March 12, 1999	Permanent	Non applicable
9	Environmental Impact Assessment of the Antamina Project and Addendum 1 and 2.	DGM - MEM	Directorial Resolution No. 169-98-EM/DGM	July 15, 1998	Permanent	Non applicable

CLOSURE PLAN

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
10	Modification of the Mine Closure Plan - Inclusion of the Auxiliary Power Line	DGAAM - MEM	Directorial Resolution No. 387-2010-MEM-AAM	November 22, 2010	Permanent	Non applicable
11	Mine Closure Plan of the Antamina Project	DGAAM - MEM	Directorial Resolution No. 373-2009-MEM-AAM	November 21, 2009	Permanent	Non applicable

BENEFICIATION CONCESSION

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
12	Modification of the Beneficiation Concession - Construction Permit (130,000 tpd case)	DGM - MEM	Resolution No. 638-2009-MEM-DGM-V	August 24, 2009	Permanent	Non applicable
13	Modification of the Beneficiation Concession - Operation Permit of the Pebble Crusher	DGM - MEM	Directorial Resolution No. 930-2008-MEM-DGM	May 6, 2008	Permanent	Non applicable
14	Modification of the Beneficiation Concession - Construction Permit of the Pebble Crusher	DGM - MEM	Resolución No. 796-2007-MEM-DGM-V	January 30, 2007	Permanent	Non applicable
15	Modification of the Beneficiation Concession - Operation License (104,000 tpd case)	DGM - MEM	Resolution No. 112-2007-MEM-DGM/V	December 07, 2004	Permanent	Non applicable
16	Beneficiation Concession HUINCUSH - Operation Permit (70,000 tpd case)	DGM - MEM	Directorial Resolution No. 076-2001-EM-DGM and Directorial Resolution No. 127-2001-EM-DGM	July 24, 2001	Permanent	Non applicable
17	Beneficiation Concession - Construction Permit of the Concentrator Plant, Plant of the Port and auxiliary facilities (70,000 tpd case)	DGM - MEM	Directorial Resolution No. 107-99-EM/DGM	June 25, 1999	Permanent	Non applicable

PIPELINE

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
18	Modification of the Mining Transport Concession - Construction Approval and Pipeline Coordinates Modification	DGM - MEM	Directorial Resolution No. 1113-2008-MEM/DGM	November 12, 2008	Permanent	Non applicable
19	Modification of the Mining Transport Concession - Construction Permit of the Variant in Canrash Zone	DGM - MEM	Resolution No. 796-2007-MEM-DGM/V	Jun 8, 2007	Permanent	Non applicable
20	Modificación of the Pipeline Easement	DGM - MEM	Supreme Resolution No. 023-2003-EM	Jun 17, 2003	Permanent	Non applicable
21	Pipeline Easement (657,241.49 m2)	DGM - MEM	Supreme Resolution No. 022-2003-EM	May 3, 2003	Permanent	Non applicable
22	Pipeline Easement (29,911.57 m2)	DGM - MEM	Supreme Resolution No. 002-2003-EM	January 15, 2003	Permanent	Non applicable
23	Mining Transport Concession "MINERODUCTO"	DGM - MEM	Directorial Resolution No. 209-99-EM/DGM	November 15, 1999	Permanent	Non applicable

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POWER LINE

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
24	Modification of the Electrical Concession of the Power Line	DGE - MEM	Supreme Resolution No. 039-2002-EM	October 16, 2002	Permanent	Non applicable
25	Power Line Easement	DGE - MEM	Ministerial Resolution No. 689-99-EM/VME	December 13, 1999	Permanent	Non applicable
26	Electrical Concession of the Power Line	DGE - MEM	Supreme Resolution No. 034-99-EM	February 24, 1999	Permanent	Non applicable

PORT

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
27	Declaration of Compliance of Port Facility (DCIP)	APN	Certificate No. 013-2009-DCIP-APN	August 6, 2009	5 years, with a yearly ratification	August 06, 2014
28	Security Certificate of Port Facility (CSIP)	APN	Certificate No. 011-2009	April 27, 2009	5 years, with a yearly ratification	April 27, 2014
29	Modification of the Concession of Usage of Aquatic Area	DICAPI	Directorial Resolution No. 0649-2003/DCG	October 17, 2003	Non applicable	Non applicable
30	Concession of Usage of Aquatic Area	DICAPI	Directorial Resolution No. 0367-2003/DCG	May 7, 2003	30 years	November 8, 2030 (R.D. 0367-2003/DCG, the term is computed as of the issuance date of the R.D. 0544-2000/DCG)

WATER USE

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
31	Authorization of Water Pumping Project - Yanacocha (Nescafe) Lake	ATDR Huari	Administrative Resolution No. 033-2005-DR.AG-ANCASH/ATDRH/AT	September 21, 2005	Permanent	Non applicable
32	License to Use Surface Water - Antamina and Huincush Streams	ATDR Pomabamba	Administrative Resolution No. 003-2001-ANCASH-DR.AG-ATDRP/AT	January 05, 2001	Permanent	Non applicable
33	License to Use Surface Water - Huincush Stream	ATDR Pomabamba	Administrative Resolution No. 002-2001-ANCASH-DR.AG-ATDRP/AT	January 05, 2001	Permanent	Non applicable
34	License to Use Ground Water - Antamina Stream	ATDR Pomabamba	Administrative Resolution No. 001-2001-ANCASH-DR.AG-ATDRP/AT	January 05, 2001	Permanent	Non applicable

TREATMENT AND DISCHARGE WASTE WATER

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
35	Authorization to Discharge Waste Water - Pampa Moruna Stream	ANA	Directorial Resolution No. 0037-2009-ANA-DCPRH	December 10, 2009	2 years	December 10, 2011
36	Authorization to Discharge Waste Water - Ayash Stream	ANA	Directorial Resolution No. 0026-2009-ANA-DCPRH	October 30, 2009	2 years	October 30, 2011
37	Authorization for Project Domestic Wastewater System Treatment - Port Camp	DIGESA	Directorial Resolution No. 0241-2000-DIGESA/SA	June 1, 2000	Permanent	Non applicable
38	Authorization for Project Domestic Wastewater System Treatment - Yanacancha Camp	DIGESA	Directorial Resolution No. 390/99/DIGESA/SA	September 22, 1999	Permanent	Non applicable

TREATMENT AND REUSE WASTE WATER

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
40	Authorization of Waste Water Reuse - Punta Lobitos Treatment Plant	ANA	Directorial Resolution No. 001-2011-ANA-DGCRH	January 4, 2011	2 years	January 04, 2013
41	Authorization for Project Domestic Wastewater System Treatment - Port Facilities	DIGESA	Directorial Resolution No. 406-2001-DIGESA/SA	May 14, 2001	Permanent	Non applicable

TREATMENT POTABLE WATER

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
42	Sanitary Authorization of System Water Treatment for Human Consumption - New Camp	DIGESA	Directorial Resolution No. 0038-2010/DSB/DIGESA/SA	April 16, 2010	Permanent	Non applicable
43	Approval of the Treatment System to Potable Water - Port Camp	DIGESA	Directorial Resolution No. 240-2000-DIGESA/SA	June 1, 2000	Permanent	Non applicable
44	Approval of the Treatment System to Potable Water - Port Facilities	DIGESA	Directorial Resolution No. 154-2000-DIGESA/SA	April 13, 2000	Permanent	Non applicable
45	Authorization for Project Water Treatment System for Human Consumption - Yanacancha Antamina Camp	DIGESA	Directorial Resolution No. 0405/99/DIGESA	September 22, 1999	Permanent	Non applicable

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HYDROCARBONS STORAGE

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
46	Hydrocarbon Register for the Lubricant Tanks of the Mine (4450) Fuel Station	OSINERGMIN	Register 0001-CDFJ-02-2010	March 3, 2010	Permanent	Non applicable
47	Hydrocarbon Register for the Lubricant Tanks of the Primary Crusher	OSINERGMIN	Register 0002-CDFJ-02-2010	March 3, 2010	Permanent	Non applicable
48	Hydrocarbon Register for the LPG Tank of the New Camp	OSINERGMIN	Register 0001-CDGL-02-2010	February 1, 2010	Permanent	Non applicable
49	Hydrocarbon Register for Mine (4450) Fuel Station (Diesel)	OSINERGMIN	Register 0001-CDFJ-02-2009	February 2, 2009	Permanent	Non applicable
50	Hydrocarbon Register for the Primary Crusher Fuel Station (Diesel)	OSINERGMIN	Register 0010-CDFJ-02-2007	December 13, 2007	Permanent	Non applicable
51	Hydrocarbon Register for the LPG Tank of the Former Camp (Sodexo Kitchen)	OSINERGMIN	Register 0002-CDGL-02-2007	September 18, 2007	Permanent	Non applicable
52	Hydrocarbon Register for the Camp Fuel Station (Diesel and Gas 90)	OSINERGMIN	Register 0003-CDFJ-02-2006	February 17, 2006	Permanent	Non applicable
53	Hydrocarbon Register for the Geology Laboratory's LPG tank	OSINERGMIN	Register 0001-CDGL-02-2004	February 26, 2003	Permanent	Non applicable
54	Hydrocarbon Register for the Concentrator's Diesel Tank	OSINERGMIN	Register 0008-CDFJ-02-2003	June 23, 2003	Permanent	Non applicable
55	Hydrocarbon Register for Port Facility Diesel Tank	OSINERGMIN	0005-CDFJ-02-2001	July 5, 2001	Permanent	Non applicable
56	Hydrocarbon Register for Port Facility Fuel Station (Diesel)	OSINERGMIN	Register 0000026-ANC	January 24, 2000	Permanent	Non applicable

EXPLOSIVES

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
57	Renewal of Operation License - 2 Explosives Magazines (Containers)	DICSCAMEC	Directorial Resolution No. 253-2011-IN-1703-2	January 21, 2011	1 year	January 21, 2012
58	Global Authorization to Use or Acquire Explosives - First Semester 2011	DICSCAMEC	Directorial Resolution No. 5523-2010-IN-1703-2	December 29, 2010	6 months	June 30, 2011
59	Certificate of Mining Operation 2011	DGM - MEM	COM 024-2011-A	December 2, 2010	1 year	December 31, 2011
60	Renewal of Operation License - 5 Explosives Magazines (Silos)	DICSCAMEC	Directorial Resolution No. 3545/2010-IN-1703-2	August 25, 2010	1 year	August 25, 2011
61	Renewal of Operation License - 1 Explosive Magazine (Ammonium Nitrate Container)	DICSCAMEC	Directorial Resolution No. 4393/2008-IN-1703-2	December 4, 2008	5 years	December 4, 2013

RESTRICTED CHEMICALS

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
62	Sole Register for the Control of Restricted Chemicals	Ministry of Production	Record 02-00186	April 09, 2010	2 years	April 09, 2010
63	User's Certificate for the Use of Restricted Chemicals for the Mine	DIVANDRO	User's Certificate 20330262428-2010-DIVANDRO-HZ	February 17, 2010	2 years	February 17, 2012
64	User's Certificate for the Use of Restricted Chemicals for the Port Facility	DIVANDRO	User's Certificate 20330262428-2010-DIVANDRO-HZ, Annex 1.	February 12, 2010	2 years	February 12, 2012



ARCHAEOLOGICAL REMAINS

No.	Permit / License	Authority	Resolution	Date	Term	Expiration Date
65	Certificate of Non Existence Archaeological Remains - Variant V6 to V8 of the Auxiliary Power Line 220 Kv. S.E. Vizcarra - S.E. Antamina: Sector 3.	INC	CIRA No. 2010-078-ANC	June 14, 2010	Permanent	Non applicable
66	Certificate of Non Existence Archaeological Remains - Variant V6 to V8 of the Auxiliary Power Line 220 Kv. S.E. Vizcarra - S.E. Antamina: Sector 2.	INC	CIRA No. 2010-077-ANC	June 14, 2010	Permanent	Non applicable
67	Certificate of Non Existence Archaeological Remains - Variant V6 to V8 of the Auxiliary Power Line 220 Kv. S.E. Vizcarra - S.E. Antamina: Sector 1.	INC	CIRA No. 2010-076-ANC	June 14, 2010	Permanent	Non applicable
68	Certificate of Non Existence Archaeological Remains - Variant V0 to V1A of the Auxiliary Power Line 220 Kv. S.E. Vizcarra - S.E. Antamina.	INC	CIRA No. 2010-075-ANC	June 14, 2010	Permanent	Non applicable
69	Certificate of Non Existence Archaeological Remains - Variant of the Auxiliary Power Line 220 Kv. S.E. Vizcarra - S.E. Antamina: Sector 2.	INC	CIRA No. 2009-05	October 5, 2009	Permanent	Non applicable
70	Certificate of Non Existence Archaeological Remains - Variant of the Auxiliary Power Line 220 Kv. S.E. Vizcarra - S.E. Antamina: Sector 1.	INC	CIRA No. 2009-04	October 5, 2009	Permanent	Non applicable
71	Certificate of Non Existence Archaeological Remains - Auxiliary Power Line 220 Kv. S.E. Vizcarra - S.E. Antamina.	INC	CIRA 2009-0415	June 15, 2009	Permanent	Non applicable
72	Certificate of Non Existence Archaeological Remains - Monitoring Wells Port Area 2	INC	CIRA No. 2008-0164	March 12, 2008	Permanent	Non applicable
73	Certificate of Non Existence Archaeological Remains - Monitoring Wells Port Area 1	INC	CIRA No. 2007-0400	October 31, 2007	Permanent	Non applicable
74	Certificate of Non Existence Archaeological Remains - Variant of the Power Line 220 Kv. Vizcarra - Sector V2 y V3	INC	CIRA No. 2002-0305	03 de setiembre de 2002	Permanent	Non applicable
75	Modification of the Certificate of Non Existence Archaeological Remains - Port Area	INC	CIRA No. 2001-085	05 de julio de 2001	Permanent	Non applicable
76	Certificado de Inexistencia de Restos Arqueológicos - Variante de la línea de transmisión 220 Kv. Vizcarra - Antamina Sector V2 y V3	INC	CIRA No. 99-0126	— setiembre de 1999	Permanent	Non applicable
77	Certificate of Non Existence Archaeological Remains - Pipeline	INC	CIRA No. 99-0101	16 de setiembre de 1999	Permanent	Non applicable
78	Certificate of Non Existence Archaeological Remains - Mine Area	INC	CIRA No. 98-0094	02 de diciembre de 1998	Permanent	Non applicable
79	Certificate of Non Existence Archaeological Remains - Port Area	INC	CIRA No. 98-0052	11 de junio de 1998	Permanent	Non applicable